

# RF and Microwave Engineering

## Complete Numerical Problems and Solutions

EX 716 (BEI, IV/I) · EX 752 (BEX, IV/II)

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All **56 solved problems** from the five problem chapters, in one volume. Every problem is a real question drawn from **all 24 past papers (2069–2082 BS)** and opens with the **question exactly as the paper prints it**, the paper's own spelling and slips included, so it can be attempted from this PDF alone.

The Chapter 2 problems are worked as **Smith chart constructions**, step by step: which circle to draw, which arc to follow, what to read off the rim, because that is what the examiner marks. The numbers beside each step are computed exactly from the line equations, so they are what a perfectly drawn chart would give; a hand-drawn reading will differ in the second decimal and that is fine.

**Chapters 1, 5 and 8 set no numerical** in 24 papers, so they do not appear here. Theory lives in the companion volume, **RF and Microwave Engineering: Complete Exam Notes**.

| Ch    | Topic                                  | Problems | Biggest family                                               |
|-------|----------------------------------------|----------|--------------------------------------------------------------|
| 2     | RF and M/W Transmission Lines          | 24       | Double-stub tuning, <b>17/24 papers</b>                      |
| 3     | RF and M/W Network Theory and Analysis | 3        | The four tests on a printed S-matrix                         |
| 4     | RF/Microwave Components and Devices    | 17       | Magic-tee family: every printed 4-port is a <b>magic tee</b> |
| 6     | RF Design Practices                    | 10       | Transistor stability and maximum gain, <b>22/24 papers</b>   |
| 7     | Microwave Antennas and Propagation     | 2        | Radiation zones, with the $D > \lambda$ validity check       |
| Total |                                        | 56       |                                                              |

**Every answer in this volume is machine checked.** `src/rf.py` solves the transmission line and both stub tuners exactly, with no chart read and no formula quoted from memory. `src/verify.py` then walks each **published** design forward through the line equations and asserts  $y_{in} = 1 + j0$ , re-deriving every susceptance from the **published stub length**, so a wrong length cannot hide behind a right  $b$ . `src/sparam.py` re-derives each device matrix from its stated properties rather than quoting it, brute-forces the three-port impossibility theorem, and reduces an  $n$ -port by terminating ports. `src/wg.py` tests every printed waveguide field component against both curl equations and the walls, and is mutation tested. `src/amp.py` and `src/filt.py` reproduce the deck's worked amplifier and filter examples and sample the stability circles. `src/ant.py` asserts every radiation-zone number, including the cases where the standard formulas are **not valid**. `src/smith.py` draws each chart as a **construction** and asserts its own geometry before rendering.

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## 2. RF and M/W Transmission Lines: Numerical Problems

24 solved PYQ constructions · **≈230 marks** across 24 papers · 1 chart reading, 6 single-stub, 17 double-stub

### READ THIS FIRST

Every problem below is worked as a **Smith chart construction**, step by step — which circle to draw, which arc to follow, what to read off the rim — because that is what the examiner marks. The numbers were computed exactly from the transmission-line equations, so they are what a perfectly drawn chart would give; your hand-drawn readings will differ in the second decimal place and that is fine.

- **Two solutions are normal.** Single-stub and double-stub tuners each have two valid answers. Quote either — but say which one you chose, and why.
- **Always check the forbidden region** before starting a double stub:  $g_1 \leq 1/\sin^2 \beta d$ . Stating this earns marks even when the load passes.
- **Never drop the S-matrix rider** — §2.8, two lines, 2–6 marks.

### M. The Three Methods, in Brief

#### M1 — Reading the chart (Problem 1)

1. Normalise  $z_L = Z_L/Z_0$  and plot.
2. Swing the **SWR circle** about the centre through that point.  $|\Gamma_L|$  is its radius;  $S$  is where it cuts the right-hand real axis.
3. Moving along the line is **rotation at constant radius**: clockwise toward the generator, one lap =  $\lambda/2$ . Read distances on the WTG rim scale.
4. The circle crosses the horizontal axis twice per lap — those are the only **purely resistive** points, at  $V_{\max}$  ( $r = S$ ) and  $V_{\min}$  ( $r = 1/S$ ).

#### M2 — Single-stub shunt tuner (Problems 2–7)

1. Plot  $z_L$ , half-turn to  $y_L$  (shunt stubs *add susceptance*, so all stub work is done in admittance).
2. Rotate clockwise to the  $g = 1$  **circle**. It is cut **twice** →  $d_1, d_2$ .
3. There  $y = 1 \pm jb$ . The stub must supply  $\mp jb$ .
4. For the length, start at the stub's own termination — **open** stub at  $y = 0$ , **short** stub at  $y = \infty$  — and run clockwise round the rim to that susceptance.  
 $\ell_{\text{short}} = \ell_{\text{open}} \pm \lambda/4$ .

#### M3 — Double-stub shunt tuner (Problems 8–24)

1. Plot  $z_L$ , half-turn to  $y_L$ ; if the first stub is offset, rotate clockwise to get  $y_1$ , the admittance **just before stub 1**.
2. **Check**  $g_1 \leq 1/\sin^2 \beta d$ . If it fails, no length of stub can match this load at this spacing.
3. Draw the **spacing circle**: the  $g = 1$  circle rotated the stub spacing  $d$  *toward the load*.
4. Stub 1 can only slide the point **along its own constant- $g$  circle**. Follow that circle to its two intersections with the spacing circle →  $y_a, y'_a$ . Then  $b_1 = \text{Im}(y_a) - \text{Im}(y_1)$ .
5. Rotate  $y_a$  clockwise by  $d$  on a new SWR circle. It lands on  $g = 1$  at  $y_2 = 1 + jb$ .
6. Stub 2 supplies  $-jb$ . Read both lengths off the rim as in M2.

#### Sign conventions that decide the answer

- **Clockwise = toward the generator.** One full lap is  $\lambda/2$ , so a quarter turn is  $\lambda/8$  and  $3\lambda/8$  is three quarter-turns.
- On the **admittance** chart the terminations swap ends:  $y = 0$  (open) is at the left,  $y = \infty$  (short) at the right — the mirror of the impedance chart. Getting this backwards corrupts every stub length by  $\lambda/4$ .
- Upper half =  $+b$  = **capacitive**; lower half =  $-b$  = inductive.

#### Analytic back-check

$$B = \frac{1 \pm \sqrt{g_1(1+t^2) - g_1^2 t^2}}{t}, \quad t = \tan \beta d, \quad b_1 = B - b_L$$

Note the leading 1 in the numerator, not  $g_1$ .

## Problem 1 — Smith chart reading of a mismatched line 10

79 Ch ·  $Z_0 = 50 \Omega$

QUESTION AS PRINTED 79 Ch [10]

A lossless  $50 \Omega$  line is terminated by  $75 + j100 \Omega$ . Using the Smith chart find (a)  $\Gamma_L$ , (b) VSWR, (c)  $Z_{in}$  at  $0.375\lambda$  from the load, (d) the shortest length of line for which the impedance is purely resistive, and (e) the value of that resistance.

Given

–  $Z_L = 75.00 + j100.00 \Omega$ ,  $Z_0 = 50 \Omega$ .

**Step 1 — normalise and plot  $z_L$**

- $z_L = Z_L/Z_0 = 75.00 + j100.00$  over  $50 = 1.50 + j2.00$ .
- Plot it where the  $r = 1.50$  circle cuts the  $x = +2.00$  arc; it reads  $0.198\lambda$  on the WTG scale.

**Step 2 — draw the SWR circle and read (a) and (b)**

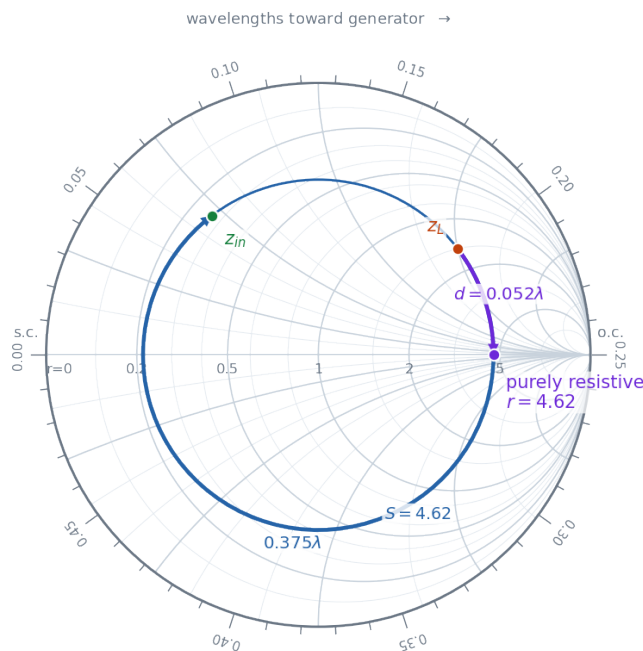
- Compass on chart centre, swing a circle through  $z_L$ : the SWR circle. Lay its radius (dividers) on the radially scaled scales below the chart.
- (a) Refl. coeff.  $E$  or  $I$  scale:  $|\Gamma_L| = 0.644$ ; straight edge through  $z_L$  to the angle of refl. coeff. scale:  $37.3^\circ$ .  $\Gamma_L = 0.644\angle 37.3^\circ$ .
- (b) SWR scale:  $S = 4.62$ , also the  $r$  where the circle cuts the right-hand real axis. Check:  $(z_L - 1)/(z_L + 1) = 0.644\angle 37.3^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 4.62$ .

**Step 3 — rotate  $0.375\lambda$  toward the generator for (c)**

- $z_L$  reads  $0.198\lambda$  WTG. Add  $0.375\lambda$ :  $0.198 + 0.375 = 0.573$ , and since this exceeds  $0.5$ , subtract half a turn  $\rightarrow 0.073\lambda$ .
- Staying on the SWR circle (the radius never changes on a lossless line), read  $z_{in} = 0.27 + j0.47$ .
- (c)  $Z_{in} = Z_0 z_{in} = 50 \times (0.27 + j0.47) = 13.33 + j23.33 \Omega$ .

**Step 4 — shortest purely resistive point, (d) and (e)**

- $z$  is purely resistive wherever the SWR circle crosses the **horizontal axis** ( $x = 0$ ) — which happens twice per lap, at the voltage maximum (right) and the voltage minimum (left).
- Rotating clockwise from  $z_L$  at  $0.198\lambda$ , the nearer crossing is the **voltage maximum**.
- (d)  $d = 0.052\lambda$ .
- (e) At a voltage maximum the normalised resistance is  $r = S = 4.62$ , so  $R = 50 \times 4.617 = 230.8 \Omega$ .



## Problem 2 — Single-stub shunt tuner 2 8

72 Ash ·  $Z_0 = 50\ \Omega$

QUESTION AS PRINTED 72 Ash [2+8]

Design single-stub shunt tuning networks (short-circuited and open-circuited) for  $Z_L = 80 + j100\ \Omega$ , and sketch the realisation in microstrip.

### Given

- $Z_L = 80.00 + j100.00\ \Omega$ ,  $Z_0 = 50\ \Omega$ .

### Step 1 — normalise and plot $z_L$

- $z_L = Z_L/Z_0 = 80.00 + j100.00$  over  $50 = 1.60 + j2.00$ .
- Plot it where the  $r = 1.60$  circle cuts the  $x = +2.00$  arc; it reads  $0.200\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.637$  on *refl. coeff. E or I*,  $S = 4.50$  on *SWR*. Check:  $(z_L - 1)/(z_L + 1) = 0.637\angle 35.7^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 4.50$ .

### Step 2 — convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection — a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.24 - j0.30$ , at  $0.450\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 — walk to the $g = 1$ circle

- Swing the SWR circle (radius  $|\Gamma_L| = 0.637$ ) through  $y_L$ .
- Rotate **clockwise** (toward the generator) to each intersection with the  $g = 1$  circle. There are always two:
  - $y_1 = 1.00 + j1.65$  at  $0.180\lambda$  WTG, so  $d_1 = (0.500 - 0.450) + 0.180 = 0.230\lambda$ .
  - $y_2 = 1.00 - j1.65$  at  $0.320\lambda$  WTG, so  $d_2 = (0.500 - 0.450) + 0.320 = 0.370\lambda$ .
- Only the *susceptance* is left to cancel: the conductance is already 1.

### Step 4 — read each stub length off the rim

- The stub must present the negative of the residual susceptance. Start at the stub's own termination and run **clockwise round the rim** until you reach that value; the arc, read on the WTG scale, is the length.
- **Short-circuited stub** — start at the short-circuit point ( $y = \infty$ , the right of the chart):
  - needs  $b = -1.651$ , giving  $\ell_1 = 0.087\lambda$ .
  - needs  $b = +1.651$ , giving  $\ell_2 = 0.413\lambda$ .
- **Open-circuited stub** — start at the open-circuit point ( $y = 0$ , the left of the chart):
  - needs  $b = -1.651$ , giving  $\ell_1 = 0.337\lambda$ .
  - needs  $b = +1.651$ , giving  $\ell_2 = 0.163\lambda$ .

### Answer

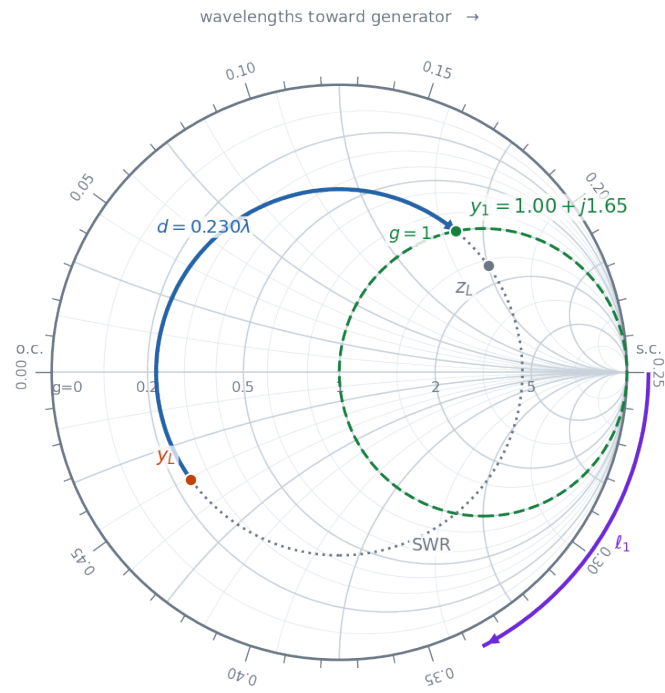
| Quantity          | Solution 1     | Solution 2     |
|-------------------|----------------|----------------|
| Stub distance $d$ | $0.230\lambda$ | $0.370\lambda$ |
| Line admittance   | $1.00 + j1.65$ | $1.00 - j1.65$ |
| Short stub $\ell$ | $0.087\lambda$ | $0.413\lambda$ |
| Open stub $\ell$  | $0.337\lambda$ | $0.163\lambda$ |

### Which to build

Solution 1 puts the stub closer to the load ( $0.230\lambda$  against  $0.370\lambda$ ), so the high-VSWR section is shorter: lower loss and wider bandwidth. Prefer it unless the layout forbids it.

### Microstrip realisation

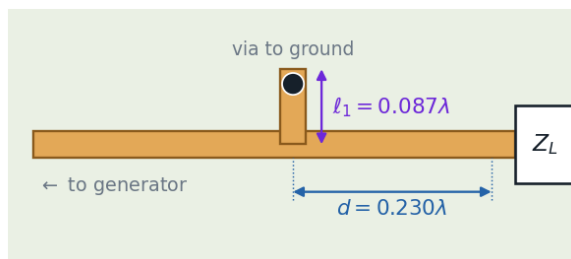
- Take FR-4,  $\epsilon_r = 4.4$ ,  $h = 1.6$  mm. Hammerstad synthesis for  $50\ \Omega$  gives  $W/h = 1.91$ , i.e.  $W = 3.06$  mm, with  $\epsilon_e = 3.33$ .
- No frequency is given, so leave the answers in wavelengths; multiply by  $\lambda_g = \lambda_0/\sqrt{\epsilon_e}$  once  $f$  is fixed.
- Draw the stubs as T-junctions off the  $W$ -wide main line. A shorted stub needs a plated via to the ground plane, drawn as a filled circle at the stub end.
- Shorten each open stub by the fringing allowance  $\Delta\ell \approx 0.3h = 0.48$  mm.



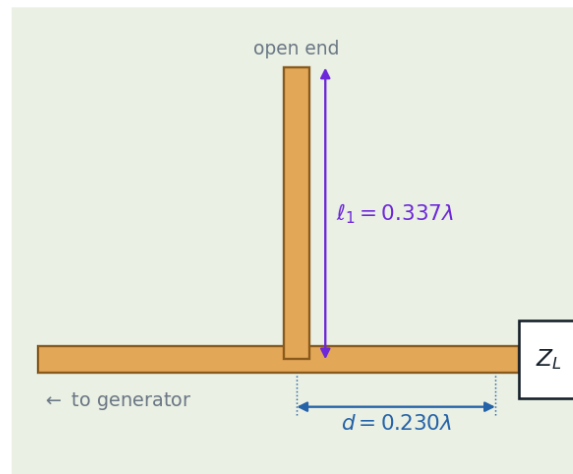
solution 1:  $d = 0.230\lambda$ , short stub  $l = 0.087\lambda$

Problem 2 — Smith chart construction. Arcs are numbered in the order of the steps above.

Short-circuited stub, solution 1



Open-circuited stub, solution 1



FR-4,  $\epsilon_r = 4.4$ ,  $h = 1.6$  mm,  $50\ \Omega$  strip  $W = 3.06$  mm,  $\epsilon_e = 3.33$   
 lengths in  $\lambda_g$  and to scale; strip width not to scale  
 cut each open end  $\Delta l = 0.48$  mm short for fringing

Problem 2 — microstrip layout, top view: stubs as T-junctions off the main line, dimensions from the answer above.

### Problem 3 — Single-stub shunt tuner 8

74 Bh ·  $Z_0 = 50 \Omega$

QUESTION AS PRINTED 74 Bh [8]

A load presents  $\Gamma_L = 0.5\angle 51^\circ$  on a  $50 \Omega$  line. Design single short-circuited and open-circuited shunt stubs using the Smith chart.

**Given**

–  $\Gamma_L = 0.500\angle 51.0^\circ$ ,  $Z_0 = 50 \Omega \Rightarrow Z_L = Z_0(1 + \Gamma_L)/(1 - \Gamma_L) = 60.42 + j62.60 \Omega$ .

**Step 1 — normalise and plot  $z_L$**

- $z_L = Z_L/Z_0 = 60.42 + j62.60$  over  $50 = 1.21 + j1.25$ .
- Plot it where the  $r = 1.21$  circle cuts the  $x = +1.25$  arc; it reads  $0.179\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.500$  on *refl. coeff. E or I*,  $S = 3.00$  on *SWR*. Check:  $(z_L - 1)/(z_L + 1) = 0.500\angle 51.0^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.00$ .

**Step 2 — convert to admittance**

- Draw the diameter through  $z_L$  and read the opposite intersection — a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.40 - j0.41$ , at  $0.429\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

**Step 3 — walk to the  $g = 1$  circle**

- Swing the SWR circle (radius  $|\Gamma_L| = 0.500$ ) through  $y_L$ .
- Rotate **clockwise** (toward the generator) to each intersection with the  $g = 1$  circle. There are always two:
  - $y_1 = 1.00 + j1.15$  at  $0.167\lambda$  WTG, so  $d_1 = (0.500 - 0.429) + 0.167 \approx 0.237\lambda$ .
  - $y_2 = 1.00 - j1.15$  at  $0.333\lambda$  WTG, so  $d_2 = (0.500 - 0.429) + 0.333 = 0.404\lambda$ .
- Only the *susceptance* is left to cancel: the conductance is already 1.

**Step 4 — read each stub length off the rim**

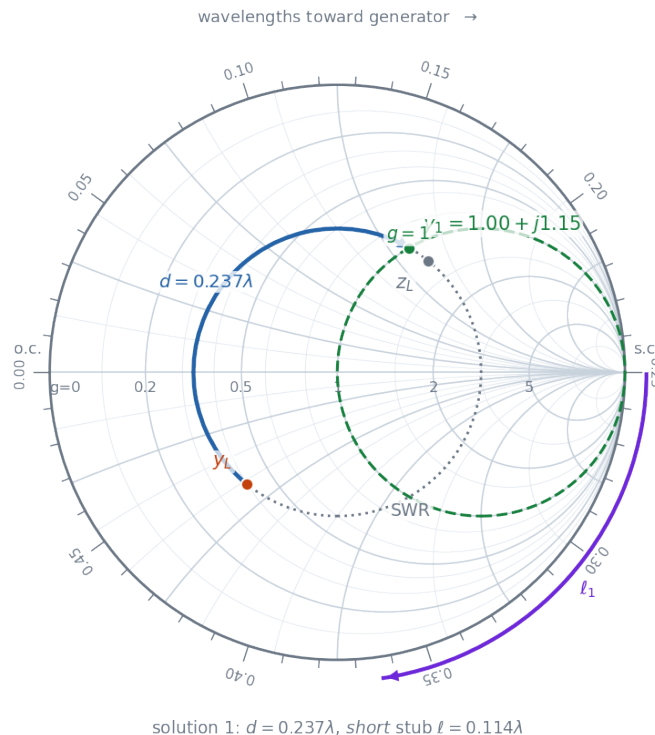
- The stub must present the negative of the residual susceptance. Start at the stub's own termination and run **clockwise round the rim** until you reach that value; the arc, read on the WTG scale, is the length.
- **Short-circuited stub** — start at the short-circuit point ( $y = \infty$ , the right of the chart):
  - needs  $b = -1.155$ , giving  $\ell_1 = 0.114\lambda$ .
  - needs  $b = +1.155$ , giving  $\ell_2 = 0.386\lambda$ .
- **Open-circuited stub** — start at the open-circuit point ( $y = 0$ , the left of the chart):
  - needs  $b = -1.155$ , giving  $\ell_1 = 0.364\lambda$ .
  - needs  $b = +1.155$ , giving  $\ell_2 = 0.136\lambda$ .

**Answer**

| Quantity          | Solution 1     | Solution 2     |
|-------------------|----------------|----------------|
| Stub distance $d$ | $0.237\lambda$ | $0.404\lambda$ |
| Line admittance   | $1.00 + j1.15$ | $1.00 - j1.15$ |
| Short stub $\ell$ | $0.114\lambda$ | $0.386\lambda$ |
| Open stub $\ell$  | $0.364\lambda$ | $0.136\lambda$ |

**Which to build**

Solution 1 puts the stub closer to the load ( $0.237\lambda$  against  $0.404\lambda$ ), so the high-VSWR section is shorter: lower loss and wider bandwidth. Prefer it unless the layout forbids it.



Problem 3 — Smith chart construction. Arcs are numbered in the order of the steps above.

## Problem 4 — Single-stub shunt tuner 5 3 2

81 Bh ·  $Z_0 = 100\ \Omega$

QUESTION AS PRINTED 81 Bh [5+3+2]

A  $100\ \Omega$  lossless line feeds a complex load  $120 - j160\ \Omega$ . Give all design steps of a single matching stub with reasoning, and the S-matrix before and after matching.

### Given

- $Z_L = 120.00 - j160.00\ \Omega$ ,  $Z_0 = 100\ \Omega$ .

### Step 1 — normalise and plot $z_L$

- $z_L = Z_L/Z_0 = 120.00 - j160.00$  over  $100 = 1.20 - j1.60$ .
- Plot it where the  $r = 1.20$  circle cuts the  $x = -1.60$  arc; it reads  $0.315\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.593$  on refl. coeff.  $E$  or  $I$ ,  $S = 3.91$  on SWR. Check:  $(z_L - 1)/(z_L + 1) = 0.593\angle -46.8^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.91$ .

### Step 2 — convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection — a  $\lambda/4$  half-turn. On an immittance chart read the admittance grid in place and skip this step.
- $y_L = 1/z_L = 0.30 + j0.40$ , at  $0.065\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 — walk to the $g = 1$ circle

- Swing the SWR circle (radius  $|\Gamma_L| = 0.593$ ) through  $y_L$ .
- Rotate **clockwise** (toward the generator) to each intersection with the  $g = 1$  circle. There are always two:
  - $y_1 = 1.00 + j1.47$  at  $0.175\lambda$  WTG, so  $d_1 = 0.175 - 0.065 = 0.110\lambda$ .
  - $y_2 = 1.00 - j1.47$  at  $0.325\lambda$  WTG, so  $d_2 = 0.325 - 0.065 \approx 0.259\lambda$ .
- Only the *susceptance* is left to cancel: the conductance is already 1.

### Step 4 — read each stub length off the rim

- The stub must present the negative of the residual susceptance. Start at the stub's own termination and run **clockwise round the rim** until you reach that value; the arc, read on the WTG scale, is the length.

- **Short-circuited stub** – start at the short-circuit point ( $y = \infty$ , the right of the chart):
  - needs  $b = -1.472$ , giving  $\ell_1 = 0.095\lambda$ .
  - needs  $b = +1.472$ , giving  $\ell_2 = 0.405\lambda$ .

### Answer

| Quantity          | Solution 1     | Solution 2     |
|-------------------|----------------|----------------|
| Stub distance $d$ | $0.110\lambda$ | $0.259\lambda$ |
| Line admittance   | $1.00 + j1.47$ | $1.00 - j1.47$ |
| Short stub $\ell$ | $0.095\lambda$ | $0.405\lambda$ |

### Which to build

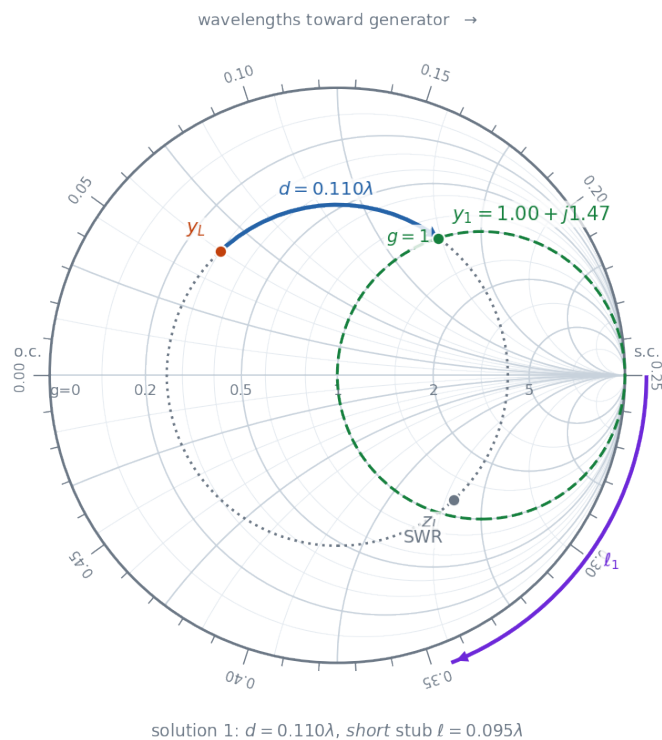
Solution 1 puts the stub closer to the load ( $0.110\lambda$  against  $0.259\lambda$ ), so the high-VSWR section is shorter: lower loss and wider bandwidth. Prefer it unless the layout forbids it.

### S-matrix before and after matching

- **Before** – the bare load is a one-port, so  $[S] = [\Gamma_L] = [0.593\angle -46.8^\circ]$ , with  $S = 3.91$ .
- **After** – the tuner makes  $Z_{in} = Z_0$ , so  $S_{11} = S_{22} = 0$ . The network is passive and reciprocal ( $S_{12} = S_{21}$ ) and lossless, so unitarity forces  $|S_{21}| = 1$ :

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the  $e^{-j\theta}$  records the through-path electrical length; either form earns the marks if you say which you mean.



Problem 4 – Smith chart construction. Arcs are numbered in the order of the steps above.

## Problem 5 – Single-stub shunt tuner 10

79 Ch ·  $Z_0 = 50\Omega$

QUESTION AS PRINTED 79 Ch [10]

Design a single short-circuited matching stub with self-defined line and load impedances, mentioning the steps.

### Given

- $Z_L = 35.00 - j60.00\Omega$ ,  $Z_0 = 50\Omega$ .

- **verified/added** The paper leaves the values to the candidate;  $Z_L = 35.00 - j60.00 \Omega$  on a  $50 \Omega$  line is assumed here. Any complex load works – the method is what earns the marks.

### Step 1 – normalise and plot $z_L$

- $z_L = Z_L/Z_0 = 35.00 - j60.00$  over  $50 = 0.70 - j1.20$ .
- Plot it where the  $r = 0.70$  circle cuts the  $x = -1.20$  arc; it reads  $0.346\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.594$  on *refl. coeff.*  $E$  or  $I$ ,  $S = 3.93$  on *SWR*. Check:  $(z_L - 1)/(z_L + 1) = 0.594 \angle -68.8^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.93$ .

### Step 2 – convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.36 + j0.62$ , at  $0.096\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 – walk to the $g = 1$ circle

- Swing the SWR circle (radius  $|\Gamma_L| = 0.594$ ) through  $y_L$ .
- Rotate **clockwise** (toward the generator) to each intersection with the  $g = 1$  circle. There are always two:
  - $y_1 = 1.00 + j1.48$  at  $0.176\lambda$  WTG, so  $d_1 = 0.176 - 0.096 = 0.080\lambda$ .
  - $y_2 = 1.00 - j1.48$  at  $0.324\lambda$  WTG, so  $d_2 = 0.324 - 0.096 \approx 0.229\lambda$ .
- Only the *susceptance* is left to cancel: the conductance is already 1.

### Step 4 – read each stub length off the rim

- The stub must present the negative of the residual susceptance. Start at the stub's own termination and run **clockwise round the rim** until you reach that value; the arc, read on the WTG scale, is the length.
- **Short-circuited stub** – start at the short-circuit point ( $y = \infty$ , the right of the chart):
  - needs  $b = -1.478$ , giving  $\ell_1 = 0.095\lambda$ .
  - needs  $b = +1.478$ , giving  $\ell_2 = 0.405\lambda$ .

### Answer

| Quantity          | Solution 1     | Solution 2     |
|-------------------|----------------|----------------|
| Stub distance $d$ | $0.080\lambda$ | $0.229\lambda$ |
| Line admittance   | $1.00 + j1.48$ | $1.00 - j1.48$ |
| Short stub $\ell$ | $0.095\lambda$ | $0.405\lambda$ |

### Which to build

Solution 1 puts the stub closer to the load ( $0.080\lambda$  against  $0.229\lambda$ ), so the high-VSWR section is shorter: lower loss and wider bandwidth. Prefer it unless the layout forbids it.

76 Bh ·  $Z_0 = 50 \Omega$ 

Design a single-stub tuner matching a lossless line to an antenna load (any assumed placement and length) and derive its S-matrix.

- $Z_L = 73.00 + j42.50 \Omega$ ,  $Z_0 = 50 \Omega$ .
- verified/added The paper leaves the values to the candidate;  $Z_L = 73.00 + j42.50 \Omega$  on a  $50 \Omega$  line is assumed here. Any complex load works – the method is what earns the marks.

- $z_L = Z_L/Z_0 = 73.00 + j42.50$  over  $50 = 1.46 + j0.85$ .
- Plot it where the  $r = 1.46$  circle cuts the  $x = +0.85$  arc; it reads  $0.191\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.371$  on *refl. coeff.* *E or I*,  $S = 2.18$  on *SWR*. Check:  $(z_L - 1)/(z_L + 1) = 0.371 \angle 42.5^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.18$ .

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.51 - j0.30$ , at  $0.441\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

- Swing the SWR circle (radius  $|\Gamma_L| = 0.371$ ) through  $y_L$ .
- Rotate **clockwise** (toward the generator) to each intersection with the  $g = 1$  **circle**. There are always two:
  - $y_1 = 1.00 + j0.80$  at  $0.155\lambda$  **WTG**, so  $d_1 = (0.500 - 0.441) + 0.155 = 0.214\lambda$ .
  - $y_2 = 1.00 - j0.80$  at  $0.345\lambda$  **WTG**, so  $d_2 = (0.500 - 0.441) + 0.345 = 0.404\lambda$ .
- Only the *susceptance* is left to cancel: the conductance is already 1.

#### Step 4 – read each stub length off the rim

- The stub must present the negative of the residual susceptance. Start at the stub's own termination and run **clockwise round the rim** until you reach that value; the arc, read on the WTG scale, is the length.
- **Short-circuited stub** — start at the short-circuit point ( $y = \infty$ , the right of the chart):
  - needs  $b = -0.800$ , giving  $\ell_1 = 0.143\lambda$ .
  - needs  $b = +0.800$ , giving  $\ell_2 = 0.357\lambda$ .

Answer

| Quantity          | Solution 1     | Solution 2     |
|-------------------|----------------|----------------|
| Stub distance $d$ | $0.214\lambda$ | $0.404\lambda$ |
| Line admittance   | $1.00 + j0.80$ | $1.00 - j0.80$ |
| Short stub $\ell$ | $0.143\lambda$ | $0.357\lambda$ |

Which to build

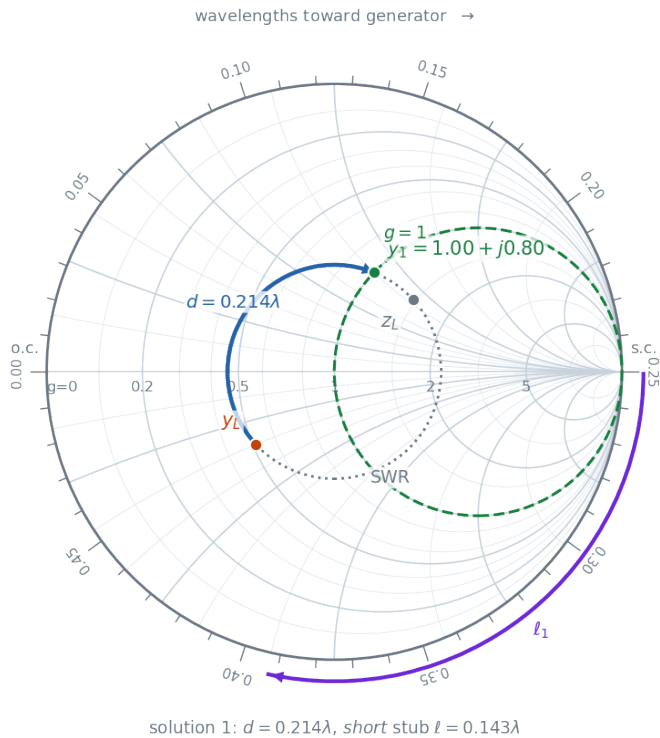
Solution 1 puts the stub closer to the load ( $0.214\lambda$  against  $0.404\lambda$ ), so the high-VSWR section is shorter: lower loss and wider bandwidth. Prefer it unless the layout forbids it.

S-matrix before and after matching

- **Before** — the bare load is a one-port, so  $[S] = [\Gamma_L] = [0.371\angle 42.5^\circ]$ , with  $S = 2.18$ .
- **After** — the tuner makes  $Z_{in} = Z_0$ , so  $S_{11} = S_{22} = 0$ . The network is passive and reciprocal ( $S_{12} = S_{21}$ ) and lossless, so unitarity forces  $|S_{21}| = 1$ :

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the  $e^{-j\theta}$  records the through-path electrical length; either form earns the marks if you say which you mean.



Problem 6 — Smith chart construction. Arcs are numbered in the order of the steps above.

## Problem 7 – Single-stub shunt tuner 4 4 2

72 Ma ·  $Z_0 = 75 \Omega$

QUESTION AS PRINTED 72 Ma [4+4+2]

Design shunt short- and open-stub matching networks using the Smith chart for  $Z_0 = 75 \Omega$ ,  $Z_L = 78.27 + j60.93 \Omega$ , and sketch the physical diagram considering microstrips.

### Given

- $Z_L = 78.27 + j60.93 \Omega$ ,  $Z_0 = 75 \Omega$ .

### Step 1 – normalise and plot $z_L$

- $z_L = Z_L/Z_0 = 78.27 + j60.93$  over  $75 = 1.04 + j0.81$ .
- Plot it where the  $r = 1.04$  circle cuts the  $x = +0.81$  arc; it reads  $0.159\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.370$  on refl. coeff.  $E$  or  $I$ ,  $S = 2.17$  on SWR. Check:  $(z_L - 1)/(z_L + 1) = 0.370 \angle 65.2^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.17$ .

### Step 2 – convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.60 - j0.46$ , at  $0.409\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 – walk to the $g = 1$ circle

- Swing the SWR circle (radius  $|\Gamma_L| = 0.370$ ) through  $y_L$ .
- Rotate **clockwise** (toward the generator) to each intersection with the  $g = 1$  circle. There are always two:
  - $y_1 = 1.00 + j0.80$  at  $0.155\lambda$  WTG, so  $d_1 = (0.500 - 0.409) + 0.155 = 0.246\lambda$ .
  - $y_2 = 1.00 - j0.80$  at  $0.345\lambda$  WTG, so  $d_2 = (0.500 - 0.409) + 0.345 \approx 0.435\lambda$ .
- Only the *susceptance* is left to cancel: the conductance is already 1.

### Step 4 – read each stub length off the rim

- The stub must present the negative of the residual susceptance. Start at the stub's own termination and run **clockwise round the rim** until you reach that value; the arc, read on the WTG scale, is the length.
- **Short-circuited stub** – start at the short-circuit point ( $y = \infty$ , the right of the chart):
  - needs  $b = -0.796$ , giving  $\ell_1 = 0.143\lambda$ .
  - needs  $b = +0.796$ , giving  $\ell_2 = 0.357\lambda$ .
- **Open-circuited stub** – start at the open-circuit point ( $y = 0$ , the left of the chart):
  - needs  $b = -0.796$ , giving  $\ell_1 = 0.393\lambda$ .
  - needs  $b = +0.796$ , giving  $\ell_2 = 0.107\lambda$ .

### Answer

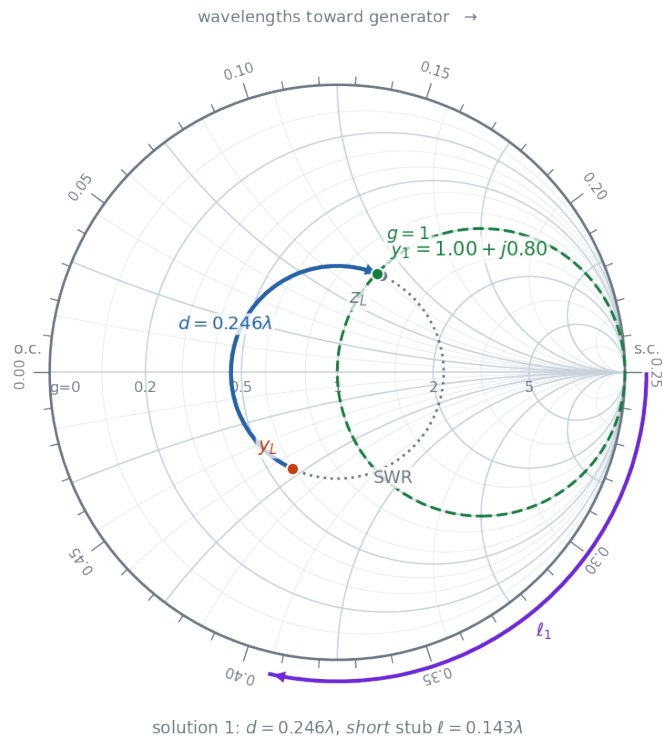
| Quantity          | Solution 1     | Solution 2     |
|-------------------|----------------|----------------|
| Stub distance $d$ | $0.246\lambda$ | $0.435\lambda$ |
| Line admittance   | $1.00 + j0.80$ | $1.00 - j0.80$ |
| Short stub $\ell$ | $0.143\lambda$ | $0.357\lambda$ |
| Open stub $\ell$  | $0.393\lambda$ | $0.107\lambda$ |

### Which to build

Solution 1 puts the stub closer to the load ( $0.246\lambda$  against  $0.435\lambda$ ), so the high-VSWR section is shorter: lower loss and wider bandwidth. Prefer it unless the layout forbids it.

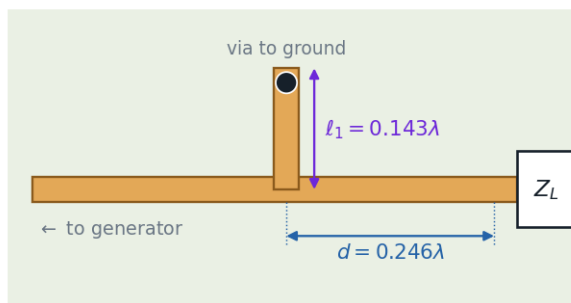
### Microstrip realisation

- Take FR-4,  $\epsilon_r = 4.4$ ,  $h = 1.6$  mm. Hammerstad synthesis for  $75 \Omega$  gives  $W/h = 0.89$ , i.e.  $W = 1.43$  mm, with  $\epsilon_e = 3.15$ .
- No frequency is given, so leave the answers in wavelengths; multiply by  $\lambda_g = \lambda_0/\sqrt{\epsilon_e}$  once  $f$  is fixed.
- Draw the stubs as T-junctions off the  $W$ -wide main line. A shorted stub needs a plated via to the ground plane, drawn as a filled circle at the stub end.
- Shorten each open stub by the fringing allowance  $\Delta\ell \approx 0.3h = 0.48$  mm.

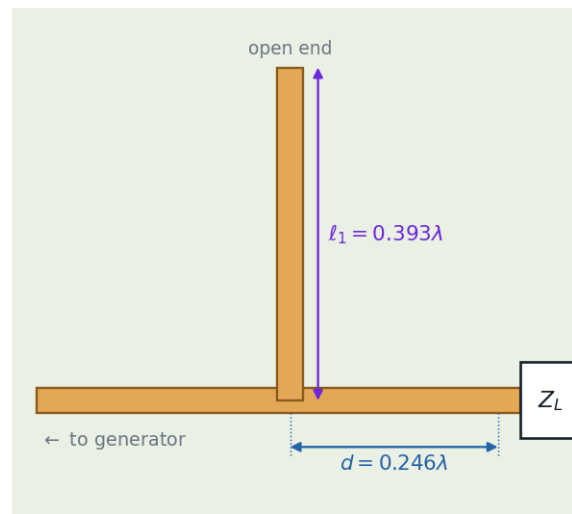


Problem 7 — Smith chart construction. Arcs are numbered in the order of the steps above.

Short-circuited stub, solution 1



Open-circuited stub, solution 1



FR-4,  $\epsilon_r = 4.4$ ,  $h = 1.6$  mm,  $75\Omega$  strip  $W = 1.43$  mm,  $\epsilon_e = 3.15$   
 lengths in  $\lambda_g$  and to scale; strip width not to scale  
 cut each open end  $\Delta l = 0.48$  mm short for fringing

Problem 7 — microstrip layout, top view: stubs as T-junctions off the main line, dimensions from the answer above.

## Problem 8 – Double-stub shunt tuner 10

71 Bh ·  $Z_0 = 300\ \Omega$

QUESTION AS PRINTED 71 Bh [10]

Design a double-stub tuner ( $3\lambda/8$  spacing) for  $Z_L = 300 + j300\ \Omega$  on a  $300\ \Omega$  line. Include the figure.

Given

$$- Z_L = 300.00 + j300.00\ \Omega, Z_0 = 300\ \Omega.$$

**Step 1 – normalise and plot  $z_L$**

- $z_L = Z_L/Z_0 = 300.00 + j300.00$  over  $300 = 1.00 + j1.00$ .
- Plot it where the  $r = 1.00$  circle cuts the  $x = +1.00$  arc; it reads  $0.162\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.447$  on *refl. coeff. E or I*,  $S = 2.62$  on SWR. Check:  $(z_L - 1)/(z_L + 1) = 0.447\angle 63.4^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.62$ .

**Step 2 – convert to admittance**

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.50 - j0.50$ , at  $0.412\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

**Step 3 – the first stub is at the load**

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.50 - j0.50$ .

**Step 4 – draw the spacing circle**

- Rotate the  $g = 1$  circle  $3\lambda/8$  **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $g_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $g_1 = 0.500$ . ✓

**Step 5 – stub 1: slide along the constant- $g$  circle**

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.500$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - **Solution A:**  $y_a = 0.50 - j1.87$ , so stub 1 must add  $b_1 = -1.866 - (-0.500) = -1.366$ .
  - **Solution B:**  $y'_a = 0.50 - j0.13$ , so  $b'_1 = +0.366$ .
- Read  $\ell_1$  off the rim, starting at the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_1 = 0.101\lambda$  (A) and  $0.306\lambda$  (B).

**Step 6 – the  $3\lambda/8$  of line between the stubs**

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j2.73$  (A),  $y'_2 = 1.00 - j0.73$  (B).

**Step 7 – stub 2 cancels what is left**

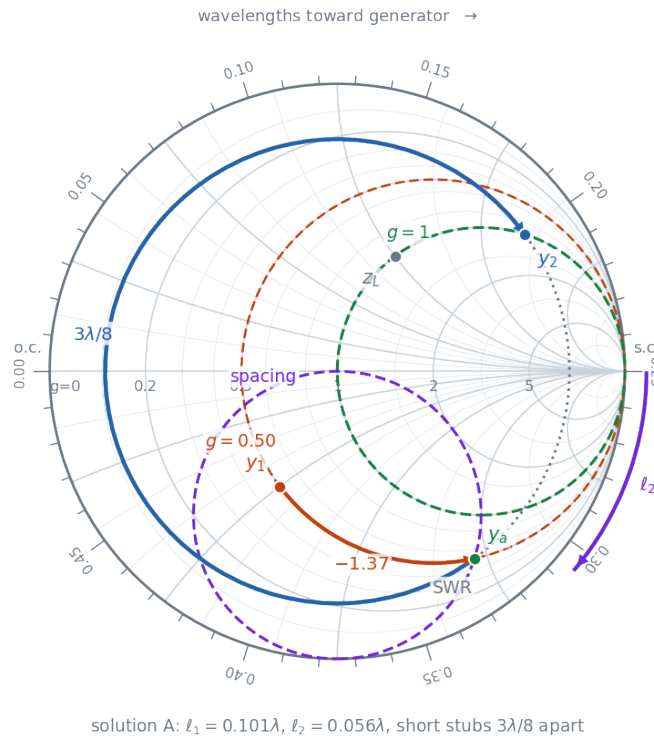
- Stub 2 supplies  $b_2 = -2.732$  (A) or  $+0.732$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 300\ \Omega$ . Matched.
- Off the rim from the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_2 = 0.056\lambda$  (A) and  $0.351\lambda$  (B).

Answer

| Quantity                 | Solution A     | Solution B     |
|--------------------------|----------------|----------------|
| Stub spacing $d$         | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load     | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1        | $0.50 - j0.50$ | $0.50 - j0.50$ |
| Stub 1 susceptance $b_1$ | $-1.366$       | $+0.366$       |
| Stub 1 length $\ell_1$   | $0.101\lambda$ | $0.306\lambda$ |
| $y$ after the spacing    | $1.00 + j2.73$ | $1.00 - j0.73$ |
| Stub 2 susceptance $b_2$ | $-2.732$       | $+0.732$       |
| Stub 2 length $\ell_2$   | $0.056\lambda$ | $0.351\lambda$ |

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



Problem 8 — Smith chart construction. Arcs are numbered in the order of the steps above.

## Problem 9 — Double-stub shunt tuner 10

70 Bh ·  $Z_0 = 100 \Omega$

QUESTION AS PRINTED 70 Bh [10]

Design a double-stub tuner ( $3\lambda/8$  spacing) for  $Z_L = 80 + j180 \Omega$  on a  $100 \Omega$  line at 3 GHz, with figure.

Given

- $Z_L = 80.00 + j180.00 \Omega$ ,  $Z_0 = 100 \Omega$ .
- $f = 3 \text{ GHz} \Rightarrow \lambda_0 = c/f = 99.9 \text{ mm}$ .

Step 1 — normalise and plot  $z_L$

- $z_L = Z_L/Z_0 = 80.00 + j180.00$  over 100 =  $0.80 + j1.80$ .
- Plot it where the  $r = 0.80$  circle cuts the  $x = +1.80$  arc; it reads  $0.179\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.711$  on *refl. coeff.* *E or I*,  $S = 5.93$  on SWR. Check:  $(z_L - 1)/(z_L + 1) = 0.711 \angle 51.3^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 5.93$ .

Step 2 — convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection — a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.21 - j0.46$ , at  $0.429\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

Step 3 — the first stub is at the load

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.21 - j0.46$ .

Step 4 — draw the spacing circle

- Rotate the  $g = 1$  circle  $3\lambda/8$  toward the load (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried onto  $g = 1$  by the  $3\lambda/8$  of line between the two stubs — which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $g_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $g_1 = 0.206$ . ✓



## Problem 10 – Double-stub shunt tuner 10

70 Ma ·  $Z_0 = 100\ \Omega$

QUESTION AS PRINTED 70 Ma [10]

Design a double-stub tuner ( $3\lambda/8$  spacing) for  $Z_L = 190 + j110\ \Omega$  on a  $100\ \Omega$  line at 10 GHz, with figure.

### Given

- $Z_L = 190.00 + j110.00\ \Omega$ ,  $Z_0 = 100\ \Omega$ .
- $f = 10\ \text{GHz} \Rightarrow \lambda_0 = c/f = 30.0\ \text{mm}$ .

### Step 1 – normalise and plot $z_L$

- $z_L = Z_L/Z_0 = 190.00 + j110.00$  over 100 = 1.90 + j1.10.
- Plot it where the  $r = 1.90$  circle cuts the  $x = +1.10$  arc; it reads  $0.208\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.458$  on *refl. coeff. E or I*,  $S = 2.69$  on SWR. Check:  $(z_L - 1)/(z_L + 1) = 0.458\angle 29.9^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.69$ .

### Step 2 – convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.39 - j0.23$ , at  $0.458\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.39 - j0.23$ .

### Step 4 – draw the spacing circle

- Rotate the  $g = 1$  circle  $3\lambda/8$  **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $g_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $g_1 = 0.394$ . ✓

### Step 5 – stub 1: slide along the constant- $g$ circle

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.394$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - **Solution A:**  $y_a = 0.39 - j1.80$ , so stub 1 must add  $b_1 = -1.796 - (-0.228) = -1.567$ .
  - **Solution B:**  $y'_a = 0.39 - j0.20$ , so  $b'_1 = +0.024$ .
- Read  $\ell_1$  off the rim, starting at the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_1 = 0.090\lambda$  (A) and  $0.254\lambda$  (B).

### Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j3.02$  (A),  $y'_2 = 1.00 - j1.02$  (B).

### Step 7 – stub 2 cancels what is left

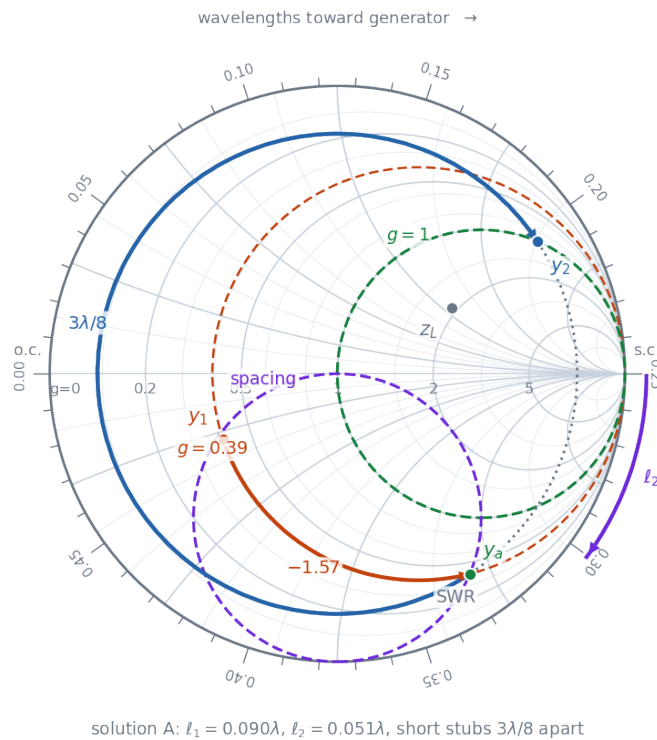
- Stub 2 supplies  $b_2 = -3.018$  (A) or  $+1.018$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 100\ \Omega$ . Matched.
- Off the rim from the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_2 = 0.051\lambda$  (A) and  $0.376\lambda$  (B).

### Answer

| Quantity                 | Solution A     | Solution B     |
|--------------------------|----------------|----------------|
| Stub spacing $d$         | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load     | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1        | $0.39 - j0.23$ | $0.39 - j0.23$ |
| Stub 1 susceptance $b_1$ | $-1.567$       | $+0.024$       |
| Stub 1 length $\ell_1$   | $0.090\lambda$ | $0.254\lambda$ |
| $y$ after the spacing    | $1.00 + j3.02$ | $1.00 - j1.02$ |
| Stub 2 susceptance $b_2$ | $-3.018$       | $+1.018$       |
| Stub 2 length $\ell_2$   | $0.051\lambda$ | $0.376\lambda$ |

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



Problem 10 — Smith chart construction. Arcs are numbered in the order of the steps above.

## Problem 11 — Double-stub shunt tuner 8

73 Ma ·  $Z_0 = 50\ \Omega$

QUESTION AS PRINTED 73 Ma [8]

Design a double-stub tuner for an inductive load on a  $50\ \Omega$  line. Include the figure.

Given

- $Z_L = 40.00 + j70.00\ \Omega$ ,  $Z_0 = 50\ \Omega$ .
- verified/added The paper leaves the values to the candidate;  $Z_L = 40.00 + j70.00\ \Omega$  on a  $50\ \Omega$  line is assumed here. Any complex load works — the method is what earns the marks.

**Step 1 — normalise and plot  $z_L$**

- $z_L = Z_L/Z_0 = 40.00 + j70.00$  over  $50 = 0.80 + j1.40$ .
- Plot it where the  $r = 0.80$  circle cuts the  $x = +1.40$  arc; it reads  $0.166\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.620$  on refl. coeff. E or I,  $S = 4.27$  on SWR. Check:  $(z_L - 1)/(z_L + 1) = 0.620\angle 60.3^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 4.27$ .

**Step 2 — convert to admittance**

- Draw the diameter through  $z_L$  and read the opposite intersection — a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.31 - j0.54$ , at  $0.416\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

**Step 3 — the first stub is at the load**

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.31 - j0.54$ .

**Step 4 — draw the spacing circle**

- Rotate the  $g = 1$  circle  $3\lambda/8$  toward the load (anticlockwise). Call the result the **spacing circle**.

- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $g_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $g_1 = 0.308$ . ✓

#### Step 5 – stub 1: slide along the constant- $g$ circle

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.308$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - **Solution A:**  $y_a = 0.31 - j1.72$ , so stub 1 must add  $b_1 = -1.722 - (-0.538) = -1.183$ .
  - **Solution B:**  $y'_a = 0.31 - j0.28$ , so  $b'_1 = +0.260$ .
- Read  $\ell_1$  off the rim, starting at the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_1 = 0.112\lambda$  (A) and  $0.290\lambda$  (B).

#### Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j3.35$  (A),  $y'_2 = 1.00 - j1.35$  (B).

#### Step 7 – stub 2 cancels what is left

- Stub 2 supplies  $b_2 = -3.345$  (A) or  $+1.345$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 50\Omega$ . Matched.
- Off the rim from the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_2 = 0.046\lambda$  (A) and  $0.398\lambda$  (B).

#### Answer

| Quantity                      | Solution A     | Solution B     |
|-------------------------------|----------------|----------------|
| Stub spacing $d$              | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load          | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1             | $0.31 - j0.54$ | $0.31 - j0.54$ |
| Stub 1 susceptance $b_1$      | $-1.183$       | $+0.260$       |
| <b>Stub 1 length</b> $\ell_1$ | $0.112\lambda$ | $0.290\lambda$ |
| $y$ after the spacing         | $1.00 + j3.35$ | $1.00 - j1.35$ |
| Stub 2 susceptance $b_2$      | $-3.345$       | $+1.345$       |
| <b>Stub 2 length</b> $\ell_2$ | $0.046\lambda$ | $0.398\lambda$ |

#### Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



### Problem 12 – Double-stub shunt tuner 3 15

QUESTION AS PRINTED 69 Bh [3+15]

**Given**

- ### Step 1 – normalise and plot $z_L$

- ### Step 2 – convert to admittance

- ### Step 3 – the first stub is at the load

- ### Step 4 – draw the spacing circle

- Step 5 – stub 1: slide along the constant- $q$  circle**

- A shunt stub adds susceptance only, so the point can move *only along its own  $g = 0.333$  circle*.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:



## Problem 13 – Double-stub shunt tuner 8 2

73 Bh ·  $Z_0 = 78 \Omega$

QUESTION AS PRINTED 73 Bh [8+2]

Design a short-circuited double-stub tuner for a complex inductive load on a  $78.0 \Omega$  line, with figure.

### Given

- $Z_L = 95.00 + j85.00 \Omega$ ,  $Z_0 = 78 \Omega$ .
- verified/added The paper leaves the values to the candidate;  $Z_L = 95.00 + j85.00 \Omega$  on a  $78 \Omega$  line is assumed here. Any complex load works – the method is what earns the marks.

### Step 1 – normalise and plot $z_L$

- $z_L = Z_L/Z_0 = 95.00 + j85.00$  over  $78 = 1.22 + j1.09$ .
- Plot it where the  $r = 1.22$  circle cuts the  $x = +1.09$  arc; it reads  $0.177\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.450$  on *refl. coeff. E or I*,  $S = 2.63$  on *SWR*. Check:  $(z_L - 1)/(z_L + 1) = 0.450 \angle 52.5^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.63$ .

### Step 2 – convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.46 - j0.41$ , at  $0.427\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.46 - j0.41$ .

### Step 4 – draw the spacing circle

- Rotate the  $g = 1$  circle  $3\lambda/8$  **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $g_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $g_1 = 0.456$ . ✓

### Step 5 – stub 1: slide along the constant- $g$ circle

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.456$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - **Solution A:**  $y_a = 0.46 - j1.84$ , so stub 1 must add  $b_1 = -1.839 - (-0.408) = -1.431$ .
  - **Solution B:**  $y'_a = 0.46 - j0.16$ , so  $b'_1 = +0.247$ .
- Read  $\ell_1$  off the rim, starting at the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_1 = 0.097\lambda$  (A) and  $0.289\lambda$  (B).

### Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j2.84$  (A),  $y'_2 = 1.00 - j0.84$  (B).

### Step 7 – stub 2 cancels what is left

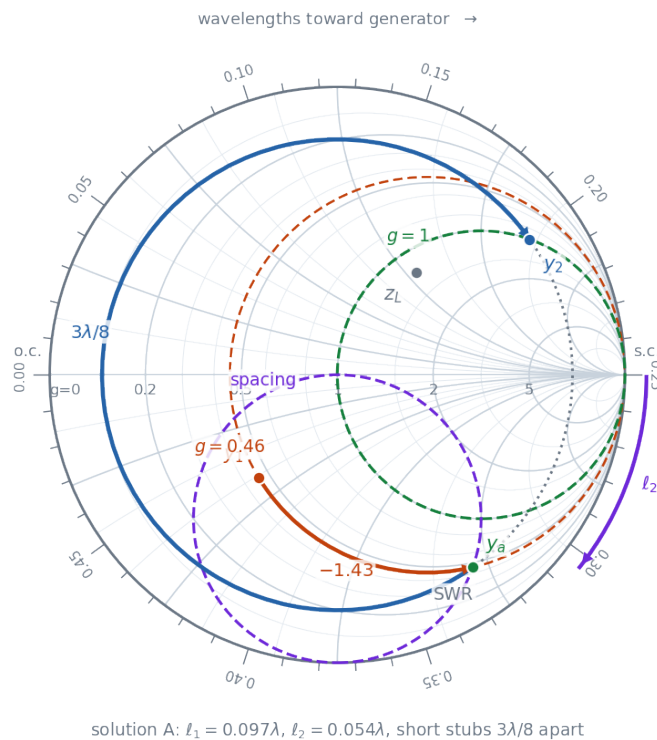
- Stub 2 supplies  $b_2 = -2.840$  (A) or  $+0.840$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 78 \Omega$ . Matched.
- Off the rim from the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_2 = 0.054\lambda$  (A) and  $0.361\lambda$  (B).

### Answer

| Quantity               | Solution A     | Solution B     |
|------------------------|----------------|----------------|
| Stub spacing $d$       | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load   | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1      | $0.46 - j0.41$ | $0.46 - j0.41$ |
| Stub 1 susceptance     | $-1.431$       | $+0.247$       |
| $b_1$                  |                |                |
| Stub 1 length $\ell_1$ | $0.097\lambda$ | $0.289\lambda$ |
| $y$ after the spacing  | $1.00 + j2.84$ | $1.00 - j0.84$ |
| Stub 2 susceptance     | $-2.840$       | $+0.840$       |
| $b_2$                  |                |                |
| Stub 2 length $\ell_2$ | $0.054\lambda$ | $0.361\lambda$ |

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



Problem 13 — Smith chart construction. Arcs are numbered in the order of the steps above.

## Problem 14 — Double-stub shunt tuner 10 2

75 Bb ·  $Z_0 = 75 \Omega$

QUESTION AS PRINTED 75 Bb [10+2]

A  $75 \Omega$  coaxial line feeds a load of normalised impedance  $z_L = 0.4 + j0.85$ . Design a short-circuited double-stub tuner.

Given

–  $Z_L = 30.00 + j63.75 \Omega$ ,  $Z_0 = 75 \Omega$ .

Step 1 — normalise and plot  $z_L$

- $z_L = Z_L/Z_0 = 30.00 + j63.75$  over  $75 = 0.40 + j0.85$ .
- Plot it where the  $r = 0.40$  circle cuts the  $x = +0.85$  arc; it reads  $0.120\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.635$  on refl. coeff. E or I,  $S = 4.48$  on SWR. Check:  $(z_L - 1)/(z_L + 1) = 0.635 \angle 94.0^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 4.48$ .

Step 2 — convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.45 - j0.96$ , at  $0.370\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.45 - j0.96$ .

### Step 4 – draw the spacing circle

- Rotate the  $g = 1$  circle  $3\lambda/8$  **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $g_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $g_1 = 0.453$ . ✓

### Step 5 – stub 1: slide along the constant- $g$ circle

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.453$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - **Solution A:**  $y_a = 0.45 - j1.84$ , so stub 1 must add  $b_1 = -1.837 - (-0.963) = -0.874$ .
  - **Solution B:**  $y'_a = 0.45 - j0.16$ , so  $b'_1 = +0.800$ .
- Read  $\ell_1$  off the rim, starting at the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_1 = 0.136\lambda$  (A) and  $0.357\lambda$  (B).

### Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j2.85$  (A),  $y'_2 = 1.00 - j0.85$  (B).

### Step 7 – stub 2 cancels what is left

- Stub 2 supplies  $b_2 = -2.847$  (A) or  $+0.847$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 75\Omega$ . Matched.
- Off the rim from the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_2 = 0.054\lambda$  (A) and  $0.362\lambda$  (B).

### Answer

| Quantity               | Solution A     | Solution B     |
|------------------------|----------------|----------------|
| Stub spacing $d$       | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load   | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1      | $0.45 - j0.96$ | $0.45 - j0.96$ |
| Stub 1 susceptance     | $-0.874$       | $+0.800$       |
| $b_1$                  |                |                |
| Stub 1 length $\ell_1$ | $0.136\lambda$ | $0.357\lambda$ |
| $y$ after the spacing  | $1.00 + j2.85$ | $1.00 - j0.85$ |
| Stub 2 susceptance     | $-2.847$       | $+0.847$       |
| $b_2$                  |                |                |
| Stub 2 length $\ell_2$ | $0.054\lambda$ | $0.362\lambda$ |

### Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



### Problem 15 – Double-stub shunt tuner 8 2

### Step 5 – stub 1: slide along the constant- $q$ circle

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.268$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - **Solution A:**  $y_a = 0.27 - j1.68$ , so stub 1 must add  $b_1 = -1.681 - (-0.208) = -1.473$ .
  - **Solution B:**  $y'_a = 0.27 - j0.32$ , so  $b'_1 = -0.111$ .
- Read  $\ell_1$  off the rim, starting at the open-circuit point ( $y = 0$ , the left of the chart):  $\ell_1 = 0.345\lambda$  (A) and  $0.482\lambda$  (B).

#### Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a *new* SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j3.54$  (A),  $y'_2 = 1.00 - j1.54$  (B).

#### Step 7 – stub 2 cancels what is left

- Stub 2 supplies  $b_2 = -3.545$  (A) or  $+1.545$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 50\Omega$ . Matched.
- Off the rim from the open-circuit point ( $y = 0$ , the left of the chart):  $\ell_2 = 0.294\lambda$  (A) and  $0.159\lambda$  (B).

#### Answer

| Quantity               | Solution A     | Solution B     |
|------------------------|----------------|----------------|
| Stub spacing $d$       | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load   | $0.400\lambda$ | $0.400\lambda$ |
| $y$ before stub 1      | $0.27 - j0.21$ | $0.27 - j0.21$ |
| Stub 1 susceptance     | $-1.473$       | $-0.111$       |
| $b_1$                  |                |                |
| Stub 1 length $\ell_1$ | $0.345\lambda$ | $0.482\lambda$ |
| $y$ after the spacing  | $1.00 + j3.54$ | $1.00 - j1.54$ |
| Stub 2 susceptance     | $-3.545$       | $+1.545$       |
| $b_2$                  |                |                |
| Stub 2 length $\ell_2$ | $0.294\lambda$ | $0.159\lambda$ |

#### Both are correct.

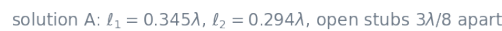
Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

#### S-matrix before and after matching

- **Before** – the bare load is a one-port, so  $[S] = [\Gamma_L] = [0.593\angle -46.8^\circ]$ , with  $S = 3.91$ .
- **After** – the tuner makes  $Z_{in} = Z_0$ , so  $S_{11} = S_{22} = 0$ . The network is passive and reciprocal ( $S_{12} = S_{21}$ ) and lossless, so unitarity forces  $|S_{21}| = 1$ :

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the  $e^{-j\theta}$  records the through-path electrical length; either form earns the marks if you say which you mean.



### Problem 16 – Double-stub shunt tuner 8 2

QUESTION AS PRINTED 80 Bh [8+2]

**Given**

- $Z_L = 109.00 + j120.00 \Omega$ ,  $Z_0 = 75 \Omega$ .

### Step 1 – normalise and plot $z_L$

- $z_L = Z_L/Z_0 = 109.00 + j120.00$  over  $75 = 1.45 + j1.60$ .
- Plot it where the  $r = 1.45$  circle cuts the  $x = +1.60$  arc; it reads  $0.193\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.568$  on *refl. coeff.* *E* or *I*,  $S = 3.63$  on *SWR*. **Check:**  $(z_L - 1)/(z_L + 1) = 0.568 \angle 41.1^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.63$ .

### Step 2 – convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.31 - j0.34$ , at  $0.443\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.31 - j0.34$ .

### Step 4 – draw the spacing circle

- Rotate the  $g = 1$  circle  $3\lambda/8$  **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $q_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $q_1 = 0.311$ . ✓

**Step 5 – stub 1: slide along the constant- $q$  circle**

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.311$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:

- **Solution A:**  $y_a = 0.31 - j1.72$ , so stub 1 must add  $b_1 = -1.725 - (-0.342) = -1.382$ .
- **Solution B:**  $y'_a = 0.31 - j0.28$ , so  $b'_1 = +0.067$ .
- Read  $\ell_1$  off the rim, starting at the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_1 = 0.100\lambda$  (A) and  $0.261\lambda$  (B).

#### Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j3.33$  (A),  $y'_2 = 1.00 - j1.33$  (B).

#### Step 7 – stub 2 cancels what is left

- Stub 2 supplies  $b_2 = -3.330$  (A) or  $+1.330$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 75\ \Omega$ . Matched.
- Off the rim from the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_2 = 0.046\lambda$  (A) and  $0.397\lambda$  (B).

#### Answer

| Quantity                 | Solution A     | Solution B     |
|--------------------------|----------------|----------------|
| Stub spacing $d$         | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load     | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1        | $0.31 - j0.34$ | $0.31 - j0.34$ |
| Stub 1 susceptance $b_1$ | $-1.382$       | $+0.067$       |
| Stub 1 length $\ell_1$   | $0.100\lambda$ | $0.261\lambda$ |
| $y$ after the spacing    | $1.00 + j3.33$ | $1.00 - j1.33$ |
| Stub 2 susceptance $b_2$ | $-3.330$       | $+1.330$       |
| Stub 2 length $\ell_2$   | $0.046\lambda$ | $0.397\lambda$ |

#### Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

#### S-matrix before and after matching

- **Before** – the bare load is a one-port, so  $[S] = [\Gamma_L] = [0.568\angle 41.1^\circ]$ , with  $S = 3.63$ .
- **After** – the tuner makes  $Z_{in} = Z_0$ , so  $S_{11} = S_{22} = 0$ . The network is passive and reciprocal ( $S_{12} = S_{21}$ ) and lossless, so unitarity forces  $|S_{21}| = 1$ :

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the  $e^{-j\theta}$  records the through-path electrical length; either form earns the marks if you say which you mean.



### Problem 17 – Double-stub shunt tuner 8 2

QUESTION AS PRINTED 81 Ba [8+2]

**Given**

- $\Gamma_L = 0.640 \angle 58.0^\circ, Z_0 = 100 \Omega \Rightarrow Z_L = Z_0(1 + \Gamma_L)/(1 - \Gamma_L) = 80.73 + j148.43 \Omega.$

### Step 1 – normalise and plot $z_L$

- $z_L = Z_L/Z_0 = 80.73 + j148.43$  over  $100 = 0.81 + j1.48$ .
- Plot it where the  $r = 0.81$  circle cuts the  $x = +1.48$  arc; it reads  $0.169\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.640$  on *refl. coeff.* *E* or *I*,  $S = 4.56$  on *SWR*. **Check:**  $(z_L - 1)/(z_L + 1) = 0.640 \angle 58.0^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 4.56$ .

### Step 2 – convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.28 - j0.52$ , at  $0.419\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.28 - j0.52$ .

### Step 4 – draw the spacing circle

- Rotate the  $g = 1$  circle  $3\lambda/8$  **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $q_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $q_1 = 0.283$ . ✓

**Step 5 – stub 1: slide along the constant- $q$  circle**

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.283$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:

- **Solution A:**  $y_a = 0.28 - j1.70$ , so stub 1 must add  $b_1 = -1.697 - (-0.520) = -1.177$ .
- **Solution B:**  $y'_a = 0.28 - j0.30$ , so  $b'_1 = +0.217$ .
- Read  $\ell_1$  off the rim, starting at the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_1 = 0.112\lambda$  (A) and  $0.284\lambda$  (B).

#### Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j3.46$  (A),  $y'_2 = 1.00 - j1.46$  (B).

#### Step 7 – stub 2 cancels what is left

- Stub 2 supplies  $b_2 = -3.464$  (A) or  $+1.464$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 100\Omega$ . Matched.
- Off the rim from the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_2 = 0.045\lambda$  (A) and  $0.405\lambda$  (B).

#### Answer

| Quantity               | Solution A     | Solution B     |
|------------------------|----------------|----------------|
| Stub spacing $d$       | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load   | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1      | $0.28 - j0.52$ | $0.28 - j0.52$ |
| Stub 1 susceptance     | $-1.177$       | $+0.217$       |
| $b_1$                  |                |                |
| Stub 1 length $\ell_1$ | $0.112\lambda$ | $0.284\lambda$ |
| $y$ after the spacing  | $1.00 + j3.46$ | $1.00 - j1.46$ |
| Stub 2 susceptance     | $-3.464$       | $+1.464$       |
| $b_2$                  |                |                |
| Stub 2 length $\ell_2$ | $0.045\lambda$ | $0.405\lambda$ |

#### Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

#### S-matrix before and after matching

- **Before** – the bare load is a one-port, so  $[S] = [\Gamma_L] = [0.640\angle 58.0^\circ]$ , with  $S = 4.56$ .
- **After** – the tuner makes  $Z_{in} = Z_0$ , so  $S_{11} = S_{22} = 0$ . The network is passive and reciprocal ( $S_{12} = S_{21}$ ) and lossless, so unitarity forces  $|S_{21}| = 1$ :

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the  $e^{-j\theta}$  records the through-path electrical length; either form earns the marks if you say which you mean.



### Problem 18 – Double-stub shunt tuner 8 2

QUESTION AS PRINTED 79 Bh [8+2]

**Given**

### Step 1 – normalise and plot $z_L$

- ### Step 2 – convert to admittance

- ### Step 3 – walk to the first stub

- ### Step 4 – draw the spacing circle

- ### Step 5 – stub 1: slide along the constant- $q$ circle

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.425$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - **Solution A:**  $y_a = 0.43 - j1.82$ , so stub 1 must add  $b_1 = -1.818 - (0.385) = -2.203$ .
  - **Solution B:**  $y'_a = 0.43 - j0.18$ , so  $b'_1 = -0.567$ .
- Read  $\ell_1$  off the rim, starting at the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_1 = 0.068\lambda$  (A) and  $0.168\lambda$  (B).

#### Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j2.92$  (A),  $y'_2 = 1.00 - j0.92$  (B).

#### Step 7 – stub 2 cancels what is left

- Stub 2 supplies  $b_2 = -2.925$  (A) or  $+0.925$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 50\Omega$ . Matched.
- Off the rim from the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_2 = 0.052\lambda$  (A) and  $0.369\lambda$  (B).

#### Answer

| Quantity               | Solution A     | Solution B     |
|------------------------|----------------|----------------|
| Stub spacing $d$       | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load   | $0.010\lambda$ | $0.010\lambda$ |
| $y$ before stub 1      | $0.43 + j0.38$ | $0.43 + j0.38$ |
| Stub 1 susceptance     | $-2.203$       | $-0.567$       |
| $b_1$                  |                |                |
| Stub 1 length $\ell_1$ | $0.068\lambda$ | $0.168\lambda$ |
| $y$ after the spacing  | $1.00 + j2.92$ | $1.00 - j0.92$ |
| Stub 2 susceptance     | $-2.925$       | $+0.925$       |
| $b_2$                  |                |                |
| Stub 2 length $\ell_2$ | $0.052\lambda$ | $0.369\lambda$ |

#### Both are correct.

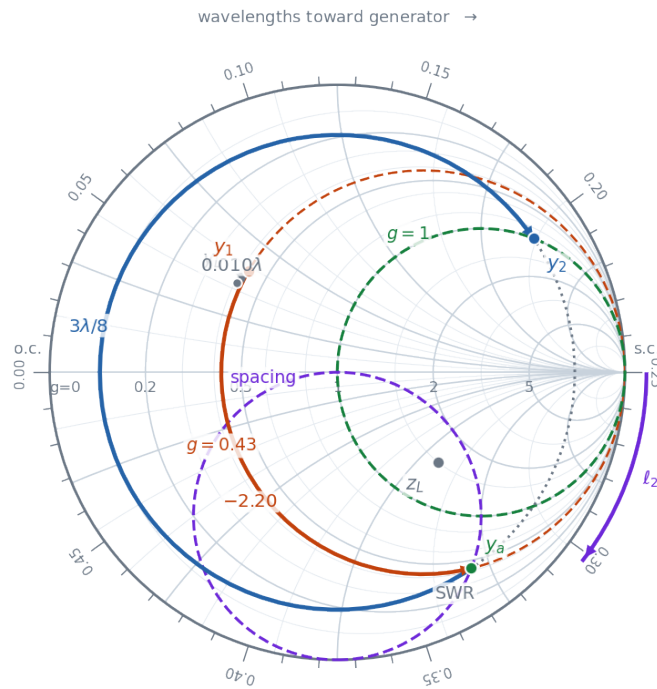
Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

#### S-matrix before and after matching

- **Before** – the bare load is a one-port, so  $[S] = [\Gamma_L] = [0.469\angle -41.7^\circ]$ , with  $S = 2.76$ .
- **After** – the tuner makes  $Z_{in} = Z_0$ , so  $S_{11} = S_{22} = 0$ . The network is passive and reciprocal ( $S_{12} = S_{21}$ ) and lossless, so unitarity forces  $|S_{21}| = 1$ :

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the  $e^{-j\theta}$  records the through-path electrical length; either form earns the marks if you say which you mean.



solution A:  $\ell_1 = 0.068\lambda$ ,  $\ell_2 = 0.052\lambda$ , short stubs  $3\lambda/8$  apart

Problem 18 – Smith chart construction. Arcs are numbered in the order of the steps above.

## Problem 19 – Double-stub shunt tuner 10

82 Bh ·  $Z_0 = 75\ \Omega$

QUESTION AS PRINTED 82 Bh [10]

A broadband microstrip antenna has load reflection coefficient  $0.33\angle 66^\circ$  on a  $75\ \Omega$  transmission patch. Design the matching stubs and express the scattering matrix of the matched network.

Given

$$- \Gamma_L = 0.330\angle 66.0^\circ, Z_0 = 75\ \Omega \Rightarrow Z_L = Z_0(1 + \Gamma_L)/(1 - \Gamma_L) = 79.52 + j53.80\ \Omega.$$

**Step 1 – normalise and plot  $z_L$**

- $z_L = Z_L/Z_0 = 79.52 + j53.80$  over  $75 = 1.06 + j0.72$ .
- Plot it where the  $r = 1.06$  circle cuts the  $x = +0.72$  arc; it reads  $0.158\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.330$  on refl. coeff. E or I,  $S = 1.99$  on SWR. Check:  $(z_L - 1)/(z_L + 1) = 0.330\angle 66.0^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 1.99$ .

**Step 2 – convert to admittance**

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.65 - j0.44$ , at  $0.408\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

**Step 3 – the first stub is at the load**

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.65 - j0.44$ .

**Step 4 – draw the spacing circle**

- Rotate the  $g = 1$  circle  $3\lambda/8$  **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $g_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $g_1 = 0.647$ . ✓

**Step 5 – stub 1: slide along the constant- $g$  circle**

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.647$  circle.

- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - **Solution A:**  $y_a = 0.65 - j1.94$ , so stub 1 must add  $b_1 = -1.936 - (-0.438) = -1.498$ .
  - **Solution B:**  $y'_a = 0.65 - j0.06$ , so  $b'_1 = +0.373$ .
- Read  $\ell_1$  off the rim, starting at the open-circuit point ( $y = 0$ , the left of the chart):  $\ell_1 = 0.344\lambda$  (A) and  $0.057\lambda$  (B).

#### Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j2.45$  (A),  $y'_2 = 1.00 - j0.45$  (B).

#### Step 7 – stub 2 cancels what is left

- Stub 2 supplies  $b_2 = -2.446$  (A) or  $+0.446$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 75\Omega$ . Matched.
- Off the rim from the open-circuit point ( $y = 0$ , the left of the chart):  $\ell_2 = 0.312\lambda$  (A) and  $0.067\lambda$  (B).

#### Answer

| Quantity               | Solution A     | Solution B     |
|------------------------|----------------|----------------|
| Stub spacing $d$       | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load   | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1      | $0.65 - j0.44$ | $0.65 - j0.44$ |
| Stub 1 susceptance     | $-1.498$       | $+0.373$       |
| $b_1$                  |                |                |
| Stub 1 length $\ell_1$ | $0.344\lambda$ | $0.057\lambda$ |
| $y$ after the spacing  | $1.00 + j2.45$ | $1.00 - j0.45$ |
| Stub 2 susceptance     | $-2.446$       | $+0.446$       |
| $b_2$                  |                |                |
| Stub 2 length $\ell_2$ | $0.312\lambda$ | $0.067\lambda$ |

#### Both are correct.

Quote either, but say which you chose. Solution B uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

#### S-matrix before and after matching

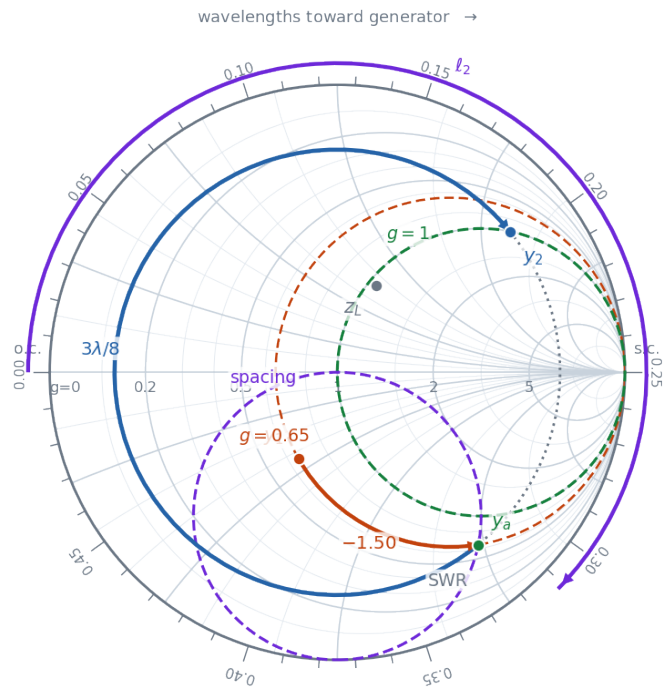
- **Before** – the bare load is a one-port, so  $[S] = [\Gamma_L] = [0.330\angle 66.0^\circ]$ , with  $S = 1.99$ .
- **After** – the tuner makes  $Z_{in} = Z_0$ , so  $S_{11} = S_{22} = 0$ . The network is passive and reciprocal ( $S_{12} = S_{21}$ ) and lossless, so unitarity forces  $|S_{21}| = 1$ :

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the  $e^{-j\theta}$  records the through-path electrical length; either form earns the marks if you say which you mean.

#### Microstrip realisation

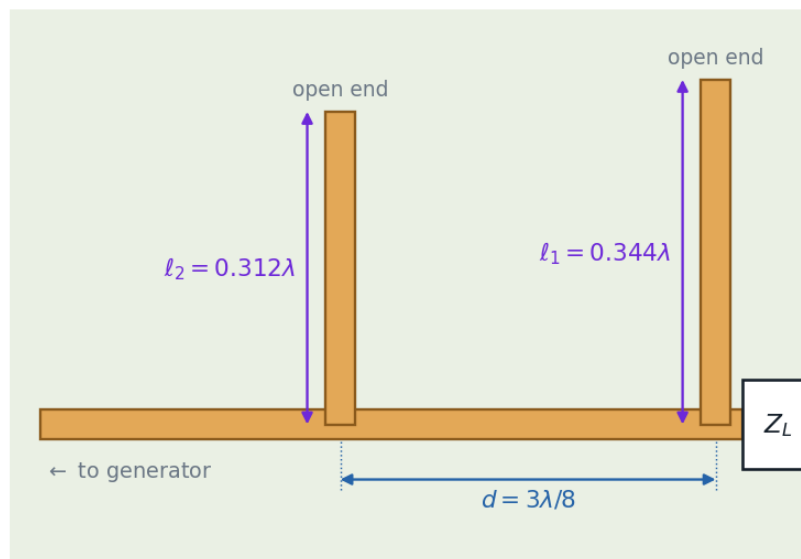
- Take FR-4,  $\epsilon_r = 4.4$ ,  $h = 1.6$  mm. Hammerstad synthesis for  $75\Omega$  gives  $W/h = 0.89$ , i.e.  $W = 1.43$  mm, with  $\epsilon_e = 3.15$ .
- No frequency is given, so leave the answers in wavelengths; multiply by  $\lambda_g = \lambda_0/\sqrt{\epsilon_e}$  once  $f$  is fixed.
- Draw the stubs as T-junctions off the  $W$ -wide main line. Open stubs are plain truncated strips (no via needed) – which is why open stubs are preferred in microstrip.
- Shorten each open stub by the fringing allowance  $\Delta\ell \approx 0.3h = 0.48$  mm.



solution A:  $l_1 = 0.344\lambda$ ,  $l_2 = 0.312\lambda$ , open stubs  $3\lambda/8$  apart

Problem 19 — Smith chart construction. Arcs are numbered in the order of the steps above.

### Double stub, open-circuited, solution A



FR-4,  $\epsilon_r = 4.4$ ,  $h = 1.6$  mm,  $75\Omega$  strip  $W = 1.43$  mm,  $\epsilon_e = 3.15$   
 lengths in  $\lambda_g$  and to scale; strip width not to scale  
 cut each open end  $\Delta l = 0.48$  mm short for fringing

Problem 19 — microstrip layout, top view: stubs as T-junctions off the main line, dimensions from the answer above.

## Problem 20 — Double-stub shunt tuner 10 2 2

82 Ba ·  $Z_0 = 100\ \Omega$

QUESTION AS PRINTED 82 Ba [10+2+2]

An antenna at 10 GHz presents  $\Gamma = |0.45|\angle 60^\circ$  on a  $100\ \Omega$  line. Design a double-stub tuner, sketch the physical diagram using microstrips, and give the S-matrices before and after matching.

### Given

- $\Gamma_L = 0.450\angle 60.0^\circ$ ,  $Z_0 = 100\ \Omega \Rightarrow Z_L = Z_0(1 + \Gamma_L)/(1 - \Gamma_L) = 105.98 + j103.58\ \Omega$ .
- $f = 10\ \text{GHz} \Rightarrow \lambda_0 = c/f = 30.0\ \text{mm}$ .

### Step 1 — normalise and plot $z_L$

- $z_L = Z_L/Z_0 = 105.98 + j103.58$  over 100 =  $1.06 + j1.04$ .
- Plot it where the  $r = 1.06$  circle cuts the  $x = +1.04$  arc; it reads  $0.167\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.450$  on *refl. coeff. E or I*,  $S = 2.64$  on *SWR*. Check:  $(z_L - 1)/(z_L + 1) = 0.450\angle 60.0^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.64$ .

### Step 2 — convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection — a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.48 - j0.47$ , at  $0.417\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 — the first stub is at the load

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.48 - j0.47$ .

### Step 4 — draw the spacing circle

- Rotate the  $g = 1$  circle  $3\lambda/8$  **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs — which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $g_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $g_1 = 0.483$ . ✓

### Step 5 — stub 1: slide along the constant- $g$ circle

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.483$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - **Solution A:**  $y_a = 0.48 - j1.86$ , so stub 1 must add  $b_1 = -1.856 - (-0.472) = -1.384$ .
  - **Solution B:**  $y'_a = 0.48 - j0.14$ , so  $b'_1 = +0.327$ .
- Read  $\ell_1$  off the rim, starting at the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_1 = 0.100\lambda$  (A) and  $0.300\lambda$  (B).

### Step 6 — the $3\lambda/8$ of line between the stubs

- Swing a *new* SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j2.77$  (A),  $y'_2 = 1.00 - j0.77$  (B).

### Step 7 — stub 2 cancels what is left

- Stub 2 supplies  $b_2 = -2.773$  (A) or  $+0.773$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 100\ \Omega$ . Matched.
- Off the rim from the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_2 = 0.055\lambda$  (A) and  $0.355\lambda$  (B).

### Answer

| Quantity               | Solution A     | Solution B     |
|------------------------|----------------|----------------|
| Stub spacing $d$       | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load   | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1      | $0.48 - j0.47$ | $0.48 - j0.47$ |
| Stub 1 susceptance     | $-1.384$       | $+0.327$       |
| $b_1$                  |                |                |
| Stub 1 length $\ell_1$ | $0.100\lambda$ | $0.300\lambda$ |
| $y$ after the spacing  | $1.00 + j2.77$ | $1.00 - j0.77$ |
| Stub 2 susceptance     | $-2.773$       | $+0.773$       |
| $b_2$                  |                |                |
| Stub 2 length $\ell_2$ | $0.055\lambda$ | $0.355\lambda$ |

Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

### S-matrix before and after matching

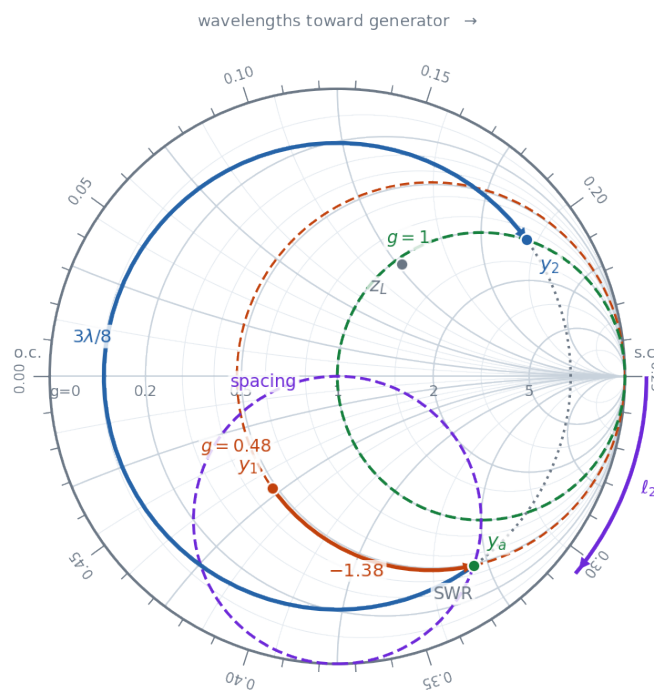
- **Before** – the bare load is a one-port, so  $[S] = [\Gamma_L] = [0.450\angle 60.0^\circ]$ , with  $S = 2.64$ .
- **After** – the tuner makes  $Z_{in} = Z_0$ , so  $S_{11} = S_{22} = 0$ . The network is passive and reciprocal ( $S_{12} = S_{21}$ ) and lossless, so unitarity forces  $|S_{21}| = 1$ :

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the  $e^{-j\theta}$  records the through-path electrical length; either form earns the marks if you say which you mean.

### Microstrip realisation

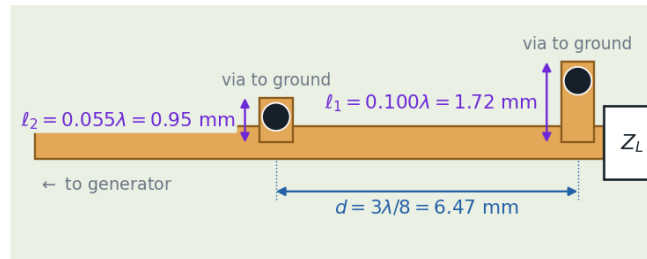
- Take FR-4,  $\epsilon_r = 4.4$ ,  $h = 1.6$  mm. Hammerstad synthesis for  $100\ \Omega$  gives  $W/h = 0.44$ , i.e.  $W = 0.71$  mm, with  $\epsilon_e = 3.02$ .
- $\lambda_g = \lambda_0/\sqrt{\epsilon_e} = 17.2$  mm at 10 GHz, so:
  - $\ell_1 = 0.100\lambda = 1.72$  mm.
  - $\ell_2 = 0.055\lambda = 0.95$  mm.
  - $d = 0.375\lambda = 6.47$  mm.
- Draw the stubs as T-junctions off the  $W$ -wide main line. A shorted stub needs a plated via to the ground plane, drawn as a filled circle at the stub end.
- Shorten each open stub by the fringing allowance  $\Delta\ell \approx 0.3h = 0.48$  mm.



solution A:  $\ell_1 = 0.100\lambda$ ,  $\ell_2 = 0.055\lambda$ , short stubs  $3\lambda/8$  apart

Problem 20 — Smith chart construction. Arcs are numbered in the order of the steps above.

### Double stub, short-circuited, solution A



FR-4,  $\epsilon_r = 4.4$ ,  $h = 1.6$  mm,  $100\ \Omega$  strip  $W = 0.71$  mm,  $\epsilon_e = 3.02$ ,  $\lambda_g = 17.2$  mm at 10 GHz; to scale

Problem 20 — microstrip layout, top view: stubs as T-junctions off the main line, dimensions from the answer above.

## Problem 21 — Double-stub shunt tuner 12

80 Ch ·  $Z_0 = 100\ \Omega$

QUESTION AS PRINTED 80 Ch [12]

Design a double-stub tuner ( $3\lambda/8$  spacing) for  $Z_L = 110 + j110\ \Omega$  on a  $100\ \Omega$  line: find the stub lengths and the spacing.

**Given**

–  $Z_L = 110.00 + j110.00\ \Omega$ ,  $Z_0 = 100\ \Omega$ .

**Step 1 — normalise and plot  $z_L$**

- $z_L = Z_L/Z_0 = 110.00 + j110.00$  over  $100 = 1.10 + j1.10$ .
- Plot it where the  $r = 1.10$  circle cuts the  $x = +1.10$  arc; it reads  $0.171\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.466$  on *refl. coeff.* *E* or *I*,  $S = 2.74$  on *SWR*. Check:  $(z_L - 1)/(z_L + 1) = 0.466\angle 57.2^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.74$ .

**Step 2 — convert to admittance**

- Draw the diameter through  $z_L$  and read the opposite intersection — a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.45 - j0.45$ , at  $0.421\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

**Step 3 — the first stub is at the load**

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.45 - j0.45$ .

**Step 4 — draw the spacing circle**

- Rotate the  $g = 1$  circle  $3\lambda/8$  **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs — which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $g_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $g_1 = 0.455$ . ✓

**Step 5 — stub 1: slide along the constant- $g$  circle**

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.455$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - **Solution A:**  $y_a = 0.45 - j1.84$ , so stub 1 must add  $b_1 = -1.838 - (-0.455) = -1.384$ .
  - **Solution B:**  $y'_a = 0.45 - j0.16$ , so  $b'_1 = +0.293$ .
- Read  $\ell_1$  off the rim, starting at the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_1 = 0.100\lambda$  (A) and  $0.295\lambda$  (B).

**Step 6 — the  $3\lambda/8$  of line between the stubs**

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j2.84$  (A),  $y'_2 = 1.00 - j0.84$  (B).

**Step 7 — stub 2 cancels what is left**

- Stub 2 supplies  $b_2 = -2.844$  (A) or  $+0.844$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 100\ \Omega$ . Matched.

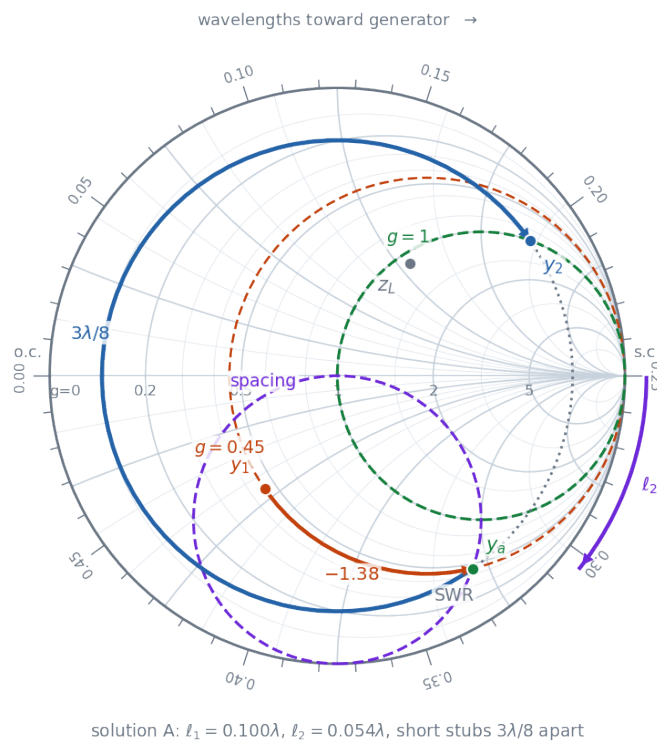
- Off the rim from the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_2 = 0.054\lambda$  (A) and  $0.362\lambda$  (B).

**Answer**

| Quantity               | Solution A     | Solution B     |
|------------------------|----------------|----------------|
| Stub spacing $d$       | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load   | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1      | $0.45 - j0.45$ | $0.45 - j0.45$ |
| Stub 1 susceptance     | $-1.384$       | $+0.293$       |
| $b_1$                  |                |                |
| Stub 1 length $\ell_1$ | $0.100\lambda$ | $0.295\lambda$ |
| $y$ after the spacing  | $1.00 + j2.84$ | $1.00 - j0.84$ |
| Stub 2 susceptance     | $-2.844$       | $+0.844$       |
| $b_2$                  |                |                |
| Stub 2 length $\ell_2$ | $0.054\lambda$ | $0.362\lambda$ |

**Both are correct.**

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.



Problem 21 – Smith chart construction. Arcs are numbered in the order of the steps above.

## Problem 22 – Double-stub shunt tuner 4 6

78 Ch ·  $Z_0 = 50\ \Omega$

QUESTION AS PRINTED 78 Ch [4+6]

Explain the different impedance matching techniques and give the resulting S-matrix of a perfectly matched double-stub network.

**Given**

- $Z_L = 100.00 + j50.00\ \Omega$ ,  $Z_0 = 50\ \Omega$ .

**Step 1 – normalise and plot  $z_L$** 

- $z_L = Z_L/Z_0 = 100.00 + j50.00$  over  $50 = 2.00 + j1.00$ .
- Plot it where the  $r = 2.00$  circle cuts the  $x = +1.00$  arc; it reads  $0.213\lambda$  on the WTG scale.

- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.447$  on *refl. coeff. E or I*,  $S = 2.62$  on *SWR*. Check:  $(z_L - 1)/(z_L + 1) = 0.447 \angle 26.6^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 2.62$ .

### Step 2 – convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.40 - j0.20$ , at  $0.463\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.40 - j0.20$ .

### Step 4 – draw the spacing circle

- Rotate the  $g = 1$  circle  $3\lambda/8$  **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $g_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $g_1 = 0.400$ . ✓

### Step 5 – stub 1: slide along the constant- $g$ circle

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.400$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - Solution A:**  $y_a = 0.40 - j1.80$ , so stub 1 must add  $b_1 = -1.800 - (-0.200) = -1.600$ .
  - Solution B:**  $y'_a = 0.40 - j0.20$ , so  $b'_1 = +0.000$ .
- Read  $\ell_1$  off the rim, starting at the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_1 = 0.089\lambda$  (A) and  $0.250\lambda$  (B).

### Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j3.00$  (A),  $y'_2 = 1.00 - j1.00$  (B).

### Step 7 – stub 2 cancels what is left

- Stub 2 supplies  $b_2 = -3.000$  (A) or  $+1.000$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 50\Omega$ . Matched.
- Off the rim from the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_2 = 0.051\lambda$  (A) and  $0.375\lambda$  (B).

### Answer

| Quantity                 | Solution A     | Solution B     |
|--------------------------|----------------|----------------|
| Stub spacing $d$         | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load     | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1        | $0.40 - j0.20$ | $0.40 - j0.20$ |
| Stub 1 susceptance $b_1$ | $-1.600$       | $+0.000$       |
| Stub 1 length $\ell_1$   | $0.089\lambda$ | $0.250\lambda$ |
| $y$ after the spacing    | $1.00 + j3.00$ | $1.00 - j1.00$ |
| Stub 2 susceptance $b_2$ | $-3.000$       | $+1.000$       |
| Stub 2 length $\ell_2$   | $0.051\lambda$ | $0.375\lambda$ |

### Both are correct.

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

### S-matrix before and after matching

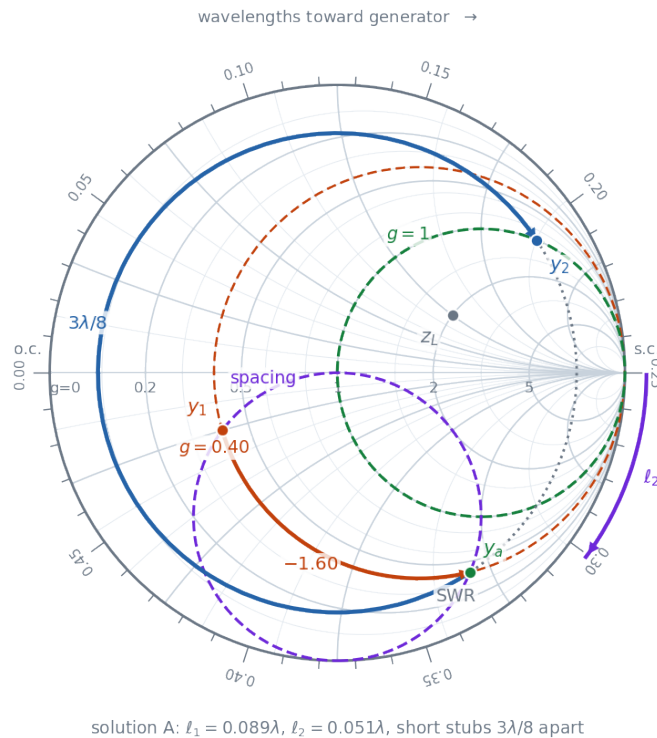
- Before** – the bare load is a one-port, so  $[S] = [\Gamma_L] = [0.447 \angle 26.6^\circ]$ , with  $S = 2.62$ .
- After** – the tuner makes  $Z_{in} = Z_0$ , so  $S_{11} = S_{22} = 0$ . The network is passive and reciprocal ( $S_{12} = S_{21}$ ) and lossless, so unitarity forces  $|S_{21}| = 1$ :

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

- Keeping the  $e^{-j\theta}$  records the through-path electrical length; either form earns the marks if you say which you mean.

### Theory half

The comparison of matching techniques and the advantages of double-stub over single-stub tuning are in §2.5; the design steps and the forbidden region are in §2.7. The worked example above supplies the “+Eg” the paper asks for.



Problem 22 – Smith chart construction. Arcs are numbered in the order of the steps above.

### Problem 23 – Double-stub shunt tuner 2 8

77 Ch ·  $Z_0 = 50 \Omega$ 

QUESTION AS PRINTED 77 Ch [2+8]

Sketch a double-stub perfectly matched network using microstrip and prepare its S-matrix.

**Given**

- $Z_L = 25.00 - j40.00 \Omega$ ,  $Z_0 = 50 \Omega$ .

### Step 1 – normalise and plot $z_L$

- $z_L = Z_L/Z_0 = 25.00 - j40.00$  over  $50 = 0.50 - j0.80$ .
- Plot it where the  $r = 0.50$  circle cuts the  $x = -0.80$  arc; it reads  $0.380\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.555$  on *refl. coeff.* *E* or *I*,  $S = 3.49$  on SWR. **Check:**  $(z_L - 1)/(z_L + 1) = 0.555 \angle -93.9^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.49$ .

### Step 2 – convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.56 + j0.90$ , at  $0.130\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.56 + j0.90$ .

#### Step 4 – draw the spacing circle

- Rotate the  $g = 1$  circle  $3\lambda/8$  **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $g_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $g_1 = 0.562$ . ✓

**Step 5 – stub 1: slide along the constant- $g$  circle**

- A shunt stub adds susceptance only, so the point can move *only along its own  $g = 0.562$  circle*.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - **Solution A:**  $y_a = 0.56 - j1.90$ , so stub 1 must add  $b_1 = -1.899 - (0.899) = -2.798$ .
  - **Solution B:**  $y'_a = 0.56 - j0.10$ , so  $b'_1 = -1.000$ .
- Read  $\ell_1$  off the rim, starting at the open-circuit point ( $y = 0$ , the left of the chart):  $\ell_1 = 0.305\lambda$  (A) and  $0.375\lambda$  (B).

**Step 6 – the  $3\lambda/8$  of line between the stubs**

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j2.60$  (A),  $y'_2 = 1.00 - j0.60$  (B).

**Step 7 – stub 2 cancels what is left**

- Stub 2 supplies  $b_2 = -2.600$  (A) or  $+0.600$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 50\Omega$ . Matched.
- Off the rim from the open-circuit point ( $y = 0$ , the left of the chart):  $\ell_2 = 0.308\lambda$  (A) and  $0.086\lambda$  (B).

**Answer**

| Quantity               | Solution A     | Solution B     |
|------------------------|----------------|----------------|
| Stub spacing $d$       | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load   | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1      | $0.56 + j0.90$ | $0.56 + j0.90$ |
| Stub 1 susceptance     | $-2.798$       | $-1.000$       |
| $b_1$                  |                |                |
| Stub 1 length $\ell_1$ | $0.305\lambda$ | $0.375\lambda$ |
| $y$ after the spacing  | $1.00 + j2.60$ | $1.00 - j0.60$ |
| Stub 2 susceptance     | $-2.600$       | $+0.600$       |
| $b_2$                  |                |                |
| Stub 2 length $\ell_2$ | $0.308\lambda$ | $0.086\lambda$ |

**Both are correct.**

Quote either, but say which you chose. Solution B uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

**S-matrix before and after matching**

- **Before** – the bare load is a one-port, so  $[S] = [\Gamma_L] = [0.555\angle -93.9^\circ]$ , with  $S = 3.49$ .
- **After** – the tuner makes  $Z_{in} = Z_0$ , so  $S_{11} = S_{22} = 0$ . The network is passive and reciprocal ( $S_{12} = S_{21}$ ) and lossless, so unitarity forces  $|S_{21}| = 1$ :

$$[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & e^{-j\theta} \\ e^{-j\theta} & 0 \end{bmatrix}$$

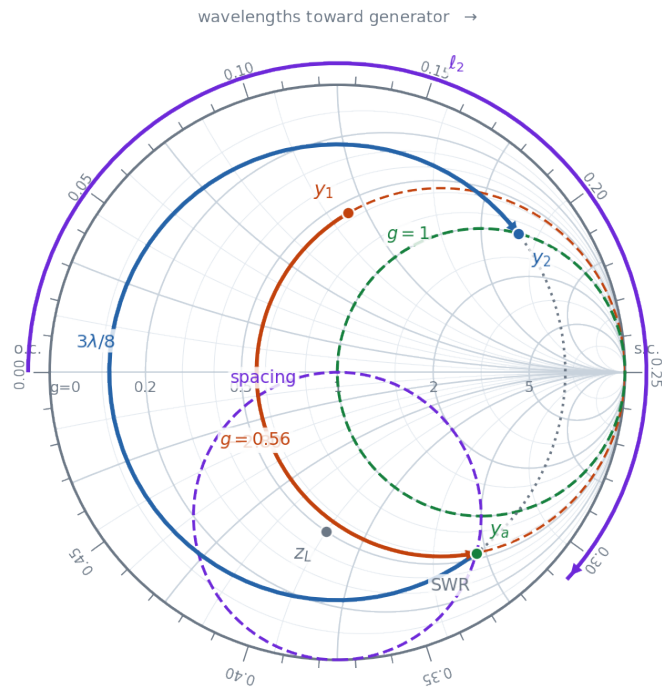
- Keeping the  $e^{-j\theta}$  records the through-path electrical length; either form earns the marks if you say which you mean.

**Microstrip realisation**

- Take FR-4,  $\epsilon_r = 4.4$ ,  $h = 1.6$  mm. Hammerstad synthesis for  $50\Omega$  gives  $W/h = 1.91$ , i.e.  $W = 3.06$  mm, with  $\epsilon_e = 3.33$ .
- No frequency is given, so leave the answers in wavelengths; multiply by  $\lambda_g = \lambda_0/\sqrt{\epsilon_e}$  once  $f$  is fixed.
- Draw the stubs as T-junctions off the  $W$ -wide main line. Open stubs are plain truncated strips (no via needed) – which is why open stubs are preferred in microstrip.
- Shorten each open stub by the fringing allowance  $\Delta\ell \approx 0.3h = 0.48$  mm.

**Theory half**

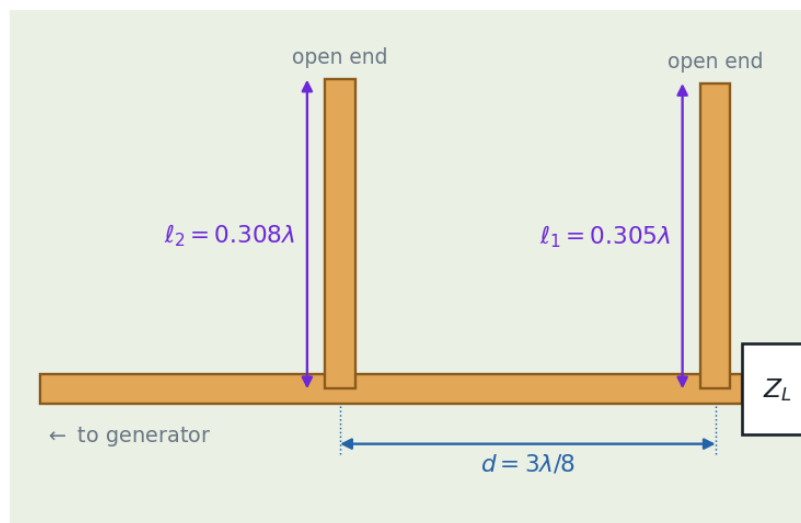
The comparison of matching techniques and the advantages of double-stub over single-stub tuning are in §2.5; the design steps and the forbidden region are in §2.7. The worked example above supplies the “+Eg” the paper asks for.



solution A:  $l_1 = 0.305\lambda$ ,  $l_2 = 0.308\lambda$ , open stubs  $3\lambda/8$  apart

Problem 23 — Smith chart construction. Arcs are numbered in the order of the steps above.

### Double stub, open-circuited, solution A



FR-4,  $\epsilon_r = 4.4$ ,  $h = 1.6$  mm,  $50\Omega$  strip  $W = 3.06$  mm,  $\epsilon_e = 3.33$   
 lengths in  $\lambda_g$  and to scale; strip width not to scale  
 cut each open end  $\Delta l = 0.48$  mm short for fringing

Problem 23 — microstrip layout, top view: stubs as T-junctions off the main line, dimensions from the answer above.

## Problem 24 – Double-stub shunt tuner 2 8

74 Ma ·  $Z_0 = 50\ \Omega$

QUESTION AS PRINTED 74 Ma [2+8]

State the advantages of double-stub matching over single-stub matching, and give the steps for matching a load with a double-stub network with an example, using the provided Smith chart.

### Given

- $Z_L = 30.00 + j40.00\ \Omega$ ,  $Z_0 = 50\ \Omega$ .

### Step 1 – normalise and plot $z_L$

- $z_L = Z_L/Z_0 = 30.00 + j40.00$  over 50 =  $0.60 + j0.80$ .
- Plot it where the  $r = 0.60$  circle cuts the  $x = +0.80$  arc; it reads  $0.125\lambda$  on the WTG scale.
- Dividers from centre to  $z_L$  (the SWR-circle radius), laid on the radially scaled scales below the chart:  $|\Gamma_L| = 0.500$  on *refl. coeff.* *E* or *I*,  $S = 3.00$  on SWR. Check:  $(z_L - 1)/(z_L + 1) = 0.500\angle 90.0^\circ$ ,  $(1 + |\Gamma_L|)/(1 - |\Gamma_L|) = 3.00$ .

### Step 2 – convert to admittance

- Draw the diameter through  $z_L$  and read the opposite intersection – a  $\lambda/4$  half-turn. On an immittance chart read the amber grid in place and skip this step.
- $y_L = 1/z_L = 0.60 - j0.80$ , at  $0.375\lambda$  on the WTG scale (exactly  $0.25\lambda$  from  $z_L$ ).

### Step 3 – the first stub is at the load

- No offset is specified, so stub 1 sits at the load and  $y_1 = y_L = 0.60 - j0.80$ .

### Step 4 – draw the spacing circle

- Rotate the  $g = 1$  circle  $3\lambda/8$  **toward the load** (anticlockwise). Call the result the **spacing circle**.
- Any point on it is carried *onto*  $g = 1$  by the  $3\lambda/8$  of line between the two stubs – which is exactly what stub 1 must aim for.
- Check the forbidden region first: matching needs  $g_1 \leq 1/\sin^2 \beta d = 2.00$ , and here  $g_1 = 0.600$ . ✓

### Step 5 – stub 1: slide along the constant- $g$ circle

- A shunt stub adds susceptance only, so the point can move *only along its own*  $g = 0.600$  circle.
- Follow that circle to where it cuts the spacing circle. It cuts twice, giving the two standard solutions:
  - **Solution A:**  $y_a = 0.60 - j1.92$ , so stub 1 must add  $b_1 = -1.917 - (-0.800) = -1.117$ .
  - **Solution B:**  $y'_a = 0.60 - j0.08$ , so  $b'_1 = +0.717$ .
- Read  $\ell_1$  off the rim, starting at the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_1 = 0.116\lambda$  (A) and  $0.349\lambda$  (B).

### Step 6 – the $3\lambda/8$ of line between the stubs

- Swing a new SWR circle through  $y_a$  and rotate **clockwise** by  $3\lambda/8$ .
- By construction it lands on the  $g = 1$  circle:  $y_2 = 1.00 + j2.53$  (A),  $y'_2 = 1.00 - j0.53$  (B).

### Step 7 – stub 2 cancels what is left

- Stub 2 supplies  $b_2 = -2.528$  (A) or  $+0.528$  (B), leaving  $y_{in} = 1 + j0$ , i.e.  $Z_{in} = Z_0 = 50\ \Omega$ . Matched.
- Off the rim from the short-circuit point ( $y = \infty$ , the right of the chart):  $\ell_2 = 0.060\lambda$  (A) and  $0.327\lambda$  (B).

### Answer

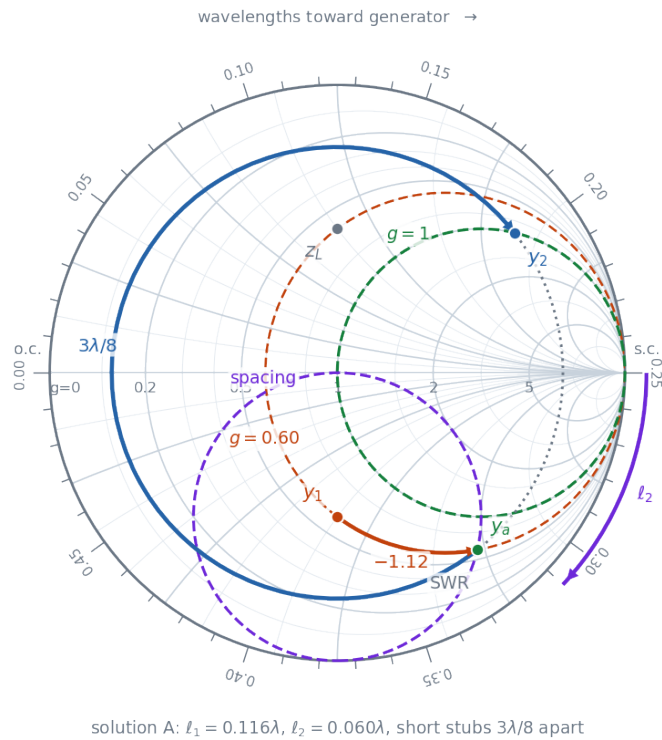
| Quantity               | Solution A     | Solution B     |
|------------------------|----------------|----------------|
| Stub spacing $d$       | $3\lambda/8$   | $3\lambda/8$   |
| First stub from load   | $0.000\lambda$ | $0.000\lambda$ |
| $y$ before stub 1      | $0.60 - j0.80$ | $0.60 - j0.80$ |
| Stub 1 susceptance     | $-1.117$       | $+0.717$       |
| $b_1$                  |                |                |
| Stub 1 length $\ell_1$ | $0.116\lambda$ | $0.349\lambda$ |
| $y$ after the spacing  | $1.00 + j2.53$ | $1.00 - j0.53$ |
| Stub 2 susceptance     | $-2.528$       | $+0.528$       |
| $b_2$                  |                |                |
| Stub 2 length $\ell_2$ | $0.060\lambda$ | $0.327\lambda$ |

**Both are correct.**

Quote either, but say which you chose. Solution A uses the shorter stubs here, which is the practical pick: less line at high VSWR means lower loss and wider bandwidth.

### Theory half

The comparison of matching techniques and the advantages of double-stub over single-stub tuning are in §2.5; the design steps and the forbidden region are in §2.7. The worked example above supplies the “+Eg” the paper asks for.



Problem 24 — Smith chart construction. Arcs are numbered in the order of the steps above.

### 3. RF and M/W Network Theory and Analysis: Numerical Problems

3 solved PYQ problems · 19 marks across 3 papers · 2 numeric matrices, 1 modulator

#### READ THIS FIRST

Chapter 3 numerals are not arithmetic problems. Every one of them is the same four tests of §3.3 applied to a matrix. **Nothing here needs a calculator beyond a square and a log.** Every number below was computed and asserted in `src/sparam.py`.

- Always run the four tests in the same order, and always write them as a table. Examiners give a mark per test; a paragraph hides them.
- A matrix that is not unitary is not wrong, it is lossy. Say so, and say how much: the shortfall in a column is the dissipated fraction.
- The tee problems live in Chapter 4. The E-plane, H-plane and magic-tee derivations, the three device identifications and the two radar problems are filed with the tee junctions, so they are the Chapter 4 companion, Problems 6 to 17. They use the same four tests.

#### M. The Four Tests, and the Conversions

**M1: the four tests (Problems 1 and 2; Ch4 companion Problems 13 to 15)**

1. **Diagonal:** which  $S_{ii} = 0$ ? Those ports are matched.
2. **Transpose:** is  $S_{ij} = S_{ji}$  for every pair? Yes means reciprocal (passive, isotropic); no means ferrite or active.
3. **Column sums:** is  $\sum_k |S_{ki}|^2 = 1$  for every column? Yes means lossless; the shortfall is the dissipated fraction of the incident power.
4. **Column dots:** is  $\sum_k S_{ki} S_{kj}^* = 0$  for every  $i \neq j$ ? This is the phase half of the unitary condition.

Always write M1 as this table

| Test        | Computation              | Conclusion                          |
|-------------|--------------------------|-------------------------------------|
| Diagonal    | every $S_{ii}$           | matched / not                       |
| Transpose   | $S_{ij}$ vs $S_{ji}$     | reciprocal / not                    |
| Column sums | $\sum_k  S_{ki} ^2$      | lossless, or lossy by the shortfall |
| Column dots | $\sum_k S_{ki} S_{kj}^*$ | phases consistent / not             |

The conversions used throughout

- $RL = -20 \log_{10} |S_{11}|$  dB,  $IL = -20 \log_{10} |S_{21}|$  dB.
- $G = +20 \log_{10} |S_{21}|$  dB,  $VSWR = \frac{1 + |S_{11}|}{1 - |S_{11}|}$ .
- $P_{\text{ref}} = |S_{11}|^2 P_{\text{inc}}$ ,  $P_{\text{trans}} = |S_{21}|^2 P_{\text{inc}}$ .
- $P_{\text{diss}} = (1 - \sum_k |S_{k1}|^2) P_{\text{inc}}$ .

**Power bookkeeping (Problem 1)**

1. Write the drive vector  $[a]$ ; multiply,  $[b] = [S][a]$ .
2. Power leaving port  $i$  is  $|b_i|^2$ , in the same units as  $|a|^2$ .
3. Check the books:  $\sum_i |b_i|^2 \leq \sum_j |a_j|^2$ , with equality for a lossless network.

#### Problem 1: The four tests on a numeric two-port matrix 3+1+1+1+2

79 Bh · the only fully numeric S-matrix in the archive

QUESTION AS PRINTED 79 Bh [3+1+1+1+2]

Why are S-parameters used in high frequencies? The S-matrix of a certain microwave network is given as  $[S] = \begin{bmatrix} 0.4 + j0.5 & j0.6 \\ j0.6 & 0.4 - j0.5 \end{bmatrix}$ .

(a) Is the network reciprocal? (b) Is the network lossless? (c) What is the return loss at the input? (d) If the input power to the network is 5 watts, what is the reflected power?

Given

- $S_{11} = 0.4 + j0.5$ ,  $S_{12} = S_{21} = j0.6$ ,  $S_{22} = 0.4 - j0.5$ .
- Input power  $P_{in} = 5$  W at port 1; port 2 terminated in  $Z_0$ .

**Step 0: the magnitudes, computed once and reused**

- $|S_{11}| = \sqrt{0.4^2 + 0.5^2} = \sqrt{0.16 + 0.25} = \sqrt{0.41} = 0.6403$ .

- $|S_{22}| = \sqrt{0.4^2 + (-0.5)^2} = \sqrt{0.41} = 0.6403$  (the same).
- $|S_{12}| = |S_{21}| = |j0.6| = 0.6$ .
- Squares, which is what every power test needs:  $|S_{11}|^2 = |S_{22}|^2 = 0.41$ ,  $|S_{12}|^2 = |S_{21}|^2 = 0.36$ .

**Step 1: run the four tests**

| Test         | Computation                                                                                                          | Result            | Conclusion                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------|
| Diagonal     | $S_{11} = 0.4 + j0.5 \neq 0$ ,<br>$S_{22} = 0.4 - j0.5 \neq 0$                                                       | both non-zero     | Neither port is matched                         |
| Transpose    | $S_{12} = j0.6$ and $S_{21} = j0.6$                                                                                  | $S_{12} = S_{21}$ | (a) Reciprocal ✓                                |
| Column 1 sum | $ S_{11} ^2 +  S_{21} ^2 = 0.41 + 0.36$                                                                              | $0.77 \neq 1$     | (b) Not lossless                                |
| Column 2 sum | $ S_{12} ^2 +  S_{22} ^2 = 0.36 + 0.41$                                                                              | $0.77 \neq 1$     | Same shortfall at port 2                        |
| Column dot   | $S_{11}S_{12}^* + S_{21}S_{22}^* = (0.4 + j0.5)(-j0.6) + (j0.6)(0.4 + j0.5)$<br>$= (0.30 - j0.24) + (-0.30 + j0.24)$ | 0                 | Phases are consistent; only the magnitudes fail |

- (a)  $S_{12} = S_{21} = j0.6$ , so the network is **reciprocal**: it is passive and contains no ferrite and no active device.
- (b) Both columns sum to 0.77, not 1, so the network is **not lossless**. It **dissipates**  $1 - 0.77 = 0.23$ , i.e. **23%** of whatever is fed into either port.
- Worth adding: the column-dot test *passes*. A network can have the right phase relations and still be lossy, which is why both halves of the unitary condition must be checked.

**Step 2: (c) return loss at the input**

- Return loss compares incident with reflected power at port 1, and  $S_{11}$  is an **amplitude** ratio, so the factor is 20:

$$RL = -20 \log_{10} |S_{11}| = -20 \log_{10}(0.6403)$$

- $\log_{10}(0.6403) = -0.19361$ , so  $RL = -20 \times (-0.19361)$ .
- (c)  **$RL = 3.87$  dB**.
- Sanity check: a low return loss means a **bad** match, and indeed  $VSWR = \frac{1 + 0.6403}{1 - 0.6403} = \frac{1.6403}{0.3597} = 4.56$ , far from 1.

**Step 3: (d) reflected power for 5 W in**

- Power scales as the **square** of the wave amplitude:

$$P_{ref} = |S_{11}|^2 P_{in} = 0.41 \times 5$$

- (d)  **$P_{ref} = 2.05$  W**.

**Step 4: complete the power budget (free marks)**

| Component           | Formula                             | Value         | Share of 5 W |
|---------------------|-------------------------------------|---------------|--------------|
| Reflected at port 1 | $ S_{11} ^2 P_{in} = 0.41 \times 5$ | 2.05 W        | 41%          |
| Delivered to port 2 | $ S_{21} ^2 P_{in} = 0.36 \times 5$ | 1.80 W        | 36%          |
| Dissipated inside   | $(1 - 0.77) \times 5$               | 1.15 W        | 23%          |
| <b>Total</b>        |                                     | <b>5.00 W</b> | <b>100%</b>  |

- The three rows sum to the input, which is the arithmetic check that the whole answer is consistent.
- Insertion loss, if asked:  $IL = -20 \log_{10}(0.6) = \mathbf{4.44$  dB.

**Answer to the theory rider (3 marks)**

- S-parameters are used at high frequencies because **total  $V$  and  $I$  are not uniquely defined** when the circuit is comparable to  $\lambda$ , because **open and short terminations cannot be realised** broadband and would **destroy or oscillate** active devices, and because **power** is what microwave instruments actually measure. See §3.1.

## Problem 2: Comparing two amplifier S-parameter sets 4

82 Bh ·  $Z_0 = 50 \Omega$ ; a reading question, not a design question

### QUESTION AS PRINTED 82 Bh [4]

Find the basic differences of the systems having the following sets of S-matrix.

- (a)  $S_{11} = 0.65\angle 146^\circ$ ,  $S_{12} = 0.12\angle 46^\circ$ ,  $S_{21} = 2.30\angle 44^\circ$ ,  $S_{22} = 0.17\angle -172^\circ$   
 (b)  $S_{11} = 0.30\angle -134^\circ$ ,  $S_{12} = 0.03\angle 46^\circ$ ,  $S_{21} = 2.20\angle 44^\circ$ ,  $S_{22} = 0.10\angle -172^\circ$

Given

- Two **two-port** sets, quoted in polar form. Only the **magnitudes** matter for the comparison the question asks for; the angles matter only for  $\Delta$  and  $K$ .

**Step 1: what kind of device is each? (the two tests that decide it)**

| Test         | Computation                                                       | Result                       | Conclusion                             |
|--------------|-------------------------------------------------------------------|------------------------------|----------------------------------------|
| Transpose    | (a) $ S_{12}  = 0.12$ vs $ S_{21}  = 2.30$ ; (b) $0.03$ vs $2.20$ | $S_{12} \neq S_{21}$ in both | <b>Both are non-reciprocal</b>         |
| Column 1 sum | (a) $0.65^2 + 2.30^2 = 0.4225 + 5.29$ ; (b) $0.30^2 + 2.20^2$     | 5.71 and 4.93, both $\gg 1$  | <b>Both are active:</b> they add power |

- $|S_{21}| > 1$  means the device **has gain**, so neither set can describe a passive component. Both are **transistor amplifiers** at the same frequency; the question is which is the better one.

**Step 2: convert every magnitude into its engineering figure**

| Quantity           | Formula                         | System (a) | System (b) | Better               |
|--------------------|---------------------------------|------------|------------|----------------------|
| Forward gain       | $20 \log_{10}  S_{21} $         | 7.23 dB    | 6.85 dB    | (a), barely          |
| Reverse isolation  | $-20 \log_{10}  S_{12} $        | 18.42 dB   | 30.46 dB   | <b>(b), by 12 dB</b> |
| Input return loss  | $-20 \log_{10}  S_{11} $        | 3.74 dB    | 10.46 dB   | <b>(b)</b>           |
| Input VSWR         | $\frac{1+ S_{11} }{1- S_{11} }$ | 4.71       | 1.86       | <b>(b)</b>           |
| Output return loss | $-20 \log_{10}  S_{22} $        | 15.39 dB   | 20.00 dB   | (b)                  |
| Output VSWR        | $\frac{1+ S_{22} }{1- S_{22} }$ | 1.41       | 1.22       | (b)                  |

- Sample line, so the arithmetic is visible:  $-20 \log_{10}(0.65) = -20 \times (-0.18709) = 3.74$  dB, and  $\text{VSWR} = (1 + 0.65)/(1 - 0.65) = 1.65/0.35 = 4.71$ .

**Step 3: stability, the one place the angles are needed**

- $\Delta = S_{11}S_{22} - S_{12}S_{21}$ , and  $K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|}$ .
- System (a):**
  - $S_{11}S_{22} = (0.65)(0.17)\angle(146^\circ - 172^\circ) = 0.1105\angle -26^\circ = 0.0993 - j0.0484$ .
  - $S_{12}S_{21} = (0.12)(2.30)\angle(46^\circ + 44^\circ) = 0.276\angle 90^\circ = j0.276$ .
  - $\Delta = 0.0993 - j0.0484 - j0.276 = 0.0993 - j0.3244$ , so  $|\Delta| = 0.3393$ .
  - $K = \frac{1 - 0.4225 - 0.0289 + 0.1151}{2 \times 0.276} = \frac{0.6637}{0.552} = 1.202$ .
- System (b):**
  - $S_{11}S_{22} = (0.30)(0.10)\angle(-134^\circ - 172^\circ) = 0.03\angle 54^\circ = 0.0176 + j0.0243$ .
  - $S_{12}S_{21} = (0.03)(2.20)\angle 90^\circ = j0.066$ .
  - $\Delta = 0.0176 - j0.0417$ , so  $|\Delta| = 0.0453$ .
  - $K = \frac{1 - 0.09 - 0.01 + 0.0021}{2 \times 0.066} = \frac{0.9021}{0.132} = 6.834$ .
- Both have  $K > 1$  and  $|\Delta| < 1$ , so both are **unconditionally stable**, but (b) is stable by a wide margin and (a) only just.

**Answer: the basic differences**

- Both are **active, non-reciprocal two-ports**, i.e. amplifiers of nearly the same gain (7.23 dB against 6.85 dB, a difference of 0.4 dB that is within normal device spread).
- The **real difference is everywhere except the gain**:

- **(b) is far better matched at the input:**  $RL$  10.46 dB against 3.74 dB, VSWR 1.86 against 4.71. System (a) reflects 42% of the drive power straight back at the source.
- **(b) is far better isolated:**  $|S_{12}| = 0.03$  against 0.12, an extra 12 dB. System (b) is close to **unilateral**, so it can be designed with the simple unilateral equations; system (a) needs the full bilateral treatment.
- **(b) is more stable:**  $K = 6.83$  against 1.20, and  $|\Delta| = 0.045$  against 0.339. System (a) sits near the edge of unconditional stability, so a mismatched load could push it into oscillation.
- **Verdict:** (b) is the better amplifier at a cost of 0.4 dB of gain: better match, better isolation, much better stability margin.
- **Note for Chapter 6:** the same paper then asks for the maximum unilateral and bilateral gains of set (a). For the record they are  $G_{TU,\max} = 9.75$  dB and  $G_{\max} = 10.11$  dB; the derivation belongs to the amplifier chapter.

### Problem 3: Evaluate the S-matrix of an amplitude modulator 7

78 Ch · verified/added no source note carries this; modelled and evaluated here

QUESTION AS PRINTED 78 Ch [3+7]

Elaborate the importance of S-parameters for microwave frequencies. Evaluate the S-matrix of an Amplitude Modulator.

#### Step 1: model the modulator as a two-port

- A microwave amplitude modulator sits **in** the RF path: the carrier enters port 1 and the modulated carrier leaves port 2. The modulating signal is a **control input**, applied as bias, not as a third RF port.
- Physically it is a **PIN-diode variable attenuator**: the diode's RF resistance is set by its DC bias, so the modulating waveform steers the transmission through the device.
- Both RF ports are designed to stay **matched** across the modulation swing, otherwise the modulation would pull the source and produce incidental phase modulation.

#### Step 2: impose the three properties

| Property                       | Equation                   | Why                                                              |
|--------------------------------|----------------------------|------------------------------------------------------------------|
| Matched at both ports          | $S_{11} = S_{22} = 0$      | The attenuator is built as a matched pad at every bias point     |
| Reciprocal                     | $S_{12} = S_{21} = \alpha$ | Passive, no ferrite and no gain: a diode resistance is bilateral |
| Lossy, and <b>time varying</b> | $0 \leq \alpha(t) \leq 1$  | The power removed from the carrier is dissipated in the diode    |

- Hence

$$[S] = \begin{bmatrix} 0 & \alpha(t) \\ \alpha(t) & 0 \end{bmatrix}, \quad \alpha(t) = \alpha_0 (1 + m \cos \omega_m t)$$

with  $\alpha_0$  the carrier-level transmission,  $m$  the modulation index and  $\omega_m$  the modulating frequency.

#### Step 3: check the condition

- Column sum:  $|S_{11}|^2 + |S_{21}|^2 = \alpha^2$ , which equals 1 only when  $\alpha = 1$ .
- So the modulator is **never lossless** except at full transmission: the dissipated fraction is  $1 - \alpha^2(t)$ , and this is unavoidable. An amplitude modulator works by **throwing power away** in step with the modulating signal.
- It is reciprocal and matched, so the only non-ideal property is the loss.

#### Step 4: evaluate with numbers

- Take  $\alpha_0 = 0.5$  and  $m = 0.6$ , a typical shallow modulation.

| Instant                           | $\alpha(t)$           | $[S]$                                              | $IL$ and $P_{out}$ for 1 W in                     |
|-----------------------------------|-----------------------|----------------------------------------------------|---------------------------------------------------|
| Peak,<br>$\cos \omega_m t = +1$   | $0.5(1 + 0.6) = 0.80$ | $\begin{bmatrix} 0 & 0.8 \\ 0.8 & 0 \end{bmatrix}$ | $IL = -20 \log 0.8 = 1.94$ dB; $P_{out} = 0.64$ W |
| Carrier,<br>$\cos \omega_m t = 0$ | 0.5                   | $\begin{bmatrix} 0 & 0.5 \\ 0.5 & 0 \end{bmatrix}$ | $IL = 6.02$ dB; $P_{out} = 0.25$ W                |
| Trough,<br>$\cos \omega_m t = -1$ | $0.5(1 - 0.6) = 0.20$ | $\begin{bmatrix} 0 & 0.2 \\ 0.2 & 0 \end{bmatrix}$ | $IL = 13.98$ dB; $P_{out} = 0.04$ W               |

- **Recovering the modulation index from the matrix**, which is the check the examiner is looking for:

$$m = \frac{|S_{21}|_{\max} - |S_{21}|_{\min}}{|S_{21}|_{\max} + |S_{21}|_{\min}} = \frac{0.80 - 0.20}{0.80 + 0.20} = \frac{0.60}{1.00} = 0.6\checkmark$$

- **Modulation depth in dB**: the transmission swings over  $13.98 - 1.94 = 12.04$  dB.

#### Notes worth adding

- **Balanced (reflection-type) realisation**: carrier into the H-arm of a magic tee, identical PIN diodes as reflecting terminations on the two collinear arms, both driven by the modulating bias, and **one diode set back by**  $\lambda_g/4$  so its round-trip reflection returns  $180^\circ$  shifted,  $\Gamma_2 = -\Gamma_1$ . A tee with loads  $\Gamma_1, \Gamma_2$  gives  $S_{44} = (\Gamma_1 + \Gamma_2)/2 = 0$  and  $S_{34} = (\Gamma_1 - \Gamma_2)/2 = \Gamma_1$  (Ch4 §4.5, asserted in `wg.py`): the input stays matched and the E-arm output is  $\Gamma_1(t)$ , which follows the bias. **Without the offset** ( $\Gamma_1 = \Gamma_2$ ) every reflection returns to the H-arm input and nothing reaches the E-arm. verified/added
- $S_{11}$  **is the figure of merit**: a good modulator keeps  $S_{11}$  small at every bias point. If it does not, the source sees a varying load and the output picks up incidental phase modulation.
- **Contrast with an ideal switch**:  $\alpha$  toggling between 1 and 0 makes the same matrix an on-off (ASK) modulator, with  $m = 1$ .

## 4. RF/Microwave Components and Devices: Numerical Problems

17 solved PYQ problems · ≈127 marks across 17 papers · 5 waveguide sets, 3 tee derivations, 4 junction networks, 3 identifications, 2 radar designs

### READ THIS FIRST

Chapter 4 numerals come in two kinds, and neither needs more than a calculator's square root.

- **Waveguide arithmetic** (Problems 1 to 5). Everything hangs off one number, the **factor**  $F = \sqrt{1 - (f_c/f)^2}$ . Compute it once, write it in a box, and every other quantity is a multiplication or a division by it:

$$\lambda_g = \frac{\lambda}{F}, \quad v_p = \frac{c}{F}, \quad v_g = cF, \quad Z_{TE} = \frac{\eta_0}{F}, \quad Z_{TM} = \eta_0 F.$$

**Always start by naming the dominant mode and checking  $f > f_c$** ; a guide below cut-off has no  $\lambda_g$  or  $v_p$  to find.

- **Tee matrices** (Problems 6 to 17). Build each tee's matrix from its physics (M2 below), then **follow one unit wave through**: out of one tee, into the next, out again. A matched termination deletes a row and column; a short reflects with  $\Gamma = -1$ ; a connection hands a leaving wave to the other tee as an arriving one. Column power sums are the self-check, and the four tests are Ch3 companion M1.
- **Watch the normalising factor**. Papers print magic-tee matrices without the  $1/\sqrt{2}$ . Spotting that it is missing is worth a mark on its own.
- **Sign, not port number, names the arm**. A column whose entries flip sign is the E-arm (difference); a column whose entries match is the H-arm (sum). Papers swap which port number carries which.
- Constants used by the papers and here:  $c = 3 \times 10^8$  m/s,  $\eta_0 = 120\pi = 376.99 \Omega$ . The guide answers are asserted in `src/wg.py`, the tee matrices in `src/sparm.py`.

### M. Three Matrix Methods for the Tee Problems

**M2: build a device matrix from its physics (Problems 6, 7, 8)**

1. Write the blank  $n \times n$  matrix.
2. Impose the **phase facts**: which arms come out in phase ( $S_{2k} = +S_{1k}$ ) and which in antiphase ( $S_{2k} = -S_{1k}$ ).
3. Impose the **matching facts**:  $S_{kk} = 0$  for every port the question calls "perfectly matched to the junction".
4. Impose **reciprocity**,  $S_{ij} = S_{ji}$ , and count what is left. Both a plane tee and a magic tee leave exactly **4 unknowns**.
5. Apply the **unitary property**  $[S][S]^* = [I]$  and read off the equations row by row. Each row-column product gives one equation; solve.

**M3: reduce an  $n$ -port by terminating ports (Problems 11, 12)**

1. A termination on port  $k$  forces  $a_k = \Gamma_k b_k$ , with  $\Gamma = -1$  for a **short**,  $+1$  for an **open**,  $0$  for a **matched load**, all measured at the junction reference plane.
2. Write out  $b_i = \sum_j S_{ij} a_j$  for every port.
3. Substitute the constraint for each terminated port and eliminate its  $a$  and  $b$ .
4. What is left is the smaller matrix seen from the remaining ports.

**M4: power bookkeeping (Problems 16, 17)**

1. Write the drive vector  $[a]$  that the connection creates. Two equal in-phase transmitters on ports 1 and 2 is  $[a] = [a, a, 0, 0]^T$ .
2. Multiply:  $[b] = [S][a]$ .
3. Power leaving port  $i$  is  $|b_i|^2$ , in the same units as  $|a|^2$ .
4. Check the books:  $\sum_i |b_i|^2 \leq \sum_j |a_j|^2$ , with equality for a lossless network.  $(1/\sqrt{2})^2 = 0.5$ : half the power.

## Problem 1: Full parameter set of an air-filled rectangular guide 6

80 Ba · width 3 cm, 6 GHz, dominant mode

QUESTION AS PRINTED 80 Ba [6+4]

For an air-filled rectangular waveguide with a width of 3 cm and a desired frequency of operation of 6 GHz (for dominant mode), determine cut-off frequency, cut-off wavelength, group velocity, phase velocity, propagation wavelength in the waveguide and the characteristic impedance. Explain the four basic parameters used to describe the performance of a directional coupler.

Given

- Air filling:  $v = c = 3 \times 10^8$  m/s,  $\eta = \eta_0 = 120\pi \Omega$ .
- Width (broad wall)  $a = 3$  cm = 0.03 m;  $f = 6$  GHz.
- Dominant mode of a rectangular guide:  $\text{TE}_{10}$  ( $m = 1, n = 0$ ), so  $b$  is not needed.

Step 1: cut-off frequency and wavelength of  $\text{TE}_{10}$

- $f_c = \frac{c}{2} \sqrt{\left(\frac{1}{a}\right)^2 + 0} = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 0.03} = \boxed{5 \text{ GHz}}$ .
- $\lambda_c = 2a = \boxed{6 \text{ cm}}$  (check:  $c/f_c = 3 \times 10^8 / 5 \times 10^9 = 0.06$  m).
- $f = 6 \text{ GHz} > f_c = 5 \text{ GHz} \Rightarrow$  **the guide propagates**, so the remaining quantities exist.

Step 2: the factor, computed once

- $f_c/f = 5/6 = 0.83333$ ;  $(f_c/f)^2 = 0.69444$ ;  $1 - 0.69444 = 0.30556$ .
- $F = \sqrt{0.30556} = \boxed{0.55277}$  (exactly  $\sqrt{11/6}$ ).
- Free-space wavelength  $\lambda = c/f = 3 \times 10^8 / 6 \times 10^9 = 5$  cm.

Step 3: every other quantity from  $F$

| Quantity         | Formula                  | Substitution                   | Value                                     |
|------------------|--------------------------|--------------------------------|-------------------------------------------|
| Group velocity   | $v_g = cF$               | $3 \times 10^8 \times 0.55277$ | <b><math>1.658 \times 10^8</math> m/s</b> |
| Phase velocity   | $v_p = c/F$              | $3 \times 10^8 / 0.55277$      | <b><math>5.427 \times 10^8</math> m/s</b> |
| Guide wavelength | $\lambda_g = \lambda/F$  | $5/0.55277$                    | <b>9.045 cm</b>                           |
| Impedance (TE)   | $Z_{TE} = \eta_0/F$      | $376.99/0.55277$               | <b>682.0 <math>\Omega</math></b>          |
| Phase constant   | $\beta = 2\pi/\lambda_g$ | $2\pi/0.090453$                | 69.46 rad/m                               |

- **Checks:**  $v_p v_g = 5.427 \times 10^8 \times 1.658 \times 10^8 = 9.0 \times 10^{16} = c^2 \checkmark$ ; and  $\frac{1}{\lambda^2} = \frac{1}{\lambda_g^2} + \frac{1}{\lambda_c^2}$ :  $400 = 122.2 + 277.8 \text{ m}^{-2} \checkmark$ .
- “Characteristic impedance” of a TE guide is its wave impedance  $Z_{TE}$ ; it is above  $\eta_0$ , as it must be for TE.
- **Rider (4 marks):** coupling, directivity, isolation, insertion loss, with  $I = C + D$ . See theory §4.8.

## Problem 2: Same parameters, with suitably assumed dimensions 6

77 Ch · the choice of dimensions is part of the marks

QUESTION AS PRINTED 77 Ch [6]

For a rectangular waveguide, with suitably assumed breadth, width and frequency, (for dominant mode), determine cut-off frequency, phase velocity, propagation wavelength in the waveguide and the characteristic impedance.

Step 0: choose the data, and justify it

- Take the standard X-band guide **WR-90**:  $a = 2.286$  cm,  $b = 1.016$  cm, air filled, at  $f = 10$  GHz.
- Why it is a good choice: it runs in **single-mode**  $\text{TE}_{10}$  at 10 GHz. The next modes are cut off:
  - $\text{TE}_{20}$ :  $f_c = c/a = 3 \times 10^8 / 0.02286 = 13.12 \text{ GHz} > 10 \text{ GHz}$ .
  - $\text{TE}_{01}$ :  $f_c = c/2b = 3 \times 10^8 / 0.02032 = 14.76 \text{ GHz} > 10 \text{ GHz}$ .
- Any other data is acceptable if it is stated and above cut-off; the method does not change.

Step 1: cut-off

- $f_c = \frac{c}{2a} = \frac{3 \times 10^8}{2 \times 0.02286} = \boxed{6.562 \text{ GHz}}$ ,  $\lambda_c = 2a = 4.572$  cm.  $10 > 6.562 \checkmark$  propagates.

Step 2: the factor

- $f_c/f = 0.65617$ ; squared 0.43056;  $1 - 0.43056 = 0.56944$ ;  $F = \sqrt{0.56944} = \boxed{0.75461}$ .
- $\lambda = c/f = 3 \text{ cm}$ .

**Step 3: the asked quantities**

| Quantity                  | Formula                 | Substitution                   | Value                                             |
|---------------------------|-------------------------|--------------------------------|---------------------------------------------------|
| Phase velocity            | $v_p = c/F$             | $3 \times 10^8 / 0.75461$      | <b><math>3.976 \times 10^8 \text{ m/s}</math></b> |
| Guide wavelength          | $\lambda_g = \lambda/F$ | $3 / 0.75461$                  | <b>3.976 cm</b>                                   |
| Impedance (TE)            | $Z_{TE} = \eta_0/F$     | $376.99 / 0.75461$             | <b>499.6 <math>\Omega</math></b>                  |
| Group velocity<br>(extra) | $v_g = cF$              | $3 \times 10^8 \times 0.75461$ | $2.264 \times 10^8 \text{ m/s}$                   |

- Check:  $1/\lambda^2 = 1111.1$  and  $1/\lambda_g^2 + 1/\lambda_c^2 = 632.7 + 478.4 = 1111.1 \text{ m}^{-2} \checkmark$ .

**Problem 3: Does the guide pass the signal? Cut-off checks** 3 2

70 Bh, 80 Bh · one method, two instances; 70 Bh worked in full

**QUESTION AS PRINTED** 70 Bh [7+3]

Define expressions for various field components of a rectangular waveguide in TE mode. Show that a 1 GHz signal cannot propagate in TE<sub>10</sub> made in a rectangular waveguide with a wall separation of 5 cm.

**QUESTION AS PRINTED** 80 Bh [1+1+2]

What is dominant mode in a waveguide? What are the dominant modes for rectangular and circular waveguides? Does a rectangular waveguide of width 2.254 cm support the propagation of a signal having frequency 6 GHz?

**Method**

1. Name the mode (TE<sub>10</sub>, the dominant one) and read the broad-wall width  $a$ .
2.  $f_c = c/2a$ .
3. Compare:  $f > f_c$  propagates;  $f < f_c$  is evanescent.
4. For "show that", add the attenuation:  $\gamma = \alpha = \sqrt{k_c^2 - k^2}$  is real.

**70 Bh worked:**  $a = 5 \text{ cm}$ ,  $f = 1 \text{ GHz}$ 

- $f_c = \frac{3 \times 10^8}{2 \times 0.05} = 3 \times 10^9 \text{ Hz} = \boxed{3 \text{ GHz}}$ .
- $f = 1 \text{ GHz} < f_c = 3 \text{ GHz} \Rightarrow$  below cut-off.
- Confirm with  $\gamma$ :
  - $k_c = \pi/a = \pi/0.05 = 62.832 \text{ rad/m}$ ,  $k_c^2 = 3947.8$ .
  - $k = 2\pi f/c = 2\pi \times 10^9 / 3 \times 10^8 = 20.944 \text{ rad/m}$ ,  $k^2 = 438.6$ .
  - $\gamma^2 = k_c^2 - k^2 = 3947.8 - 438.6 = 3509.2 > 0 \Rightarrow \gamma$  is real:  $\gamma = \alpha = \sqrt{3509.2} = 59.24 \text{ Np/m}$ ,  $\beta = 0$ .
- The field decays as  $e^{-59.24z}$ ; to  $1/e$  in  $1/59.24 = 1.69 \text{ cm}$ , i.e.  $20 \log_{10} e \times 59.24 = 514.5 \text{ dB/m}$ . **No wave travels: the 1 GHz signal cannot propagate.**

**Both instances**

| Paper | Given                          | $f_c = c/2a$                                 | Compare    | Answer                                                 |
|-------|--------------------------------|----------------------------------------------|------------|--------------------------------------------------------|
| 70 Bh | $a = 5 \text{ cm}$ , 1 GHz     | 3.00 GHz                                     | $1 < 3$    | <b>cannot propagate</b> ; $\alpha = 59.2 \text{ Np/m}$ |
| 80 Bh | $a = 2.254 \text{ cm}$ , 6 GHz | $3 \times 10^8 / 0.04508 = 6.65 \text{ GHz}$ | $6 < 6.65$ | <b>not supported</b>                                   |

- **80 Bh theory part:** dominant mode = lowest cut-off mode; rectangular TE<sub>10</sub>, circular TE<sub>11</sub> (§4.2).
- The handwritten local example (Waveguide.pdf p6) runs the same 2.254 cm guide at 1 GHz and reaches the same verdict with  $f_c = 6.65 \text{ GHz}$ .

## Problem 4: Ranking TM modes when $a = 3b$ 2+2

72 Ma

QUESTION AS PRINTED 72 Ma [6+2+2]

Describe magic Tee based on S-parameters. Differentiate between dominant and degenerate modes. Consider a rectangular waveguide having dimension of  $a = 3b$  and find the dominant mode among  $TM_{01}$ ,  $TM_{10}$ ,  $TM_{11}$ ,  $TM_{21}$ ,  $TM_{12}$ ,  $TM_{02}$  and  $TM_{20}$ .

**Step 1: put the cut-off formula in terms of  $a$  only**

$$- f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \text{ and } b = a/3 \Rightarrow n/b = 3n/a:$$

$$f_c = \frac{c}{2a} \sqrt{m^2 + 9n^2}$$

- So each mode is ranked by the number  $\sqrt{m^2 + 9n^2}$  (in units of  $c/2a$ ).

**Step 2: test every listed mode**

| Mode      | $(m, n)$ | Exists? (TM needs $m, n \geq 1$ ) | $\sqrt{m^2 + 9n^2}$ | Rank               |
|-----------|----------|-----------------------------------|---------------------|--------------------|
| $TM_{01}$ | (0, 1)   | no, $E_z \propto \sin 0 = 0$      | n/a                 | excluded           |
| $TM_{10}$ | (1, 0)   | no                                | n/a                 | excluded           |
| $TM_{11}$ | (1, 1)   | yes                               | $\sqrt{10} = 3.162$ | <b>1, dominant</b> |
| $TM_{21}$ | (2, 1)   | yes                               | $\sqrt{13} = 3.606$ | 2                  |
| $TM_{12}$ | (1, 2)   | yes                               | $\sqrt{37} = 6.083$ | 3                  |
| $TM_{02}$ | (0, 2)   | no                                | n/a                 | excluded           |
| $TM_{20}$ | (2, 0)   | no                                | n/a                 | excluded           |

- **Answer:** among the listed modes the dominant one is  $TM_{11}$ ,  $f_c = 3.162 c/2a$ .
- Worth one line: the guide's overall dominant mode is  $TE_{10}$  at  $1 \times c/2a$ , which the list does not include.
- Riders: magic tee §4.5; dominant vs degenerate §4.2.

## Problem 5: Dominant TE and TM modes of a circular guide, with numbers 2

75 Bh · plus the local worked example as practice

QUESTION AS PRINTED 75 Bh [8+2]

Derive the expression for the field strength for TM waves for a air-filled circular waveguide. Check the dominant mode in TE and TM modes.

**Step 1: cut-off in terms of the roots**

- $TE_{nm}$ :  $f_c = \frac{p'_{nm} c}{2\pi a}$ ;  $TM_{nm}$ :  $f_c = \frac{p_{nm} c}{2\pi a}$ .
- Same  $a$  and  $c$  for every mode, so **the smallest root is the dominant mode.**

**Step 2: compare the smallest roots**

| Family | Smallest root                             | Mode      | Verdict                 |
|--------|-------------------------------------------|-----------|-------------------------|
| TE     | $p'_{11} = 1.841$ (first zero of $J'_1$ ) | $TE_{11}$ | <b>dominant TE mode</b> |
| TE     | $p'_{01} = 3.832$ (first zero of $J'_0$ ) | $TE_{01}$ | higher                  |
| TM     | $p_{01} = 2.405$ (first zero of $J_0$ )   | $TM_{01}$ | <b>dominant TM mode</b> |
| TM     | $p_{11} = 3.832$ (first zero of $J_1$ )   | $TM_{11}$ | higher                  |

- $1.841 < 2.405 \Rightarrow$   **$TE_{11}$  is the dominant mode of the guide**,  $TM_{01}$  the dominant TM mode. The derivation part is theory §4.3.

**Practice (Waveguide.pdf p13): air-filled, radius 2 cm**

| Mode      | Substitution                                       | $f_c$           | $\lambda_c = 2\pi a/p$ |
|-----------|----------------------------------------------------|-----------------|------------------------|
| $TE_{11}$ | $1.8412 \times 3 \times 10^8 / (2\pi \times 0.02)$ | <b>4.40 GHz</b> | 6.83 cm                |
| $TM_{01}$ | $2.4048 \times 3 \times 10^8 / (2\pi \times 0.02)$ | <b>5.74 GHz</b> | 5.23 cm                |

- Does 3 GHz propagate?  $3 < 4.40 \text{ GHz} \Rightarrow$  **no**, not even in the dominant mode.

**Problem 6: Derive the S-matrix of an E-plane Tee** 4 8

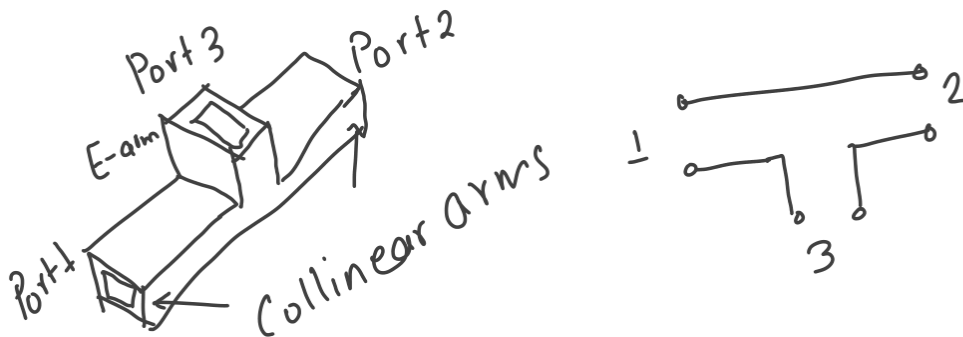
80 Bh, 81 Ba · the model 3-port derivation; learn this and Problems 7 and 8 are the same script

QUESTION AS PRINTED 80 Bh [4+4]

Briefly explain the need of S-parameters and their properties in microwave analysis. Explain three-port network parameters using an example of an E-plane Tee.

**Given**

- An E-plane (series) tee: ports 1 and 2 are the **collinear arms**, port 3 is the **E-arm** side arm, assumed **perfectly matched to the junction**.
- The junction is a passive, lossless, isotropic waveguide structure.



E-plane tee: 3D view and schematic, ports labelled as used below. Source: RF Pulchowk, S-parameter derivation sheets.

**Step 1: write the blank matrix**

- A 3-port has  $3^2 = 9$  coefficients:

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} \quad (\text{i})$$

**Step 2: impose the three physical facts**

| Physical fact                                                       | Equation                | Why                                                                     |
|---------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------|
| An input at the E-arm leaves ports 1 and 2 $180^\circ$ out of phase | $S_{23} = -S_{13}$ (ii) | The E-field of the side arm drives the two collinear arms in opposition |
| Port 3 is perfectly matched to the junction                         | $S_{33} = 0$ (iii)      | Nothing incident on the E-arm is reflected                              |
| The junction is passive and isotropic                               | $S_{ij} = S_{ji}$ (iv)  | Reciprocity: $S_{12} = S_{21}$ , $S_{23} = S_{32}$ , $S_{31} = S_{13}$  |

- Substituting (ii), (iii) and (iv) into (i):

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix} \quad (\text{v})$$

- Four unknowns remain:**  $S_{11}$ ,  $S_{12}$ ,  $S_{22}$ ,  $S_{13}$ . Four equations are needed, and the unitary property supplies them.

**Step 3: apply the unitary (lossless) property**

- A lossless network satisfies  $[S][S]^*{}^T = [I]$ , and since (v) is symmetric this is simply  $[S][S]^* = [I]$ :

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & -S_{13} \\ S_{13} & -S_{13} & 0 \end{bmatrix} \begin{bmatrix} S_{11}^* & S_{12}^* & S_{13}^* \\ S_{12}^* & S_{22}^* & -S_{13}^* \\ S_{13}^* & -S_{13}^* & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

#### Step 4: read off one equation per row-column product

| Product         | Equation                                                                                             | What it gives                                                   |       |
|-----------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-------|
| $R_3 \cdot C_3$ | $ S_{13} ^2 +  S_{13} ^2 = 1 \Rightarrow 2 S_{13} ^2 = 1$                                            | $S_{13} = \frac{1}{\sqrt{2}}$                                   | (vi)  |
| $R_3 \cdot C_1$ | $S_{13}S_{11}^* - S_{13}S_{12}^* + 0 = 0 \Rightarrow S_{11}^* = S_{12}^*$                            | $S_{11} = S_{12}$                                               | (vii) |
| $R_1 \cdot C_1$ | $ S_{11} ^2 +  S_{12} ^2 +  S_{13} ^2 = 1$ , and with (vi) and (vii) $2 S_{11} ^2 = 1 - \frac{1}{2}$ | $ S_{11} ^2 = \frac{1}{4}$ , so $S_{11} = S_{12} = \frac{1}{2}$ |       |
| $R_2 \cdot C_2$ | $ S_{12} ^2 +  S_{22} ^2 +  S_{13} ^2 = 1$ , i.e. $\frac{1}{4} +  S_{22} ^2 + \frac{1}{2} = 1$       | $ S_{22} ^2 = \frac{1}{4}$ , so $S_{22} = \frac{1}{2}$          |       |

#### Answer

- Substituting all four values back into (v):

$$[S]_{\text{E-tee}} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

#### Reading the answer back (this is the “explain” half of the marks)

- **Power split:**  $|S_{13}|^2 = |S_{23}|^2 = \frac{1}{2}$ , so an input at the E-arm divides **equally** between the collinear arms.
- **Phase:**  $S_{23} = -S_{13}$ , so those two outputs are **antiphase**; this is what makes the E-tee a **difference** junction.
- **Matching:**  $S_{33} = 0$  but  $S_{11} = S_{22} = \frac{1}{2}$ , so the collinear arms are **not matched**: a quarter of the power incident on port 1 comes straight back. That is the 3-port theorem of §3.5 appearing as arithmetic.
- **Isolation:**  $S_{12} = \frac{1}{2} \neq 0$ , so the collinear arms are **not isolated** from each other either. Adding the H-arm to fix both defects is exactly what turns a tee into a magic tee (Problem 8).
- **Check:** every column sums to  $\frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1 \checkmark$  lossless.

#### Problem 7: Derive the S-matrix of an H-plane Tee 8

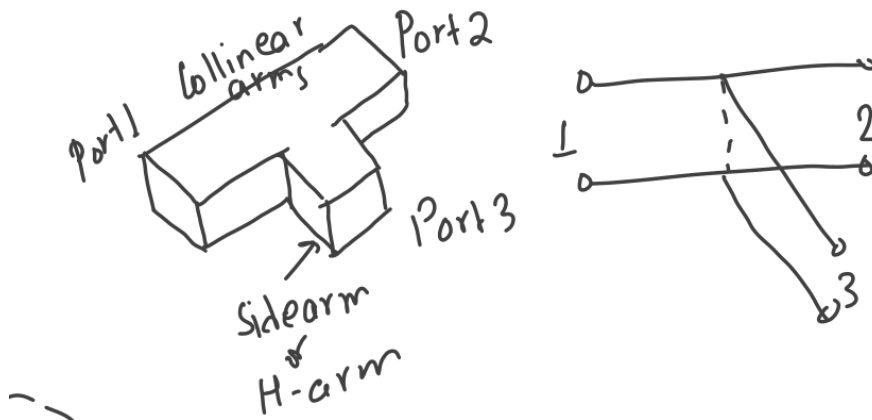
81 Ba · the same script as Problem 6 with one sign changed

QUESTION AS PRINTED 81 Ba [8]

Prepare S-matrices of a perfectly working E-plane Tee, H-plane Tee and a Duplexer, and explore their uses.

#### Given

- An H-plane (shunt) tee: ports 1 and 2 collinear, port 3 the **H-arm** side arm, perfectly matched to the junction.



H-plane tee: the side arm lies in the plane of the magnetic field. Source: RF Pulchowk, S-parameter derivation sheets.

### Step 1: the one fact that differs from Problem 6

- In an H-plane tee an input at the side arm reaches the two collinear arms **in phase**:

$$S_{13} = S_{23} \quad (\text{ii}')$$

- Everything else is unchanged:  $S_{33} = 0$  (iii) and  $S_{ij} = S_{ji}$  (iv).
- So the working matrix is

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & S_{13} \\ S_{13} & S_{13} & 0 \end{bmatrix} \quad (\text{v}')$$

with the same four unknowns.

### Step 2: unitary property, row by row

| Product         | Equation                                                                          | What it gives                                       |
|-----------------|-----------------------------------------------------------------------------------|-----------------------------------------------------|
| $R_3 \cdot C_3$ | $ S_{13} ^2 +  S_{13} ^2 = 1$                                                     | $S_{13} = \frac{1}{\sqrt{2}}$                       |
| $R_3 \cdot C_1$ | $S_{13}S_{11}^* + S_{13}S_{12}^* = 0 \Rightarrow S_{11}^* = -S_{12}^*$            | $S_{11} = -S_{12}$                                  |
| $R_1 \cdot C_1$ | $ S_{11} ^2 +  S_{12} ^2 + \frac{1}{2} = 1 \Rightarrow 2 S_{11} ^2 = \frac{1}{2}$ | $S_{11} = \frac{1}{2}$ , so $S_{12} = -\frac{1}{2}$ |
| $R_2 \cdot C_2$ | $\frac{1}{4} +  S_{22} ^2 + \frac{1}{2} = 1$                                      | $S_{22} = \frac{1}{2}$                              |

- **The sign moved.** In the E-tee the minus sign sat in the side-arm column; here it sits in  $S_{12}$  instead. That single difference is the whole distinction between the two tees.

### Answer, and the duplexer the same question asks for

- **H-plane tee:**

$$[S]_{\text{H-tee}} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

- **Duplexer**, taken as an ideal 3-port ferrite circulator with port 1 the transmitter, port 2 the antenna and port 3 the receiver:

$$[S]_{\text{duplexer}} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Matched at every port ( $S_{ii} = 0$ ), lossless (every column sums to 1) and **deliberately non-reciprocal** ( $S_{21} = 1$  but  $S_{12} = 0$ ), which is the only way a 3-port can be both matched and lossless (§3.5).

### Uses, which the question asks for explicitly

| Device             | Behaviour                                                                                       | Uses                                                                                                         |
|--------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>E-plane tee</b> | Equal split, outputs in antiphase; difference junction                                          | Balanced mixers, phase comparators, the difference arm of a magic tee, feeding balanced antennas             |
| <b>H-plane tee</b> | Equal split, outputs in phase; sum junction                                                     | In-phase power dividers and combiners, feeding two antennas from one source, the sum arm of a magic tee      |
| <b>Duplexer</b>    | One antenna shared by transmitter and receiver, with the receiver isolated from the transmitter | Radar, and any single-antenna transceiver; protects the sensitive receiver front end from the transmit pulse |

## Problem 8: Derive the S-matrix of a Magic Tee 3+3 8 10

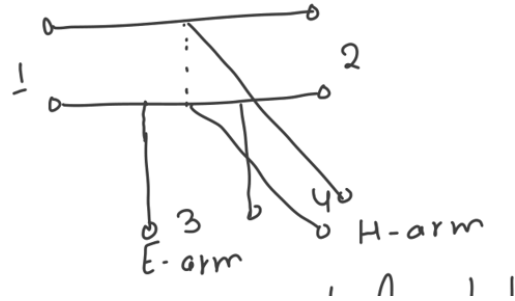
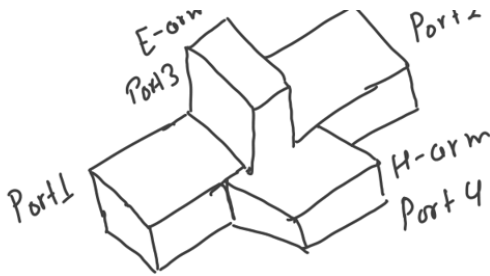
72 Ash, 74 Bh, 75 Bh, 78 Ch, 81 Bh · the single highest-value derivation in the chapter

**QUESTION AS PRINTED** The device behind five identification questions

Describe the operational principles of a magic tee based on its S-parameters, and derive its scattering matrix.

**Given**

- A hybrid (magic) tee: ports 1 and 2 are the **collinear arms**, port 3 is the **E-arm** and port 4 is the **H-arm**. Both side arms are perfectly matched to the junction.



Magic tee: an E-plane tee and an H-plane tee sharing the same junction. Source: RF Pulchowk, S-parameter derivation sheets.

**Step 1: write the blank**  $4 \times 4$

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{21} & S_{22} & S_{23} & S_{24} \\ S_{31} & S_{32} & S_{33} & S_{34} \\ S_{41} & S_{42} & S_{43} & S_{44} \end{bmatrix} \quad (\text{i})$$

**Step 2: impose the five physical facts**

| Physical fact                                                                                       | Equation              | Why                                                                                                                      |
|-----------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------|
| It contains an E-plane tee, so an input at the E-arm (port 3) leaves ports 1 and 2 <b>antiphase</b> | $S_{23} = -S_{13}$    | (ii) Same argument as Problem 6                                                                                          |
| It contains an H-plane tee, so an input at the H-arm (port 4) leaves ports 1 and 2 <b>in phase</b>  | $S_{24} = +S_{14}$    | (iii) Same argument as Problem 7                                                                                         |
| An input at the E-arm produces no output at the H-arm, and the reverse                              | $S_{34} = S_{43} = 0$ | (iv) The two side arms are geometrically orthogonal; the E-arm excites an antisymmetric field, the H-arm a symmetric one |
| Both side arms are perfectly matched to the junction                                                | $S_{33} = S_{44} = 0$ | (v) Stated in the question                                                                                               |
| No output at port 2 for an input at port 1, and the reverse                                         | $S_{12} = S_{21} = 0$ | (vi) The collinear arms are decoupled, which is the “magic”                                                              |

- With reciprocity  $S_{ij} = S_{ji}$  (vii), equations (ii) to (vii) reduce (i) to

$$[S] = \begin{bmatrix} S_{11} & 0 & S_{13} & S_{14} \\ 0 & S_{22} & -S_{13} & S_{14} \\ S_{13} & -S_{13} & 0 & 0 \\ S_{14} & S_{14} & 0 & 0 \end{bmatrix} \quad (\text{viii})$$

- Four unknowns:**  $S_{11}, S_{22}, S_{13}, S_{14}$ .

**Step 3: unitary property, row by row**

| Product         | Equation                                                                                              | What it gives                                        |
|-----------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| $R_3 \cdot C_3$ | $ S_{13} ^2 +  -S_{13} ^2 + 0 + 0 = 1$ , i.e.<br>$2 S_{13} ^2 = 1$                                    | $S_{13} = \frac{1}{\sqrt{2}}$                        |
| $R_4 \cdot C_4$ | $ S_{14} ^2 +  S_{14} ^2 + 0 + 0 = 1$ , i.e.<br>$2 S_{14} ^2 = 1$                                     | $S_{14} = \frac{1}{\sqrt{2}}$                        |
| $R_1 \cdot C_1$ | $ S_{11} ^2 + 0 +  S_{13} ^2 +  S_{14} ^2 = 1$ , i.e.<br>$ S_{11} ^2 + \frac{1}{2} + \frac{1}{2} = 1$ | $ S_{11} ^2 = 0$ , so $\mathbf{S}_{11} = \mathbf{0}$ |
| $R_2 \cdot C_2$ | $0 +  S_{22} ^2 +  S_{13} ^2 +  S_{14} ^2 = 1$                                                        | $\mathbf{S}_{22} = \mathbf{0}$                       |

- **This is the striking result:** matching the two side arms and decoupling the collinear pair **forces** the collinear arms to be matched as well. Nothing was assumed about  $S_{11}$  or  $S_{22}$ ; the unitary condition produced them.

**Answer**

$$[S]_{\text{magic tee}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

**Operational principles, read straight off the matrix**

| Drive                         | Outputs                                                                   | Consequence                                               |
|-------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>E-arm</b> (port 3)         | $b_1 = \frac{1}{\sqrt{2}}a_3, b_2 = -\frac{1}{\sqrt{2}}a_3,$<br>$b_4 = 0$ | Equal split, antiphase, nothing into the H-arm            |
| <b>H-arm</b> (port 4)         | $b_1 = b_2 = \frac{1}{\sqrt{2}}a_4, b_3 = 0$                              | Equal split, in phase, nothing into the E-arm             |
| <b>Port 1</b>                 | $b_3 = \frac{1}{\sqrt{2}}a_1, b_4 = \frac{1}{\sqrt{2}}a_1,$<br>$b_2 = 0$  | Splits into both side arms, <b>nothing reaches port 2</b> |
| <b>Ports 1 and 2 together</b> | $b_4 \propto (a_1 + a_2),$<br>$b_3 \propto (a_1 - a_2)$                   | <b>H-arm is the sum arm, E-arm the difference arm</b>     |

- **Why “magic”:** all four ports are matched *and* two pairs (1–2 and 3–4) are completely isolated, which a 3-port can never achieve.
- **Uses:** power combining and splitting (§4.7), balanced mixers, duplexers, impedance bridges, monopulse radar comparators.
- **In practice** the ideal matrix needs matching posts or irises at the junction; without them  $S_{11}$  and  $S_{22}$  are not zero.

**Problem 9: Two magic tees joined at their E-arms** 10

77 Ch · derived by connecting two 4-port matrices; asserted in `wg.py`

QUESTION AS PRINTED 77 Ch [10]

Explain the properties of two Magic Tees if one connects their E-arms and derive its S-matrix.

**Given: the two tees**

- Each tee, ports (1, 2) collinear, 3 = E-arm, 4 = H-arm,  $r = 1/\sqrt{2}$  (§4.5):  $[S_T] = r \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$ .
- Tee A ports 1, 2, 3, 4; tee B ports 1', 2', 3', 4'. The E-arms 3 and 3' are joined, so they disappear as external ports.
- Joining means: the wave **leaving** arm 3 **enters** arm 3', and vice versa:  $a_{3'} = b_3, a_3 = b_{3'}$ .
- External ports left: 1, 2, 4 (tee A) and 1', 2', 4' (tee B)  $\Rightarrow$  a **6-port**,  $6 \times 6$  matrix.

**Step 1: drive port 1 with a unit wave ( $a_1 = 1$ , all other inputs zero)**

- Inside tee A:  $b_3 = S_{31} = r, b_4 = S_{41} = r, b_2 = S_{21} = 0$ .
- $b_3 = r$  enters tee B's E-arm:  $a_{3'} = r$ .
- Inside tee B:  $b_{1'} = S_{1'3'} a_{3'} = r \cdot r = \frac{1}{2}, b_{2'} = S_{2'3'} a_{3'} = -r \cdot r = -\frac{1}{2}, b_{4'} = S_{4'3'} a_{3'} = 0$ .
- Nothing reflects back into arm 3, because  $S_{3'3'} = 0$  (matched E-arm).

- Column for port 1:  $(b_1, b_2, b_4, b_{1'}, b_{2'}, b_{4'}) = (0, 0, r, \frac{1}{2}, -\frac{1}{2}, 0)$ .
- Power check:  $\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 1 \checkmark$ .

### Step 2: the other drives, same way

- Port 2:  $b_3 = S_{32} = -r, b_4 = r$ ; tee B receives  $-r \Rightarrow b_{1'} = -\frac{1}{2}, b_{2'} = +\frac{1}{2}$ .
- Port 4 (H-arm A):  $b_1 = b_2 = r$ , and  $b_3 = S_{34} = 0 \Rightarrow$  **nothing crosses** to tee B.
- Ports 1', 2', 4': mirror images, by symmetry of the connection.

### Step 3: the 6-port S-matrix

$$[S] = \begin{matrix} & \begin{matrix} 1 & 2 & 4 & 1' & 2' & 4' \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 4 \\ 1' \\ 2' \\ 4' \end{matrix} & \begin{bmatrix} 0 & 0 & r & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & r & -\frac{1}{2} & \frac{1}{2} & 0 \\ r & r & 0 & 0 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & r \\ -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & r \\ 0 & 0 & 0 & r & r & 0 \end{bmatrix} \end{matrix}$$

$r = 1/\sqrt{2}$ . Identical to the `connect()` result in `wg.py`.

### Step 4: properties

- **Matched**: all six diagonal entries are 0.
- **Reciprocal**:  $[S] = [S]^T$ .
- **Lossless**: every column's powers sum to 1.
- **Collinear isolation kept**:  $S_{12} = S_{1'2'} = 0$ .
- **H-arms isolated** from each other and from the far tee:  $S_{44'} = 0$ .
- **Difference channel crosses**: an antiphase pair at (1, 2) appears entirely at (1', 2'), again antiphase.
- **Sum channel stays**: an in-phase pair at (1, 2) leaves entirely by 4.
- Use: a sum/difference separator, e.g. two monopulse channels feeding a common difference path.

## Problem 10: Two magic tees connected in the H-plane 5

78 Ch · a short note; the Problem 9 method with the H-arms joined

QUESTION AS PRINTED 78 Ch [5]

Write short notes on: Combinational effect of two magic tees connected in H-plane.

The paper's typos ("Combinational", "magic tees") are reproduced as printed; read "combinational effect of two magic tees".

### Step 1: join H-arm 4 to H-arm 4' ( $a_{4'} = b_4, a_4 = b_{4'}$ ) and follow port 1

- Tee A:  $b_3 = S_{31} = r, b_4 = S_{41} = r$ .
- $b_4 = r$  enters tee B's H-arm:  $b_{1'} = S_{14} r = \frac{1}{2}, b_{2'} = S_{24} r = +\frac{1}{2}$  (in phase),  $b_{3'} = S_{34} r = 0$ .
- From port 2:  $b_4 = S_{42} = +r$  as well, so tee B again gets  $+\frac{1}{2}, +\frac{1}{2}$ .

### Step 2: the 6-port S-matrix

$$[S] = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 1' & 2' & 3' \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 1' \\ 2' \\ 3' \end{matrix} & \begin{bmatrix} 0 & 0 & r & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -r & \frac{1}{2} & \frac{1}{2} & 0 \\ r & -r & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & r \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & -r \\ 0 & 0 & 0 & r & -r & 0 \end{bmatrix} \end{matrix}$$

### Short-note points

- Matched at all six ports, reciprocal, lossless.
- The **sum** (in-phase) signal crosses between the tees through the joined H-arms and re-appears **in phase** at 1', 2'.
- The **difference** signal stays in its own tee's E-arm.
- Collinear ports of each tee stay isolated; the two E-arms are isolated.
- The exact mirror of Problem 9: E-joining passes the difference, H-joining passes the sum.

## Problem 11: Hybrid tee with a terminated arm 4+4+2

82 Ba · 70 Ma's "hybrid tee" [8] is answered by §4.5 plus this

QUESTION AS PRINTED 82 Ba [4+4+2]

Using proper S-Matrices, mention the properties of a Hybrid-Tee in the situation of (a) terminated E-arm, (b) terminated H-arm and (c) terminated co-planner arms.

**Step 0: what “terminated” does to a matrix**

- Terminated = a **matched load**,  $\Gamma = 0$ : whatever leaves that arm is absorbed, nothing comes back, so  $a_k = 0$ .
- Therefore row  $k$  and column  $k$  simply **drop out**; the rest of the matrix is unchanged.
- Start from the magic tee, ports 1, 2 collinear (the co-planar arms), 3 = E, 4 = H.
- If the paper meant a **short**,  $\Gamma = -1$ , the answer is different; that is the 82 Bh case, Problem 12.

| Case                                             | Reduced S-matrix                                                      | Properties                                                                                                                                                                                                                                                                        |
|--------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) E-arm terminated, ports (1, 2, 4)            | $\begin{bmatrix} 0 & 0 & r \\ 0 & 0 & r \\ r & r & 0 \end{bmatrix}$   | Matched and reciprocal. From the H-arm: <b>equal, in-phase</b> halves at 1 and 2, no loss. Ports 1 and 2 <b>isolated</b> . From port 1: only half reaches port 4, the other half is absorbed in the E-arm load, so the 3-port is <b>not lossless</b> . An in-phase power divider. |
| (b) H-arm terminated, ports (1, 2, 3)            | $\begin{bmatrix} 0 & 0 & r \\ 0 & 0 & -r \\ r & -r & 0 \end{bmatrix}$ | Matched and reciprocal. From the E-arm: <b>equal, antiphase</b> halves at 1 and 2. Ports 1 and 2 isolated. From port 1: half to port 3, half lost in the H-arm load. An antiphase (180°) power divider.                                                                           |
| (c) Both co-planar arms terminated, ports (3, 4) | $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$                        | E and H arms both <b>matched</b> and <b>completely isolated</b> . Any power entering either side arm splits into the two loads and nothing comes out anywhere: the tee acts as a matched termination for both.                                                                    |

- Why (a) and (b) **must** be lossy: they are matched, reciprocal 3-ports, and §3.5 proves no such 3-port can be lossless. The load is where the missing half goes.
- Column check, (a): column 1 =  $0 + 0 + \frac{1}{2} = \frac{1}{2}$  (half absorbed); column 4 =  $\frac{1}{2} + \frac{1}{2} = 1$  (lossless from the H-arm).

**Problem 12: Magic tee with both E- and H-arms short-circuited** 4+4

82 Bh · verified/added no source note carries this; derived here and asserted in `sparam.py`

**QUESTION AS PRINTED** 82 Bh [4+4]

Design a model of a three-port network and define its characteristic parameters. Prepare the S-matrix of a rectangular magic tee having both E- and H-arms shorted, and explain how it behaves.

**Given**

- The ideal magic tee of Problem 8, with ports 3 (E-arm) and 4 (H-arm) **short-circuited at the junction reference plane**.
- A short reflects with  $\Gamma = -1$ , so  $a_3 = -b_3$  and  $a_4 = -b_4$ .

**Step 1: write out the four scattering equations**

- From  $[b] = [S][a]$  with the magic-tee matrix:

$$\begin{aligned} b_1 &= \frac{1}{\sqrt{2}}(a_3 + a_4), & b_2 &= \frac{1}{\sqrt{2}}(-a_3 + a_4) \\ b_3 &= \frac{1}{\sqrt{2}}(a_1 - a_2), & b_4 &= \frac{1}{\sqrt{2}}(a_1 + a_2) \end{aligned}$$

**Step 2: substitute the short-circuit conditions**

- $a_3 = -b_3 = -\frac{1}{\sqrt{2}}(a_1 - a_2)$ .
- $a_4 = -b_4 = -\frac{1}{\sqrt{2}}(a_1 + a_2)$ .

**Step 3: eliminate and collect**

- Into the expression for  $b_1$ :

$$b_1 = \frac{1}{\sqrt{2}} \left[ -\frac{1}{\sqrt{2}}(a_1 - a_2) - \frac{1}{\sqrt{2}}(a_1 + a_2) \right] = -\frac{1}{2} [(a_1 - a_2) + (a_1 + a_2)] = -\frac{1}{2}(2a_1) = -a_1$$

- Into the expression for  $b_2$ :

$$b_2 = \frac{1}{\sqrt{2}} \left[ +\frac{1}{\sqrt{2}}(a_1 - a_2) - \frac{1}{\sqrt{2}}(a_1 + a_2) \right] = \frac{1}{2} [(a_1 - a_2) - (a_1 + a_2)] = \frac{1}{2}(-2a_2) = -a_2$$

- The  $a_2$  terms cancel in  $b_1$  and the  $a_1$  terms cancel in  $b_2$ : the two reflections arriving from the shorted arms add in the driven arm and subtract in the other one.

**Answer**

- Seen from the two collinear arms, the shorted magic tee is the two-port

$$[S]_{\text{shorted}} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

**How it behaves**

- **Total reflection:**  $|S_{11}| = |S_{22}| = 1$ , so every watt fed into port 1 comes back out of port 1. Return loss = 0 dB, VSWR =  $\infty$ .
- **Phase reversal:** the  $-1$  is a  $180^\circ$  shift, exactly what an ideal short does. The tee behaves as a **pair of independent short circuits**, one at each collinear arm.
- **Complete isolation survives:**  $S_{12} = S_{21} = 0$ , so the two arms still cannot see each other even though both are totally reflecting.
- **Still lossless:** each column sums to  $|-1|^2 = 1$ ; nothing is dissipated, everything is returned.
- **Still reciprocal:**  $[S] = [S]^T$ .
- **Use:** a reflective phase shifter or a calibration short. Moving the shorts a distance  $\ell$  out from the junction changes  $\Gamma$  from  $-1$  to  $-e^{-2j\beta\ell}$ , and the same algebra then gives  $S_{11} = S_{22} = -e^{-2j\beta\ell}$  with  $S_{12}$  still zero, so a tunable phase with a constant unity magnitude.

**Problem 13: Identify the device from a numeric 4-port matrix** 10

78 Ch · the printed matrix is missing its normalising factor

**QUESTION AS PRINTED** 78 Ch [10]

Identify and explain the properties of a microwave passive device having the following S-matrix, with necessary diagram.  $[S] =$

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

**Given**

- A  $4 \times 4$  matrix, so the device has **four ports**.
- **The paper prints the first row as “0 .0 1 1”;** the stray point is a scanner artefact over a zero, and the matrix is the standard form.

**Step 1: run the four tests**

| Test               | Computation                                                                                                     | Result        | Conclusion                                                              |
|--------------------|-----------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------|
| <b>Diagonal</b>    | $S_{11} = S_{22} = S_{33} = S_{44} = 0$                                                                         | all zero      | <b>Matched at every port</b>                                            |
| <b>Transpose</b>   | $S_{13} = S_{31} = 1, S_{23} = S_{32} = -1,$<br>$S_{14} = S_{41} = 1, S_{24} = S_{42} = 1$                      | $[S] = [S]^T$ | <b>Reciprocal</b> , so passive and non-ferrite                          |
| <b>Column sums</b> | $ 0 ^2 +  0 ^2 +  1 ^2 +  1 ^2 = 2$ for every column                                                            | $2 \neq 1$    | <b>Not unitary as printed:</b> the $1/\sqrt{2}$ factor has been omitted |
| <b>Column dots</b> | $C_1 \cdot C_2^* = (0)(0) + (0)(0) + (1)(-1) + (1)(1) = 0;$<br>$C_3 \cdot C_4^* = (1)(1) + (-1)(1) + 0 + 0 = 0$ | all zero      | Phases are consistent; only the scale is wrong                          |

- **Judging the condition:** divide the printed matrix by  $\sqrt{2}$ . Then every column sums to  $\frac{1}{2} + \frac{1}{2} = 1$  and the device is **lossless**. **Say this explicitly:** it is the “judge the condition” mark.

**Step 2: map the zeros and read the signs**

- **Zero blocks:**  $S_{12} = S_{21} = 0$ , so **ports 1 and 2 are isolated**;  $S_{34} = S_{43} = 0$ , so **ports 3 and 4 are isolated**. Two isolated pairs on a matched, lossless, reciprocal 4-port.
- **Column 3** reads  $(1, -1)$  down the collinear ports: the signs **flip**, so port 3 is the **difference (E) arm**.
- **Column 4** reads  $(1, 1)$ : the signs **match**, so port 4 is the **sum (H) arm**.
- **Equal magnitudes** everywhere off the zero blocks, so every split is 50/50.

**Answer**

- The device is a magic tee (hybrid tee), with ports 1 and 2 the collinear arms, port 3 the E-arm and port 4 the H-arm. Normalised:

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

- Diagram: reproduce the 3D sketch of Problem 8 with all four arms labelled.

#### Properties, one line each

- **Matched** at all four ports, so no port reflects:  $S_{ii} = 0$ , return loss infinite in the ideal case.
- **Reciprocal**,  $[S] = [S]^T$ : passive waveguide, no ferrite, no gain.
- **Lossless** once normalised: all incident power is redistributed, none dissipated.
- **Two isolated pairs**: 1-2 and 3-4. Two sources on ports 1 and 2 cannot see each other, and the E-arm and H-arm are decoupled.
- **Equal power division**,  $|S_{13}|^2 = \frac{1}{2}$ : every drive splits in half, i.e. a 3 dB hybrid.
- **Phase**: E-arm drive gives antiphase outputs, H-arm drive gives in-phase outputs.
- **Uses**: power combiner and splitter, duplexer, balanced mixer, monopulse comparator, impedance bridge.

### Problem 14: Identify and complete a symbolic 4-port matrix 5 8

74 Bh, 81 Bh · the same device as Problem 13, printed before the unitary step

QUESTION AS PRINTED 81 Bh [5] and 74 Bh [8]

Identify and explain, along with a neat diagram, the properties of the passive microwave device described by the following S-matrix:

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix}$$

#### Step 1: read the structure before doing any algebra

| Feature                         | Where              | What it means                                                       |
|---------------------------------|--------------------|---------------------------------------------------------------------|
| $S_{33} = S_{44} = 0$           | bottom-right block | Ports 3 and 4 are matched to the junction                           |
| $S_{34} = S_{43} = 0$           | bottom-right block | Ports 3 and 4 are isolated from each other                          |
| Column 3 is $(S_{13}, S_{13})$  | signs <b>match</b> | Port 3 drives the collinear arms <b>in phase</b> : the <b>H-arm</b> |
| Column 4 is $(S_{14}, -S_{14})$ | signs <b>flip</b>  | Port 4 drives them <b>in antiphase</b> : the <b>E-arm</b>           |
| $[S] = [S]^T$                   | everywhere         | Reciprocal, hence passive and isotropic                             |

- Four ports, two matched side arms, one sum arm and one difference arm: **this is a magic tee**. Note that the port labels are the reverse of Problem 13: here **port 3 is the H-arm and port 4 the E-arm**. **Read the signs, not the numbers**.

#### Step 2: finish the job the printed matrix leaves undone

- The matrix still carries  $S_{11}, S_{12}, S_{22}, S_{13}, S_{14}$  as unknowns. Apply the unitary property exactly as in Problem 8:

| Product         | Equation                                                                                     | Result                                                                     |
|-----------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| $R_3 \cdot C_3$ | $ S_{13} ^2 +  S_{13} ^2 = 1$                                                                | $S_{13} = \frac{1}{\sqrt{2}}$                                              |
| $R_4 \cdot C_4$ | $ S_{14} ^2 +  -S_{14} ^2 = 1$                                                               | $S_{14} = \frac{1}{\sqrt{2}}$                                              |
| $R_3 \cdot C_4$ | $S_{13}S_{14}^* - S_{13}S_{14}^* = 0$                                                        | satisfied identically, no new information                                  |
| $R_1 \cdot C_2$ | $S_{11}S_{12}^* + S_{12}S_{22}^* +  S_{13} ^2 -  S_{14} ^2 = 0$ ,<br>and the last two cancel | $S_{11}S_{12}^* + S_{12}S_{22}^* = 0$                                      |
| $R_1 \cdot C_1$ | $ S_{11} ^2 +  S_{12} ^2 + \frac{1}{2} + \frac{1}{2} = 1$                                    | $ S_{11} ^2 +  S_{12} ^2 = 0$ , so <b><math>S_{11} = S_{12} = 0</math></b> |
| $R_2 \cdot C_2$ | $ S_{12} ^2 +  S_{22} ^2 + \frac{1}{2} + \frac{1}{2} = 1$                                    | <b><math>S_{22} = 0</math></b>                                             |

**Answer**

- Magic tee, with the completed matrix

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & -1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 \end{bmatrix}$$

which is Problem 13's matrix with columns 3 and 4 interchanged, as the port labelling demands.

- **The strongest single line in this answer:** nothing was assumed about  $S_{11}$ ,  $S_{12}$  or  $S_{22}$ , yet all three came out zero. Matching the two side arms of a 4-port **forces** the collinear arms to be matched and isolated. That is precisely what a 3-port cannot do (§3.5).
- Properties and uses: as listed in Problem 13.

**Problem 15: Judge the condition of a given S-matrix** 6

75 Bh · the isolation is already printed in, so only the scale is left to judge

QUESTION AS PRINTED 75 Bh [6]

Which passive microwave device is explained by this S-matrix? Judge the condition and explain its characteristics.  $[S] = \begin{bmatrix} S_{11} & 0 & S_{13} & S_{14} \\ 0 & S_{22} & -S_{13} & S_{14} \\ S_{13} & -S_{13} & 0 & 0 \\ S_{14} & S_{14} & 0 & 0 \end{bmatrix}$

**Step 1: identify**

- Four ports;  $S_{33} = S_{44} = 0$  and  $S_{34} = S_{43} = 0$ , so ports 3 and 4 are matched and mutually isolated.
- $S_{12} = S_{21} = 0$  is **already written in**, so ports 1 and 2 are isolated too: the collinear pair.
- Column 3 is  $(S_{13}, -S_{13})$ , signs flipping → **E-arm**. Column 4 is  $(S_{14}, S_{14})$ , signs matching → **H-arm**.
- $[S] = [S]^T \rightarrow$  reciprocal, so passive.
- Magic tee, with the port labelling of Problem 13.

**Step 2: judge the condition (what the question actually rewards)**

| Test            | Computation                                                 | Result                        | Conclusion                            |
|-----------------|-------------------------------------------------------------|-------------------------------|---------------------------------------|
| Diagonal        | $S_{11}, S_{22}$ left as unknowns;<br>$S_{33} = S_{44} = 0$ | side arms<br>matched          | Collinear match not yet stated        |
| Transpose       | every $S_{ij} = S_{ji}$                                     | symmetric                     | Reciprocal                            |
| Column 3<br>sum | $ S_{13} ^2 +  -S_{13} ^2 = 1$                              | $S_{13} = \frac{1}{\sqrt{2}}$ | Equal, antiphase split from the E-arm |
| Column 4<br>sum | $ S_{14} ^2 +  S_{14} ^2 = 1$                               | $S_{14} = \frac{1}{\sqrt{2}}$ | Equal, in-phase split from the H-arm  |
| Column 1<br>sum | $ S_{11} ^2 + 0 + \frac{1}{2} + \frac{1}{2} = 1$            | $S_{11} = 0$                  | Port 1 is matched after all           |
| Column 2<br>sum | $0 +  S_{22} ^2 + \frac{1}{2} + \frac{1}{2} = 1$            | $S_{22} = 0$                  | Port 2 likewise                       |

- **The condition:** the matrix describes a **lossless, reciprocal, matched-at-all-four-ports** device, and this is **consistent** only because it has four ports. The same set of demands on a 3-port is impossible (§3.5).
- Completed:

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

**Characteristics**

- **3 dB equal division:**  $|S_{13}|^2 = |S_{14}|^2 = \frac{1}{2}$ , so every input splits half and half.
- **Double isolation:** 1-2 and 3-4. Two transmitters on the collinear arms cannot load each other; the E-arm and H-arm cannot leak into each other.
- **Phase discrimination:** sum at the H-arm, difference at the E-arm, which is what makes it usable as a comparator.

- **Bidirectional:** being reciprocal, it combines as readily as it splits.
- **Limitation:** the ideal matrix assumes matching irises at the junction and holds over a limited bandwidth.

### Problem 16: Two radar transmitters, twice the power into one antenna 6 8

73 Bh, 74 Ma · a design question answered entirely by one matrix multiplication

QUESTION AS PRINTED 73 Bh [8] and 74 Ma [6]

Suppose there are two identical radar transmitters and a few passive devices in equipment stock. A particular application requires twice more input power to an antenna than either transmitter can deliver. As an RF engineer, give your appropriate solution with necessary figures, mathematics and sufficient explanation.

#### Given

- Two identical transmitters, each able to deliver  $P$  watts.
- One antenna, which must receive  $2P$  watts.

#### Step 1: choose the device, and say why

- **Choice:** a **magic tee** used as a power **combiner** (a 3 dB hybrid coupler works identically).
- **Why not a simple T-junction or Y-junction:**
  - A plain junction gives  $S_{12} \neq 0$ , so each transmitter drives power into the other one: mutual loading, pulling of the oscillators, and possible damage.
  - Its collinear ports are not matched ( $S_{11} = S_{22} = \frac{1}{2}$  for a tee, Problem 6), so both transmitters see a high VSWR.
- The magic tee cures both:  $S_{12} = 0$  (perfect isolation) and  $S_{ii} = 0$  (matched at every port).

#### Step 2: the connection

| Port          | Connected to       | Reason                                                                   |
|---------------|--------------------|--------------------------------------------------------------------------|
| 1 (collinear) | Transmitter A      | Isolated from port 2                                                     |
| 2 (collinear) | Transmitter B      | Isolated from port 1                                                     |
| 3 (E-arm)     | Matched dummy load | Takes the <b>difference</b> , which is zero when the drives are balanced |
| 4 (H-arm)     | Antenna            | Takes the <b>sum</b> of the two drives                                   |

- The two feed lines must be cut to the **same electrical length**, so that the transmitters arrive in phase.

#### Step 3: the mathematics

- Both transmitters equal and in phase, nothing incident on the side arms:

$$[a] = [a \quad a \quad 0 \quad 0]^T, \quad |a|^2 = P$$

- Multiply  $[b] = [S][a]$  with the magic-tee matrix:

$$\begin{aligned}
 b_3 &= \frac{1}{\sqrt{2}}(a_1 - a_2) = \frac{1}{\sqrt{2}}(a - a) = 0 \\
 b_4 &= \frac{1}{\sqrt{2}}(a_1 + a_2) = \frac{1}{\sqrt{2}}(2a) = \sqrt{2}a \\
 b_1 &= \frac{1}{\sqrt{2}}(a_3 + a_4) = 0, \quad b_2 = \frac{1}{\sqrt{2}}(-a_3 + a_4) = 0
 \end{aligned}$$

#### Step 4: read the powers

| Port                  | Wave              | Power                   | Meaning                                             |
|-----------------------|-------------------|-------------------------|-----------------------------------------------------|
| 4, antenna            | $b_4 = \sqrt{2}a$ | $ b_4 ^2 = 2 a ^2 = 2P$ | <b>The requirement, met exactly</b>                 |
| 3, dummy load         | $b_3 = 0$         | 0                       | Nothing wasted when the drives are balanced         |
| 1 and 2, transmitters | $b_1 = b_2 = 0$   | 0                       | No reflection back into either transmitter          |
| <b>Total out</b>      |                   | <b>2P</b>               | Equals the total in, as a lossless network requires |

**Answer**

- Feed the two transmitters into the collinear arms of a magic tee, antenna on the H-arm, matched load on the E-arm. The antenna then receives  $2P$  watts, twice what either transmitter alone can deliver.

**Explanation, which carries the remaining marks**

- **Why the sum appears at the H-arm:** an H-plane junction excites the two collinear arms in phase, so by reciprocity two in-phase drives add there.
- **Why the dummy load matters:** if the transmitters are unequal or out of phase,  $b_3 = \frac{1}{\sqrt{2}}(a_1 - a_2) \neq 0$  and the imbalance is **absorbed** by the load instead of being reflected into the transmitters. The load must be rated for the worst-case imbalance, i.e. the full power of one transmitter.
- **Phase error costs power:** with a phase error  $\phi$  between the drives, the antenna receives  $2P \cos^2(\phi/2)$  and the load takes the rest. A  $20^\circ$  error already loses 0.13 dB.
- **Alternatives:** a 3 dB branch-line or rat-race hybrid does the same job in microstrip; a Wilkinson combiner does it with an isolation resistor when the device must be planar.

**Problem 17: ASR duplexer delivering half the transmitter power** 8 10

73 Ma, 76 Bh · the same magic tee, driven from the other end

QUESTION AS PRINTED 76 Bh [10] and 73 Ma [8]

A radar installation engineer is given the responsibility of installing an Airport Surveillance Radar that requires half the input power to the antenna than the transmitter can deliver, using a duplexer. Prepare its S-matrix. (73 Ma: choose a suitable passive microwave device to split power in half and explain its properties.)

**Given**

- A transmitter delivering  $P$  watts.
- The antenna must receive  $P/2$  watts, and the receiver must share the same antenna while staying isolated from the transmitter.

**Step 1: choose the device**

- **Choice:** a **magic tee used as a duplexer**. It is the only passive device that splits power exactly in half *and* provides the transmitter-to-receiver isolation a duplexer must have.
- A resistive splitter would also halve the power but would burn the other half and give no isolation.

**Step 2: the connection**

| Port          | Connected to       | Reason                                                          |
|---------------|--------------------|-----------------------------------------------------------------|
| 4 (H-arm)     | Transmitter        | Drives both collinear arms equally and in phase                 |
| 1 (collinear) | Antenna            | Receives half the transmitter power                             |
| 2 (collinear) | Matched dummy load | Absorbs the other half, and balances the bridge                 |
| 3 (E-arm)     | Receiver           | Isolated from the H-arm, so the transmit pulse never reaches it |

**Step 3: transmit path**

- Drive vector  $[a] = [0 \ 0 \ 0 \ a]^T$  with  $|a|^2 = P$ .
- From  $[b] = [S][a]$ :

$$b_1 = \frac{1}{\sqrt{2}}a_4 = \frac{1}{\sqrt{2}}a, \quad b_2 = \frac{1}{\sqrt{2}}a_4 = \frac{1}{\sqrt{2}}a, \quad b_3 = 0$$

- Powers:

$$|b_1|^2 = \frac{1}{2}|a|^2 = \frac{P}{2}, \quad |b_2|^2 = \frac{P}{2}, \quad |b_3|^2 = 0$$

- **The antenna receives exactly  $P/2$** , which is the requirement, and the receiver on the E-arm receives **nothing** from the transmitter, which is the point of a duplexer.

**Step 4: receive path**

- The echo arrives at port 1 alone:  $[a] = [a_1 \ 0 \ 0 \ 0]^T$  with  $|a_1|^2 = P_{echo}$ .
- $b_3 = \frac{1}{\sqrt{2}}a_1$  and  $b_4 = \frac{1}{\sqrt{2}}a_1$ , so

$$|b_3|^2 = \frac{P_{echo}}{2} \quad \text{to the receiver,} \quad |b_4|^2 = \frac{P_{echo}}{2} \quad \text{back into the transmitter}$$

- **Nothing reaches port 2**, since  $S_{21} = 0$ .

### Answer

- The duplexer is the magic tee, whose S-matrix is

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

with port 4 the transmitter, port 1 the antenna, port 2 the matched load and port 3 the receiver.

### Properties and the honest limitation

- **Equal split:**  $|S_{14}|^2 = |S_{24}|^2 = \frac{1}{2}$ , a 3 dB hybrid.
- **Isolation:** transmitter (port 4) to receiver (port 3) is  $S_{34} = 0$ , so the receiver front end is protected from the full transmit pulse.
- **Matched at every port**, so neither the transmitter nor the antenna sees a standing wave.
- **The 3 dB penalty is real and should be stated:** half the transmit power goes into the dummy load, and half the echo goes back to the transmitter. The magic-tee duplexer therefore costs 3 dB in each direction.
- **Better in practice:** a **ferrite circulator** duplexer,  $[S] = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ , routes the *whole* transmit power to the antenna and the *whole* echo to the receiver, with no 3 dB loss. Quote it as the improvement whenever the question does not explicitly demand the half-power split, as **76 Bh** does.

## 6. RF Design Practices: Numerical Problems

10 solved problems covering **every amplifier S-parameter set in 22 papers** and the 82 Bh filter · 13 transistor sets, 1 filter design

### READ THIS FIRST

Every amplifier question in the archive is one procedure run on four numbers. The papers attach the formula sheet ( $\Delta$ ,  $K$ ,  $\mu$ ,  $B_1$ ,  $C_1$ ,  $\Gamma_S$ ,  $\Gamma_L$ ,  $C_S$ ,  $R_S$ ,  $C_L$ ,  $R_L$ ,  $G_{T,max}$ ), so the marks are in doing the steps in order and getting the complex arithmetic right.

- **The procedure, always in this order**, always tabulated: (1) magnitudes and products, (2)  $\Delta$ , (3)  $K$  and  $\mu$ , verdict, (4) if stable,  $B_1$ ,  $C_1$ ,  $B_2$ ,  $C_2$ ,  $\Gamma_S$ ,  $\Gamma_L$ ,  $G_{T,max}$ , (5) unilateral  $G_S$ ,  $G_0$ ,  $G_L$ ,  $G_{TU,max}$ , (6) figure of merit  $M$  and the error bound, (7) compare. Problem 1 shows every line.
- **If  $K < 1$  stop and say so.** The simultaneous conjugate match does not exist (the square root goes imaginary or  $|\Gamma| > 1$ ). Draw stability circles instead and give the unilateral gain with its design points checked to be stable (Problems 3 to 5).
- **Work in polar form for products, rectangular for sums.** Most slips are a sum done in polar form. Keep four significant figures until the end.
- **“Compare” means a sentence:** which is larger, by how many dB, and whether  $M$  says the unilateral figure can be trusted.
- Every number below is from `src/amp.py`, which cross-checks  $G_{T,max}$  against  $|S_{21}/S_{12}|(K - \sqrt{K^2 - 1})$ , confirms  $\Gamma_S = \Gamma_{in}^*$  and  $\Gamma_L = \Gamma_{out}^*$ , and walks each matching network forward.

### Problem 1: Stability and maximum gain, bilateral and unilateral (the full method) 6+3+3 8+4 15

71 Ma, 81 Ba, 82 Ba · the same transistor in three papers; 71 Bh prints it with  $S_{11}$  at  $+150^\circ$  (Problem 2)

#### QUESTION AS PRINTED 71 Ma [15]

Using the following S-parameters of  $S_{11}=0.55\angle 150^\circ$ ,  $S_{12}=0.04\angle 20^\circ$ ,  $S_{21}=2.82\angle 180^\circ$  and  $S_{22}=0.45\angle 30^\circ$ , calculate and compare maximum power gain for unilateral and bilateral modes.

81 Ba [6+3+3] asks the same plus a design flowchart (§6.4); 82 Ba [8+4] asks to “design a microwave amplifier for maximum power gain” for both cases (add the matching networks of Step 6).

#### Step 1: magnitudes and products

| Quantity                 | Working                                         | Value                                         |
|--------------------------|-------------------------------------------------|-----------------------------------------------|
| $ S_{11} ^2,  S_{22} ^2$ | $0.55^2, 0.45^2$                                | $0.3025, 0.2025$                              |
| $ S_{21} ^2$             | $2.82^2$                                        | $7.9524$                                      |
| $S_{11}S_{22}$           | $0.55 \times 0.45\angle(-150^\circ - 30^\circ)$ | $0.2475\angle 180^\circ = -0.2475$            |
| $S_{12}S_{21}$           | $0.04 \times 2.82\angle(20^\circ + 180^\circ)$  | $0.1128\angle -160^\circ = -0.1060 - j0.0386$ |

#### Step 2 and 3: $\Delta$ , $K$ , $\mu$

- $\Delta = S_{11}S_{22} - S_{12}S_{21} = -0.2475 - (-0.1060 - j0.0386) = -0.1415 + j0.0386 = \boxed{0.1467\angle 164.7^\circ}$ ,  $|\Delta|^2 = 0.0215$ .
- $K = \frac{1 - 0.3025 - 0.2025 + 0.0215}{2 \times 0.1128} = \frac{0.5165}{0.2256} = \boxed{2.290}$ .
- $\Delta S_{11}^* = 0.1467\angle 164.7^\circ \times 0.55\angle 150^\circ = 0.0807\angle -45.3^\circ$ ;  $S_{22} - \Delta S_{11}^* = 0.3728\angle -26.7^\circ$ ;  $\mu = \frac{1 - 0.3025}{0.3728 + 0.1128} = \boxed{1.436}$ .
- $K > 1$ ,  $|\Delta| < 1$  (and  $\mu > 1$ )  $\Rightarrow$  **unconditionally stable**: any passive source and load are safe, so the conjugate match may be used.

**Step 4: bilateral maximum gain (simultaneous conjugate match)**

| Quantity     | Formula and substitution                                                                                                          | Value                                         |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| $B_1$        | $1 +  S_{11} ^2 -  S_{22} ^2 -  \Delta ^2 =$<br>$1 + 0.3025 - 0.2025 - 0.0215$                                                    | 1.0785                                        |
| $C_1$        | $S_{11} - \Delta S_{22}^*$                                                                                                        | $0.4866 \angle -148.0^\circ$                  |
| $B_2$        | $1 +  S_{22} ^2 -  S_{11} ^2 -  \Delta ^2$                                                                                        | 0.8785                                        |
| $C_2$        | $S_{22} - \Delta S_{11}^*$                                                                                                        | $0.3728 \angle -26.7^\circ$                   |
| Discriminant | $B_1^2 - 4 C_1 ^2 = 1.1632 - 0.9471$ (same for $B_2, C_2$ )                                                                       | 0.2159, root 0.4646                           |
| $\Gamma_S$   | $\frac{B_1 - \sqrt{\cdot}}{2C_1} = \frac{1.0785 - 0.4646}{2 \times 0.4866 \angle -148.0^\circ}$                                   | <b><math>0.6307 \angle 148.0^\circ</math></b> |
| $\Gamma_L$   | $\frac{B_2 - \sqrt{\cdot}}{2C_2} = \frac{0.8785 - 0.4646}{2 \times 0.3728 \angle -26.7^\circ}$                                    | <b><math>0.5551 \angle 26.7^\circ</math></b>  |
| $G_{T,max}$  | $\frac{1}{1 -  \Gamma_S ^2}  S_{21} ^2 \frac{1 -  \Gamma_L ^2}{ 1 - S_{22}\Gamma_L ^2} =$<br>$1.6606 \times 7.9524 \times 1.2275$ | <b><math>16.21 = 12.10</math> dB</b>          |
| Check        | $\frac{ S_{21} }{ S_{12} } (K - \sqrt{K^2 - 1}) =$<br>$70.5 \times (2.290 - 2.060)$                                               | 16.21 ✓                                       |

Take the minus sign because  $B_1, B_2 > 0$ ; the plus sign gives  $|\Gamma| > 1$ .

**Step 5 and 6: unilateral maximum gain and its validity**

| Quantity     | Formula and substitution                                           | Value                                |
|--------------|--------------------------------------------------------------------|--------------------------------------|
| $G_{S,max}$  | $1/(1 -  S_{11} ^2) = 1/0.6975$                                    | 1.434 = 1.56 dB                      |
| $G_0$        | $ S_{21} ^2$                                                       | 7.952 = 9.00 dB                      |
| $G_{L,max}$  | $1/(1 -  S_{22} ^2) = 1/0.7975$                                    | 1.254 = 0.98 dB                      |
| $G_{TU,max}$ | $1.434 \times 7.952 \times 1.254$                                  | <b><math>14.30 = 11.55</math> dB</b> |
| $M$          | $\frac{0.6975 \times 0.7975}{1}$                                   | 0.0502                               |
| Error bound  | $\frac{1}{(1.0502)^2} < \frac{G_T}{G_{TU}} < \frac{1}{(0.9498)^2}$ | -0.43 to +0.45 dB                    |

**Step 7: compare**

- $G_{T,max} = 12.10$  dB (bilateral) vs  $G_{TU,max} = 11.55$  dB (unilateral): the bilateral design gives **0.55 dB more**.
- The unilateral error bound is  $\pm 0.45$  dB, and the true bilateral figure lies just outside it: the unilateral assumption is **approximately valid** ( $M = 0.05$  is at the usual limit) but slightly pessimistic, because  $S_{12}$  feedback here adds gain.

**Design (82 Ba): matching networks for the bilateral case**

- Input: open shunt stub  $0.1622\lambda$  at the  $50\ \Omega$  side, then  $0.1152\lambda$  of  $50\ \Omega$  line to the gate  $\rightarrow$  presents  $\Gamma_S = 0.631 \angle 148.0^\circ$ .
- Output: open stub  $0.1477\lambda$ , line  $0.2910\lambda \rightarrow \Gamma_L = 0.555 \angle 26.7^\circ$ . Method in Problem 6.

**Problem 2: The same method on the other unconditionally stable sets** 10 12 18 4+4+4+4 5+5 7+8

69 Bh, 70 Bh, 71 Bh, 72 Ma, 74 Bh, 74 Ma, 77 Ch, 80 Bh, 80 Ch, 81 Bh, 82 Bh · identical steps; answers tabulated with the intermediates a marker checks

**QUESTION AS PRINTED** 80 Bh [10]

A GaAs MESFET has the following S-parameters measured with a  $50\ \Omega$  resistance at 10 GHz:  $S_{11} = 0.55 \angle -160^\circ$ ,  $S_{12} = 0.04 \angle 10^\circ$ ,  $S_{21} = 4.82 \angle 180^\circ$ ,  $S_{22} = 0.45 \angle -30^\circ$ . Determine the stability and compare maximum power gains for unilateral and bilateral modes using supplied formulas.

Other statements, verbatim in the Sorted PYQ document: 70 Bh “find the maximum gain” [10]; 74 Bh adds a flowchart and “trace  $Z_{in}$ ,  $Z_{out}$  on the Smith chart” [4+4+4+4]; 69 Bh “design an amplifier to attain maximum gain at 4.0 GHz” [18]; 77 Ch “Ku-band satellite transmit amplifier, HEMT at 4 GHz” [5+5]; 80 Ch “check the stability and find the unilateral gain” [7+8]; 72 Ma “check the stability and find the maximum power gain” [10]; 74 Ma [12] and 82 Bh [12] “stability and compare”; 81 Bh “investigate with neat flow diagram the stability” [3+8]; 71 Bh as Problem 1 with  $S_{11} = 0.55 \angle 150^\circ$ .

## Stability

| Paper              | $S_{11}, S_{12}, S_{21}, S_{22}$                                                        | $\Delta$              | $K$   | $\mu$ | Verdict                 |
|--------------------|-----------------------------------------------------------------------------------------|-----------------------|-------|-------|-------------------------|
| 80 Bh              | 0.55 $\angle$ -160, 0.04 $\angle$ 10,<br>4.82 $\angle$ 180, 0.45 $\angle$ -30           | 0.094 $\angle$ 125.2  | 1.306 | 1.147 | uncond. stable          |
| 70 Bh, 74 Bh       | 0.656 $\angle$ 146.7, 0.122 $\angle$ 46.1,<br>2.30 $\angle$ 44.7, 0.172 $\angle$ -117.1 | 0.247 $\angle$ -65.6  | 1.071 | 1.080 | uncond. stable (barely) |
| 69 Bh, 77 Ch       | 0.72 $\angle$ -116, 0.03 $\angle$ 57, 2.60 $\angle$ 76,<br>0.73 $\angle$ -54            | 0.488 $\angle$ -162.3 | 1.195 | 1.040 | uncond. stable          |
| 82 Bh              | 0.65 $\angle$ 146, 0.12 $\angle$ 46, 2.30 $\angle$ 44,<br>0.17 $\angle$ -172            | 0.339 $\angle$ -73.0  | 1.202 | 1.317 | uncond. stable          |
| 80 Ch              | 0.45 $\angle$ -160, 0.04 $\angle$ 20,<br>3.12 $\angle$ 180, 0.35 $\angle$ -40           | 0.101 $\angle$ 107.7  | 2.746 | 1.776 | uncond. stable          |
| 72 Ma (79 Ch)      | 0.45 $\angle$ 163, 0.04 $\angle$ 40,<br>2.55 $\angle$ -106, 0.46 $\angle$ -65           | 0.306 $\angle$ 103.3  | 3.332 | 1.877 | uncond. stable          |
| 74 Ma              | 0.78 $\angle$ -113, 0.028 $\angle$ 247,<br>2.60 $\angle$ 76, 0.81 $\angle$ -54          | 0.681 $\angle$ -171.7 | 1.367 | 1.097 | uncond. stable          |
| 81 Bh              | 0.6 $\angle$ -140, 0.03 $\angle$ 62, 2.40 $\angle$ 54,<br>0.70 $\angle$ -58             | 0.374 $\angle$ 170.0  | 2.011 | 1.161 | uncond. stable          |
| 71 Bh (as printed) | 0.55 $\angle$ +150, 0.04 $\angle$ 20,<br>2.82 $\angle$ 180, 0.45 $\angle$ -30           | 0.254 $\angle$ 94.0   | 2.479 | 1.574 | uncond. stable          |

## Bilateral and unilateral maximum gain

| Paper              | $\Gamma_S$            | $\Gamma_L$          | $G_{T,max}$     | $G_S, G_0, G_L$ (dB) | $M$   | $G_{TU,max}, \text{bound}$    |
|--------------------|-----------------------|---------------------|-----------------|----------------------|-------|-------------------------------|
| 80 Bh              | 0.736 $\angle$ 156.7  | 0.683 $\angle$ 25.0 | <b>17.49 dB</b> | 1.56, 13.66, 0.98    | 0.086 | <b>16.21 dB</b> , -0.71/+0.78 |
| 70 Bh, 74 Bh       | 0.850 $\angle$ -150.4 | 0.655 $\angle$ 76.3 | <b>11.13 dB</b> | 2.44, 7.23, 0.13     | 0.057 | <b>9.81 dB</b> , -0.48/+0.51  |
| 69 Bh, 77 Ch       | 0.872 $\angle$ 123.4  | 0.876 $\angle$ 61.0 | <b>16.71 dB</b> | 3.17, 8.30, 3.31     | 0.182 | <b>14.78 dB</b> , -1.45/+1.75 |
| 82 Bh              | 0.743 $\angle$ -150.0 | 0.378 $\angle$ 88.9 | <b>10.11 dB</b> | 2.38, 7.23, 0.13     | 0.054 | <b>9.75 dB</b> , -0.46/+0.49  |
| 80 Ch              | 0.503 $\angle$ 156.2  | 0.419 $\angle$ 33.6 | <b>11.68 dB</b> | 0.98, 9.88, 0.57     | 0.028 | <b>11.43 dB</b> , -0.24/+0.25 |
| 72 Ma (79 Ch)      | 0.401 $\angle$ -160.6 | 0.413 $\angle$ 67.2 | <b>9.91 dB</b>  | 0.98, 8.13, 1.03     | 0.034 | <b>10.15 dB</b> , -0.29/+0.30 |
| 74 Ma              | 0.752 $\angle$ 101.9  | 0.789 $\angle$ 45.2 | <b>16.06 dB</b> | 4.07, 8.30, 4.64     | 0.342 | <b>17.01 dB</b> , -2.55/+3.63 |
| 81 Bh              | 0.699 $\angle$ 146.1  | 0.772 $\angle$ 61.7 | <b>13.28 dB</b> | 1.94, 7.60, 2.92     | 0.093 | <b>12.47 dB</b> , -0.77/+0.84 |
| 71 Bh (as printed) | 0.582 $\angle$ -156.4 | 0.490 $\angle$ 19.3 | <b>11.72 dB</b> | 1.56, 9.00, 0.98     | 0.050 | <b>11.55 dB</b> , -0.43/+0.45 |

- **Reading the comparison:** bilateral exceeds unilateral in every set except 72 Ma / 79 Ch and 74 Ma, where the  $S_{12}$  feedback **subtracts** gain.
- **74 Ma and 69 Bh / 77 Ch:**  $M = 0.34$  and  $0.18$ , bounds of several dB: the unilateral model is **not** acceptable for these transistors; quote the bilateral figure as the design gain.
- **70 Bh / 74 Bh:**  $K = 1.07$ , only just stable, and  $|\Gamma_S| = 0.85$ : a high-Q, narrow-band input match.
- **71 Bh:** the other three papers print  $S_{11}$  at  $-150^\circ$ ; the  $+150^\circ$  is probably a misprint. Both give the same unilateral gain (magnitudes only) but differ by 0.38 dB bilaterally; say which one you used.
- **74 Bh, trace  $Z_{in}$  and  $Z_{out}$ :** at the conjugate match  $\Gamma_{in} = \Gamma_S^* = 0.850\angle 150.4^\circ \Rightarrow Z_{in} = 4.32 + j13.13 \Omega$  ( $z = 0.086 + j0.263$ , near the short-circuit end, upper half);  $\Gamma_{out} = \Gamma_L^* = 0.655\angle -76.3^\circ \Rightarrow Z_{out} = 25.5 - j56.9 \Omega$  ( $z = 0.51 - j1.14$ , lower half, capacitive).
- **81 Bh** asks stability only: stop after the first table, and draw the flowchart of §6.4.

Problem 3: A conditionally stable transistor 5+5 10

70 Ma, 72 Ash, 80 Ba, 79 Bh ·  $S_{11} = 0.894\angle -60.6^\circ$ : the most repeated single set

## QUESTION AS PRINTED 72 Ash [5+5]

Justify that a transistor having following S-parameters  $S_{11} = 0.894\angle -60.6^\circ$ ,  $S_{12} = 0.020\angle 62.4^\circ$ ,  $S_{21} = 3.122\angle 123.6^\circ$  and  $S_{22} = 0.781\angle -27.6^\circ$  is conditionally stable while designing an amplifier. Considering unilateral model calculate maximum gain.

70 Ma [10] asks only the justification; 80 Ba [5+5] “investigate the stability” at 6 GHz; 79 Bh [5+5] “determine the stability and compare maximum power gains for bilateral and unilateral modes”.

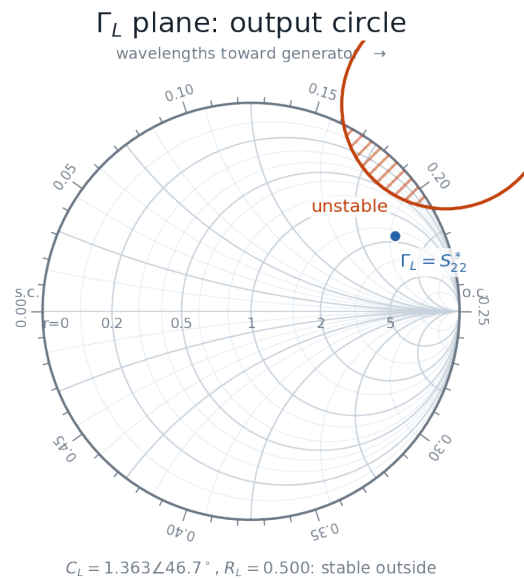
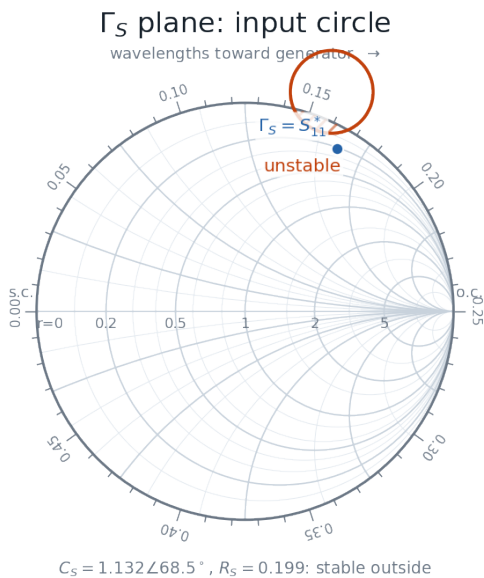
## Step 1 to 3: stability

- $|S_{11}|^2 = 0.7992$ ,  $|S_{22}|^2 = 0.6100$ ,  $|S_{21}|^2 = 9.747$ ,  $|S_{12}S_{21}| = 0.06244$ .
- $S_{11}S_{22} = 0.6982\angle -88.2^\circ$ ,  $S_{12}S_{21} = 0.0624\angle -174.0^\circ \Rightarrow \Delta = \boxed{0.6964\angle -83.1^\circ}$ ,  $|\Delta|^2 = 0.4850$ .
- $K = \frac{1 - 0.7992 - 0.6100 + 0.4850}{0.1249} = \frac{0.0758}{0.1249} = \boxed{0.607} < 1$ ;  $\mu = 0.863 < 1$ .
- $|\Delta| < 1$  but  $K < 1 \Rightarrow$  **potentially unstable**: some passive terminations make it oscillate.

**Step 4: stability circles, and the proof that it is conditionally stable**

| Circle                | Centre and radius                                                                                           | Nearest point to origin                      | Stable side                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------|-----------------------------------------------------------|
| Output ( $\Gamma_L$ ) | $C_L = \frac{(S_{22} - \Delta S_{11}^*)^*}{ S_{22} ^2 -  \Delta ^2} = 1.363 \angle 46.7^\circ, R_L = 0.500$ | $1.363 - 0.500 = 0.863 < 1$ : cuts the chart | origin outside, $ S_{11}  < 1 \Rightarrow$ <b>outside</b> |
| Input ( $\Gamma_S$ )  | $C_S = 1.132 \angle 68.5^\circ, R_S = 0.199$                                                                | $0.933 < 1$ : cuts the chart                 | <b>outside</b>                                            |

- Both circles cut the unit circle, so part of the chart is unstable and the rest is stable: the device is **conditionally stable**. Pick  $\Gamma_S, \Gamma_L$  outside the circles (deck figure, §6.4).



The two circles drawn from the table: each cuts the chart, so a sliver of the chart (hatched) is unstable; the unilateral design points  $S_{11}^*, S_{22}^*$  used in Step 5 sit on the stable side. Generated from the S-parameters by `ampfig.py`, which stops if these centres and radii differ from the table.

**Step 5: unilateral maximum gain, and whether its design points are safe**

- $G_{TU,max} = \frac{1}{1 - 0.7992} \times 9.747 \times \frac{1}{1 - 0.6100} = 4.981 \times 9.747 \times 2.564 = \boxed{124.5 = 20.95 \text{ dB}}$ .
- Design points  $\Gamma_S = S_{11}^* = 0.894 \angle 60.6^\circ, \Gamma_L = S_{22}^* = 0.781 \angle 27.6^\circ$ : both lie **outside** their circles, and together give  $|\Gamma_{in}| = 0.912, |\Gamma_{out}| = 0.848$ , both  $< 1$ . The design is stable. **verified/added**
- But  $M = 0.557$ : the error bound is  $-3.84$  to  $+7.07$  dB, so 20.95 dB is only a rough estimate for this strongly bilateral device.

**Step 6 (79 Bh): the bilateral comparison**

- With  $K < 1$  there is **no simultaneous conjugate match** and no  $G_{T,max}$ .
- The limit is the **maximum stable gain**  $MSG = |S_{21}|/|S_{12}| = 3.122/0.020 = 156.1 = \boxed{21.9 \text{ dB}}$ , approached only with stabilising loss.
- Compare: unilateral estimate 20.95 dB against an MSG ceiling of 21.9 dB.

**Problem 4: A potentially unstable set, and “modify the S-parameters if necessary”** 12

73 Ma ·  $|S_{21}| = 10.11$  makes this one borderline

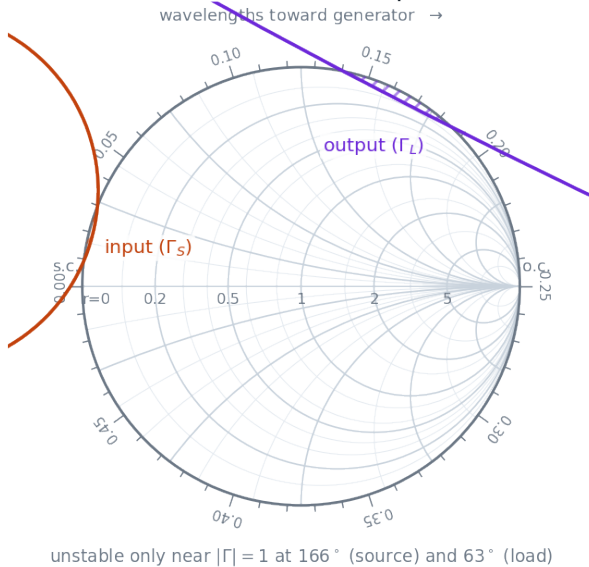
**QUESTION AS PRINTED** 73 Ma [12]

Check the stability and find the maximum gain a transistor amplifier having  $S_{11} = 0.64 \angle -169^\circ, S_{12} = 0.03 \angle 50^\circ, S_{21} = 10.11 \angle 91^\circ, S_{22} = 0.22 \angle -82^\circ$ . Consider both bilateral and unilateral model. Modify the S-parameters if necessary.

- $|S_{12}S_{21}| = 0.3033; \Delta = 0.1985 \angle -16.9^\circ, |\Delta|^2 = 0.0394$ .
- $K = \frac{1 - 0.4096 - 0.0484 + 0.0394}{0.6066} = \frac{0.5814}{0.6066} = \boxed{0.958} < 1, \mu = 0.960$ : **potentially unstable**, narrowly.

- Circles:  $C_S = 1.800 \angle 166.0^\circ$ ,  $R_S = 0.819$  (nearest 0.981);  $C_L = 34.60 \angle 62.7^\circ$ ,  $R_L = 33.64$  (nearest 0.960). Both just clip the chart edge; stable outside. Only terminations with  $|\Gamma|$  near 1 in those directions are dangerous.
- **Bilateral**: no conjugate match exists;  $MSG = 10.11/0.03 = 337 = 25.3$  dB.
- **Unilateral**:  $G_{TU,max} = 1.694 \times 102.2 \times 1.051 = \boxed{181.9 = 22.6 \text{ dB}}$ ;  $S_{11}^*$  and  $S_{22}^*$  give  $|\Gamma_{in}| = 0.70$ ,  $|\Gamma_{out}| = 0.53$ : stable.  $M = 0.076$ , bound  $-0.64$  to  $+0.69$  dB.
- **“Modify if necessary”**: for a bilateral maximum-gain design the device must be made unconditionally stable first, e.g. by a small series or shunt resistor at a port, which lowers  $|S_{21}|$  slightly and raises  $K$  above 1; a gain of around 20 dB is then realistic. State the modification you assume and recompute  $K$ .

### 73 Ma: both circles clip the rim



- Both stability circles drawn on one chart, since each only **clips the rim**: the input circle near  $166^\circ$ , the output circle (radius 33.6, so nearly a straight line) near  $63^\circ$ .
- Everything inside  $|\Gamma| \approx 0.96$  is stable: this is why “potentially unstable” here is narrow, and why  $S_{11}^*$  and  $S_{22}^*$  are safe.
- From `ampfig.py`, checked against the centres and radii above.

## Problem 5: Gains of the input and output networks for given terminations 6+4

76 Bh

QUESTION AS PRINTED 76 Bh [6+4]

A microwave transistor at 1.0 GHz with  $50 \Omega$  has  $S_{11} = 0.38 \angle -158^\circ$ ,  $S_{12} = 0.11 \angle 54^\circ$ ,  $S_{21} = 3.50 \angle 80^\circ$ ,  $S_{22} = 0.40 \angle -43^\circ$ . The source impedance is  $Z_s = 25 \Omega$  and load impedance is  $Z_L = 40 \Omega$ . Compute the gains of input and output matching networks. Also, draw a design a flowchart.

### Step 1: reflection coefficients of the given terminations

$$- \Gamma_S = \frac{25 - 50}{25 + 50} = -0.3333, \quad \Gamma_L = \frac{40 - 50}{40 + 50} = -0.1111.$$

### Step 2: stability first

- $\Delta = 0.2555 \angle -60.6^\circ$ ,  $K = 0.988 < 1$ : potentially unstable. Check the actual terminations:  $|\Gamma_{in}(\Gamma_L)| = 0.365$ ,  $|\Gamma_{out}(\Gamma_S)| = 0.545$ , both  $< 1 \Rightarrow$  **stable with these terminations.** verified/added

### Step 3: the three gain blocks (unilateral)

| Block                        | Formula and substitution                                                              | Value                   |
|------------------------------|---------------------------------------------------------------------------------------|-------------------------|
| Input network $G_S$          | $\frac{1 -  \Gamma_S ^2}{ 1 - S_{11}\Gamma_S ^2} = \frac{1 - 0.1111}{1 - 0.1111}$     | <b>1.138 = +0.56 dB</b> |
| Transistor $G_0$             | $\frac{ 1 - 0.38\angle-158^\circ \times (-0.333) ^2}{ S_{21} ^2} = 3.50^2$            | 12.25 = 10.88 dB        |
| Output network $G_L$         | $\frac{1 -  \Gamma_L ^2}{ 1 - S_{22}\Gamma_L ^2} = \frac{1 - 0.0123}{1 - 0.0123}$     | <b>0.926 = -0.34 dB</b> |
| Total $G_{TU}$               | $\frac{ 1 - 0.40\angle-43^\circ \times (-0.111) ^2}{1.138 \times 12.25 \times 0.926}$ | 12.90 = 11.11 dB        |
| Exact $G_T$ (with $S_{12}$ ) | full formula, $\Gamma_{in}$ in place of $S_{11}$                                      | 12.60 = 11.01 dB        |

- The input termination **adds** 0.56 dB (it moves toward  $S_{11}^*$ ); the output termination **costs** 0.34 dB. Maxima would be  $G_{S,max} = 0.68$  dB and  $G_{L,max} = 0.76$  dB. Flowchart: §6.4.

## Problem 6: Designing the amplifier: matching networks and total gain 10 18

79 Ch, 69 Bh · the only questions that ask for the networks themselves

### QUESTION AS PRINTED 79 Ch [10]

A GaAs power FET has the S-parameters at 5 GHz with 50-ohm line measured as  $S_{11} = 0.45\angle 163^\circ$ ,  $S_{12} = 0.04\angle 40^\circ$ ,  $S_{21} = 2.55\angle -106^\circ$  and  $S_{22} = 0.46\angle -65^\circ$ . Using these parameters, design an amplifier and find the total gain power gain.

### Step 1 to 4: as Problem 1 (values in Problem 2's tables, 72 Ma row)

- $K = 3.332$ ,  $|\Delta| = 0.306$ : unconditionally stable.
- $\Gamma_S = 0.4011\angle -160.6^\circ$ ,  $\Gamma_L = 0.4131\angle 67.2^\circ$ ,  $G_{T,max} = 1.1917 \times 6.5025 \times 1.2636 = \boxed{9.79 = 9.91 \text{ dB}}$  (total power gain).

### Step 5: input matching network (open shunt stub, then a line)

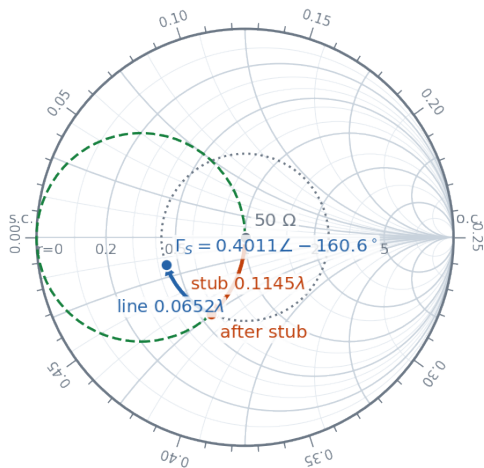
1. Start at the  $50\Omega$  source:  $y = 1$ , the chart centre.
2. A shunt stub moves along the  $g = 1$  circle to  $y = 1 + jb$ ; this must land on the circle  $|\Gamma| = |\Gamma_S| = 0.4011$ :  $b = \frac{2|\Gamma_S|}{\sqrt{1 - |\Gamma_S|^2}} = 0.8757$ .
3. Open stub with  $b = +0.8757$ :  $l_{stub} = \frac{\tan^{-1} 0.8757}{2\pi} = \boxed{0.1145\lambda}$ .
4. At the stub  $\Gamma = \frac{1 - y}{1 + y} = 0.4011\angle -113.6^\circ$ ; a line of length  $l$  rotates it clockwise by  $720l$  degrees to  $\angle -160.6^\circ$ :  $l = \frac{160.6 - 113.6}{720} = \boxed{0.0652\lambda}$ .

### Step 6: output network, the same way

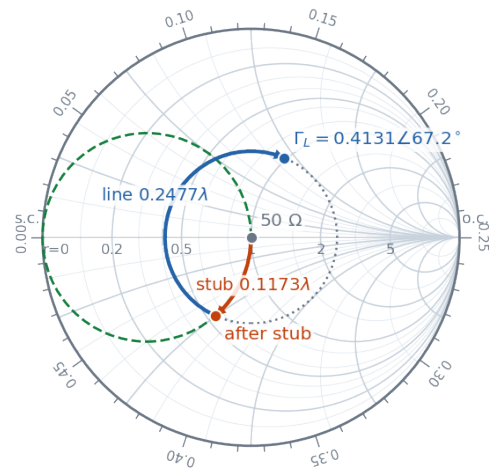
- $b = 0.9072$ , open stub  $\boxed{0.1173\lambda}$ , line  $\boxed{0.2477\lambda}$  to present  $\Gamma_L = 0.4131\angle 67.2^\circ$ .
- Circuit: source  $\rightarrow$  stub  $\rightarrow$  line  $\rightarrow$  FET  $\rightarrow$  line  $\rightarrow$  stub  $\rightarrow$  load, as in the deck's final diagram (§6.4).
- `amp.py` walks both networks forward and recovers  $\Gamma_S$ ,  $\Gamma_L$  exactly.

79 Ch: input network, to  $\Gamma_S$ 

wavelengths toward generator →

 $b = +0.8757$  (open stub), then the line rotates clockwise by  $47.0^\circ$ 79 Ch: output network, to  $\Gamma_L$ 

wavelengths toward generator →

 $b = +0.9072$  (open stub), then the line rotates clockwise by  $178.4^\circ$ 

Steps 5 and 6 on the chart. From the  $50\ \Omega$  centre the open stub moves the point along the admittance  $g = 1$  circle (green) onto the  $|\Gamma_S|$  or  $|\Gamma_L|$  circle (dotted); the line then rotates it clockwise to the required  $\Gamma$ . Drawn by `ampfig.py` from the same numbers, which it checks against the boxed lengths.

## 69 Bh [18], the same design for the 4 GHz GaAs FET

- $K = 1.195$ ;  $\Gamma_S = 0.872\angle 123.4^\circ$ ,  $\Gamma_L = 0.876\angle 61.0^\circ$ ,  $G_{T,max} = 46.9 = 16.71\text{ dB}$ .
- Input: open stub  $0.2064\lambda$ , line  $0.1193\lambda$ . Output: open stub  $0.2073\lambda$ , line  $0.2052\lambda$ . High  $|\Gamma| \rightarrow$  long stubs and a narrow bandwidth.

Problem 7: Stability from given circles, and synthesis on a sketched Smith chart 5 8 10

71 Bh, 73 Bh, 74 Bh, 75 Bh

QUESTION AS PRINTED 71 Bh [5+5]

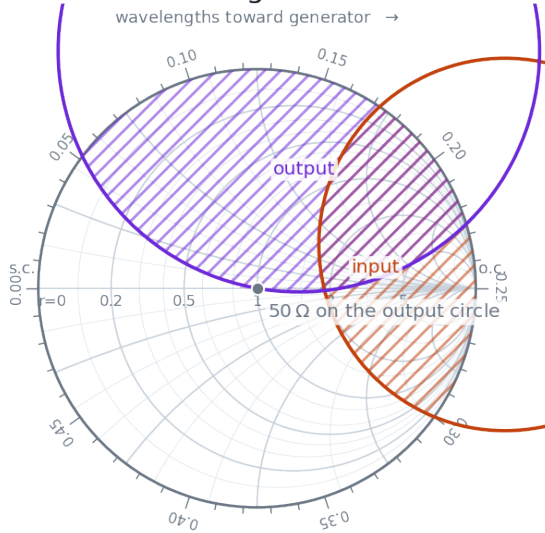
Draw a flow diagram to describe the design procedure of a microwave amplifier. Define the stability of an amplifier having  $C_S = 1.15\angle 10^\circ$ ,  $R_S = 0.85$ ,  $C_L = 1.10\angle 80^\circ$ ,  $R_L = 1.10$ .

## 71 Bh: read the circles against the unit chart

| Circle             | Span of $ \Gamma $ it covers   | Origin                       | Conclusion                                                         |
|--------------------|--------------------------------|------------------------------|--------------------------------------------------------------------|
| Input, $C_S, R_S$  | $1.15 - 0.85 = 0.30$ to $2.00$ | outside<br>( $1.15 > 0.85$ ) | cuts the chart; stable <b>outside</b> (if $ S_{22}  < 1$ )         |
| Output, $C_L, R_L$ | $1.10 - 1.10 = 0$ to $2.20$    | on the circle                | cuts the chart; the $50\ \Omega$ load sits exactly on the boundary |

- Both circles intersect the unit Smith chart  $\Rightarrow$  the amplifier is **conditionally stable**.
- The output circle passing through the centre means  $\Gamma_L = 0$  gives  $|\Gamma_{in}| = 1$  exactly, i.e.  $|S_{11}| = 1$ : a plain  $50\ \Omega$  load is on the edge of oscillation. Choose  $\Gamma_L$  well on the stable side, away from the direction of  $C_L$  ( $80^\circ$ ). verified/added

## 71 Bh: the given circles



hatched = unstable side, taking  $|S_{11}|, |S_{22}| < 1$  (stable outside)

- The sketch the question wants: the unit chart and the two given circles, each **cutting** it. Hatch the unstable side and say which side that is.
- The output circle passes through the centre: that is the  $50\ \Omega$  load sitting on the stability boundary.
- Drawn by `ampfig.py` from the printed  $C$  and  $R$ .

## 73 Bh, 74 Bh, 75 Bh: synthesis from a sketched chart (assume your own numbers)

1. Read each circle's centre  $C$  (distance and angle from the chart centre) and radius  $R$ .
2. Decide the stable side by testing the centre (§6.4).
3. Mark the unstable overlap (hatch it), as in the deck's 800 MHz example:  $C_S = 1.79\angle 122^\circ$ ,  $R_S = 1.04$ ;  $C_L = 1.30\angle 48^\circ$ ,  $R_L = 0.45$ ,  $K = 0.547$ .
4. Choose  $\Gamma_S$  and  $\Gamma_L$  in the stable part, and for gain on the relevant constant-gain circles; confirm  $|\Gamma_{in}|, |\Gamma_{out}| < 1$ .
5. "Stability parameters of the input matching network" (74 Bh) = the input circle's  $C_S$ ,  $R_S$  and the resulting allowed  $\Gamma_S$  region.

Problem 8: Unilateral design for a chosen gain, with assumed S-parameters 10

78 Ch · use the deck's 8 GHz MESFET as the assumed transistor

## QUESTION AS PRINTED 78 Ch [10]

Assuming suitable S-parameters calculate maximum gain of an amplifier working in an unilateral model. Also illustrate a flow chart showing the designing steps.

**Assumed: GaAs MESFET, 8 GHz,**  $S_{11} = 0.52\angle -145^\circ$ ,  $S_{12} = 0.03\angle 20^\circ$ ,  $S_{21} = 2.56\angle 170^\circ$ ,  $S_{22} = 0.48\angle -20^\circ$

| Step                        | Working                                                                                    | Value                   |
|-----------------------------|--------------------------------------------------------------------------------------------|-------------------------|
| Stability                   | $\Delta = 0.173\angle -162.8^\circ$ ;<br>$K = \frac{1 - 0.2704 - 0.2304 + 0.0300}{0.1536}$ | $K = 3.445$ , stable    |
| $G_{S,max}, G_0, G_{L,max}$ | $\frac{1}{0.7296}, 2.56^2, \frac{1}{0.7696}$                                               | 1.371, 6.554, 1.299     |
| $G_{TU,max}$                | $1.371 \times 6.554 \times 1.299$                                                          | <b>11.67 = 10.67 dB</b> |
| $M$                         | $\frac{0.03 \times 2.56 \times 0.52 \times 0.48}{0.7296 \times 0.7696}$                    | 0.0341                  |
| Bound                       | $1/(1.0341)^2$ to $1/(0.9659)^2$                                                           | -0.29 to +0.30 dB       |

- **Deck corrections:** the deck prints  $|\Delta| = 0.168$ ,  $K = 3.53$  and  $M = 0.04$  (bound -0.36 to +0.37 dB); the exact values are those above. The conclusions stand. verified/added
- **For a 10 dB design (deck):**  $g_S = 1.25$ ,  $g_L = 10/(1.25 \times 6.554) = 1.221$ ; constant-gain circles  $C_{gs} = 0.486\angle 145^\circ$ ,  $r_{gs} = 0.222$  and  $C_{gl} = 0.457\angle 20^\circ$ ,  $r_{gl} = 0.192$ ; nearest points  $\Gamma_S = 0.264\angle 145^\circ$ ,  $\Gamma_L = 0.265\angle 20^\circ$  give exactly 10.0 dB; input network open stub  $0.080\lambda$  + line  $0.152\lambda$ , output  $0.080\lambda$  +  $0.326\lambda$  (deck: 0.08, 0.154, 0.082, 0.326).
- Flowchart: §6.4.

## Problem 9: Negative-resistance oscillator numbers

practice from the deck (slides 64 and 66); the theory asks are 70 Ma, 71 Ma, §6.5

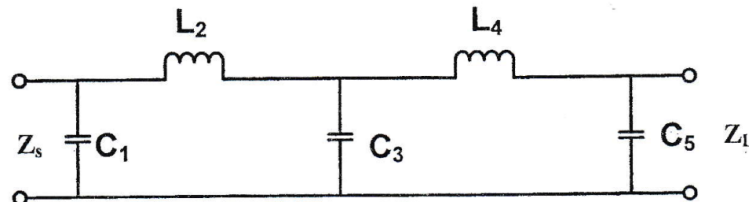
- $Z_{out} = -10\ \Omega$ :  $\Gamma_{out} = \frac{-10 - 50}{-10 + 50} = -1.5 = 1.5\angle 180^\circ$ . Plot  $1/\Gamma_{out}^* = 0.667\angle 180^\circ$ , read  $r = 0.2$ , negate:  $Z_{out} = -10\ \Omega$  ✓.
- **Gunn diode**,  $\Gamma_{out} = 1.24\angle 30^\circ$  at 10 GHz:  $Z_{out} = 50 \frac{1 + \Gamma}{1 - \Gamma} = -68.9 + j159.0\ \Omega$ .
  - Resonate:  $X_L = -159\ \Omega$ ,  $C = \frac{1}{2\pi \times 10^{10} \times 159} = 0.100\ \text{pF}$ .
  - Start-up:  $R_L = |R_{out}|/3 = 23\ \Omega$  (Pozar), or at most  $|R_{out}|/1.2 = 57.4\ \Omega$  by the deck's rule. The deck's  $60\ \Omega$  is too large.

## Problem 10: Design steps of a given fifth-order ladder, in microstrip 3+8+3

82 Bh · the figure gives topology only; specify the rest

QUESTION AS PRINTED 82 Bh [3+8+3]

Explain, why microwave filters are designed using the Insertion Loss Method? Explain the detail design steps of the following microwave filter, and implement the circuit using microstrip.



### Step 1: specification (assumed, and say so)

- The ladder is shunt  $C_1$ , series  $L_2$ , shunt  $C_3$ , series  $L_4$ , shunt  $C_5$ : a **fifth-order low-pass** filter, first element shunt.
- Assume Butterworth,  $f_c = 2.5\ \text{GHz}$ ,  $R_0 = Z_S = Z_L = 50\ \Omega$ , substrate  $\epsilon_r = 4.2$ ,  $h = 1.58\ \text{mm}$ .

### Step 2: prototype and Step 3: scaling

| Element | $g_k$ | Scaling           | Lumped value | Microstrip ( $Z, \beta l, W, l$ )   |
|---------|-------|-------------------|--------------|-------------------------------------|
| $C_1$   | 0.618 | $g/(R_0\omega_c)$ | 0.787 pF     | 20 $\Omega$ , 14.2°, 11.27, 2.49 mm |
| $L_2$   | 1.618 | $R_0g/\omega_c$   | 5.15 nH      | 120 $\Omega$ , 38.6°, 0.43, 7.64 mm |
| $C_3$   | 2.000 | $g/(R_0\omega_c)$ | 2.55 pF      | 20 $\Omega$ , 45.8°, 11.27, 8.07 mm |
| $L_4$   | 1.618 | $R_0g/\omega_c$   | 5.15 nH      | 120 $\Omega$ , 38.6°, 0.43, 7.64 mm |
| $C_5$   | 0.618 | $g/(R_0\omega_c)$ | 0.787 pF     | 20 $\Omega$ , 14.2°, 11.27, 2.49 mm |

$\omega_c = 2\pi \times 2.5 \times 10^9 = 1.571 \times 10^{10}\ \text{rad/s}$ . Capacitor lines:  $\beta l = g Z_l / R_0$ ; inductor lines:  $\beta l = g R_0 / Z_h$ .  $W$  from Hammerstad synthesis,  $l = (\beta l / 360^\circ) \lambda_g$ .

- **Check:** the lumped ladder gives  $|S_{21}| = -3.01\ \text{dB}$  at 2.5 GHz (filt.py).
- **Step 4: implementation:** the double- $\pi$  stepped-impedance layout of §6.3 (three pads, two thin lines), with  $50\ \Omega$  feeds ( $W = 3.13\ \text{mm}$ ) at both ends.
- **Why ILM** (3 marks): §6.1.

## 7. Microwave Antennas and Propagation: Numerical Problems

2 solved problems covering every radiation-zone calculation in 24 papers · 81 Bh, 82 Ba, and the microwave-oven zones of 74 Ma and 77 Ch

READ THIS FIRST

Chapter 7 has one calculation: where the three radiation zones of an antenna begin and end. It is always a 3 or 4 mark part of a longer theory question.

- **The procedure:** (1)  $\lambda = c/f$ , (2) check  $D > \lambda$ , (3)  $R_1 = 0.62\sqrt{D^3/\lambda}$ , (4)  $R_2 = 2D^2/\lambda$ , (5) state the three zones and which are hazardous. Problem 1 shows every line.
- **Do step 2.** When the antenna is smaller than a wavelength (82 Ba’s quarter-wave antennas) the formulas put the “far field” inside the reactive field. Give the formula result, then the small-antenna boundaries  $\lambda/2\pi$  and  $2\lambda$  (Problem 2).
- **Units:** keep  $\lambda$  and  $D$  in metres; the answers come out in metres.
- Every number below is asserted in `src/ant.py`, which also cross-checks the 81 Bh zone answer against an ICNIRP compliance distance.

Problem 1: Hazardous and safe zones of an antenna larger than a wavelength (the full method) 4 3

81 Bh solved in full · 74 Ma, 77 Ch (microwave oven) in the sibling table

QUESTION AS PRINTED 81 Bh [4+4]

Approximate the hazardous and safe zone for BTS tower operating at 2600MHz with antenna size of 0.5 m. Explain, how non-radiating radiation is hazardous to the human body?

The second part is theory: §7.2, “how is non-radiating radiation hazardous”.

Given

- $f = 2600\text{ MHz} = 2.6 \times 10^9\text{ Hz}$ , antenna largest dimension  $D = 0.5\text{ m}$ ,  $c = 3 \times 10^8\text{ m/s}$ .

Formulas

$$\lambda = \frac{c}{f}, \quad R_1 = 0.62\sqrt{\frac{D^3}{\lambda}}, \quad R_2 = \frac{2D^2}{\lambda}$$

$R < R_1$  reactive near field,  $R_1 < R < R_2$  radiating near field (Fresnel),  $R > R_2$  far field. Valid when  $D > \lambda$ .

Step-by-step

| # | Quantity             | Substitution                                         | Value                                 |
|---|----------------------|------------------------------------------------------|---------------------------------------|
| 1 | wavelength $\lambda$ | $\frac{3 \times 10^8}{2.6 \times 10^9}$              | 0.1154 m                              |
| 2 | validity check       | $D = 0.5\text{ m vs } \lambda = 0.1154\text{ m}$     | $D = 4.3\lambda > \lambda \checkmark$ |
| 3 | $D^3/\lambda$        | $\frac{0.5^3}{0.1154} = \frac{0.125}{0.1154}$        | 1.0833 m <sup>2</sup>                 |
| 4 | $R_1$                | $0.62\sqrt{1.0833} = 0.62 \times 1.0408$             | <b>0.645 m</b>                        |
| 5 | $R_2$                | $\frac{2 \times 0.5^2}{0.1154} = \frac{0.5}{0.1154}$ | <b>4.33 m</b>                         |

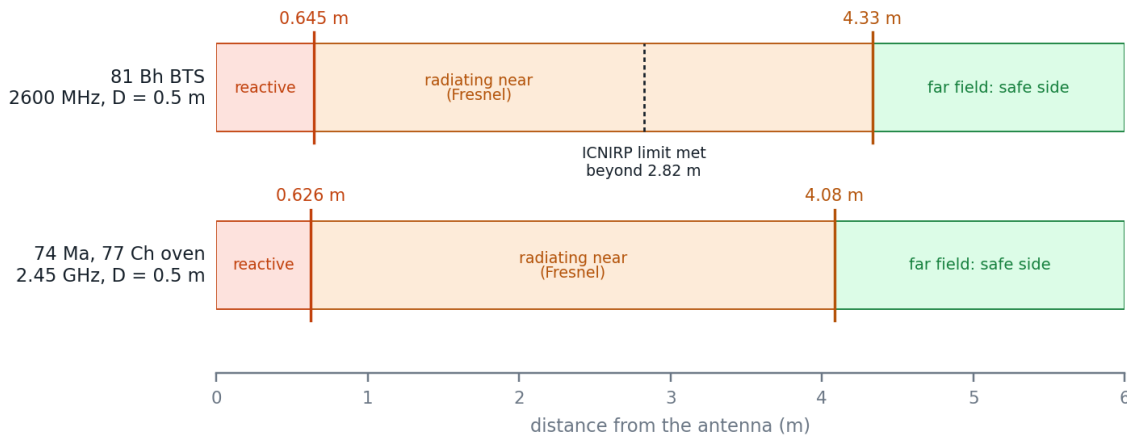
Why step 2: the formulas describe an aperture larger than a wavelength. If  $D < \lambda$  they must not be used (Problem 2).

## Final answer

| Zone                 | Distance from antenna | Verdict                                               |
|----------------------|-----------------------|-------------------------------------------------------|
| Reactive near field  | $R < 0.645$ m         | <b>hazardous</b> (stored-energy coupling)             |
| Radiating near field | $0.645 < R < 4.33$ m  | <b>hazardous</b> (full beam power, not yet spreading) |
| Far field            | $R > 4.33$ m          | <b>safe side:</b> power density falls as $1/R^2$      |

|                                              |                          |
|----------------------------------------------|--------------------------|
| Hazardous zone: within 4.33 m of the antenna | Safe zone: beyond 4.33 m |
|----------------------------------------------|--------------------------|

## The zone figure, to scale



What “approximate the zones” and the oven’s “illustrate the radiation fields” want drawn: the three regions along the distance from the source, boundaries marked. The dashed line is the ICNIRP compliance distance from the cross-check below. Generated by `zonefig.py`, which checks every boundary against this page.

Cross-check against the exposure limit verified/added

- The paper gives no power, so assume a typical sector:  $P_t = 20$  W into a  $G = 17$  dBi = 50.1 panel  $\rightarrow$  EIRP =  $20 \times 50.1 = 1002$  W.
- ICNIRP 1998 public limit above 2 GHz:  $S_{lim} = 10$  W/m<sup>2</sup> (§7.4).
- Compliance distance  $R = \sqrt{\frac{\text{EIRP}}{4\pi S_{lim}}} = \sqrt{\frac{1002}{4\pi \times 10}} = \sqrt{7.98} = 2.82$  m (occupational limit 50 W/m<sup>2</sup>: 1.26 m).
- At the far-field boundary  $S = \frac{1002}{4\pi \times 4.33^2} = 4.25$  W/m<sup>2</sup>, under half the limit.
- So the zone answer (4.33 m) is the **more conservative** of the two: anyone kept beyond the far-field boundary is also within the limit. Every value: `ant.py`.

## Sibling instances: same method, only the numbers change

| Paper         | Given                                                             | $\lambda$ | Reactive / Fresnel / far                       | OK? |
|---------------|-------------------------------------------------------------------|-----------|------------------------------------------------|-----|
| 81 Bh (above) | BTS, 2600 MHz, $D = 0.5$ m                                        | 0.1154 m  | $< 0.645$ m / $0.645$ to $4.33$ m / $> 4.33$ m | yes |
| 74 Ma, 77 Ch  | microwave oven, 2.45 GHz, $D = 0.5$ m (assumed, as the deck does) | 0.1224 m  | $< 0.626$ m / $0.626$ to $4.08$ m / $> 4.08$ m | yes |

## QUESTION AS PRINTED 74 Ma [4+4]

What are the different types of electromagnetic radiation hazard? Illustrate radiation fields of a microwave oven.

## QUESTION AS PRINTED 77 Ch [3+3]

How are microwaves hazardous to humans? Define different radiation zones of a microwave oven.

- Oven working, step by step:  $\lambda = 3 \times 10^8 / 2.45 \times 10^9 = 0.1224$  m;  $D^3 / \lambda = 0.125 / 0.1224 = 1.021$ ;  $R_1 = 0.62 \times 1.010 = 0.626$  m;  $R_2 = 0.5 / 0.1224 = 4.08$  m. The deck prints 0.6266 m and 4.085 m from the same numbers.
- Add the oven-specific point: the metal cavity and door mesh keep leakage below 1 mW/cm<sup>2</sup> at 5 cm, so the zones apply to **leakage** only (§7.1). Draw the zone figure with the oven at the source end.

## Problem 2: Radiation fields of a small (quarter-wave) antenna, 900 MHz vs 1800 MHz 4

82 Ba · solved separately because the Problem-1 formulas break down for  $D < \lambda$

**QUESTION AS PRINTED** 82 Ba [4+2+2]

Compare radiation fields around a mobile BTS operating at 900 MHz and 1800 MHz frequency bands with their quarter-wavelength antennas. Explain how microwave EMR becomes hazardous to humans; and mention its IEEE safety standard.

Parts two and three are theory: §7.2 and §7.4 (IEEE C95.1).

### Given

- $f_1 = 900$  MHz,  $f_2 = 1800$  MHz, antenna length  $D = \lambda/4$  at each frequency.

### Method

- First apply the textbook formulas, as the examiner expects. With  $D = \lambda/4$  they reduce to

$$R_1 = 0.62 \sqrt{\frac{(\lambda/4)^3}{\lambda}} = \frac{0.62 \lambda}{8} = 0.0775 \lambda, \quad R_2 = \frac{2(\lambda/4)^2}{\lambda} = \frac{\lambda}{8} = 0.125 \lambda$$

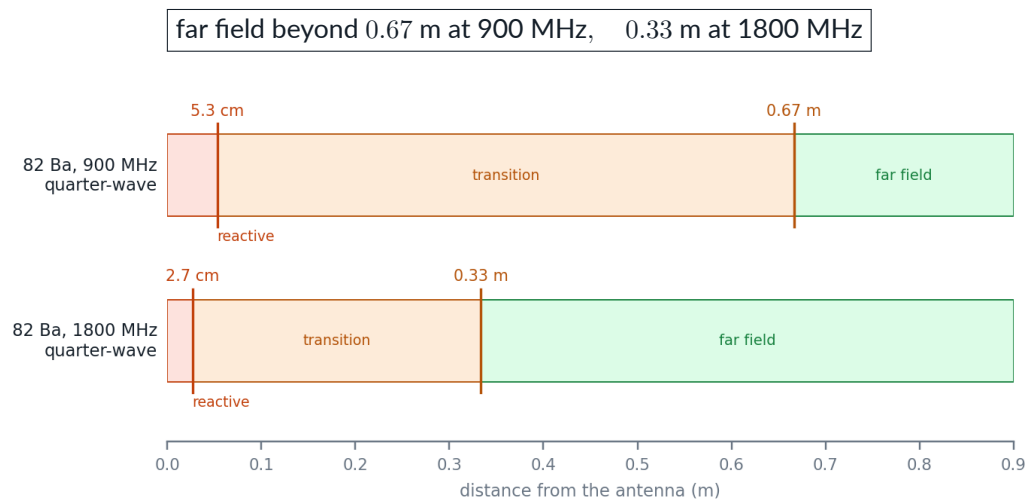
- Then check validity:  $D < \lambda$ , and  $R_2 = 0.125 \lambda$  is inside  $\lambda/2\pi = 0.159 \lambda$ , where the reactive field of any small antenna still dominates. So the “far field” of the formula is not a far field. Use the small-antenna boundaries verified/added : reactive out to  $\lambda/2\pi$ , far field beyond about  $2\lambda$ , a transition region between.

### Step-by-step

| Quantity                    | Formula                              | 900 MHz                | 1800 MHz               |
|-----------------------------|--------------------------------------|------------------------|------------------------|
| wavelength                  | $\lambda = c/f$                      | 0.3333 m               | 0.1667 m               |
| antenna length              | $D = \lambda/4$                      | 8.33 cm                | 4.17 cm                |
| textbook reactive boundary  | $R_1 = 0.0775 \lambda$               | 2.58 cm                | 1.29 cm                |
| textbook far-field boundary | $R_2 = 0.125 \lambda$                | 4.17 cm                | 2.08 cm                |
| validity                    | $D > \lambda?$                       | no                     | no                     |
| reactive near field (use)   | $R < \lambda/2\pi$                   | $R < 5.3$ cm           | $R < 2.7$ cm           |
| transition (radiating near) | $\lambda/2\pi$ to $2\lambda$         | 5.3 to 66.7 cm         | 2.7 to 33.3 cm         |
| far field (use)             | $R > 2\lambda$                       | $R > 0.67$ m           | $R > 0.33$ m           |
| public limit, FCC           | $f/1500$ or $1.0$ mW/cm <sup>2</sup> | 0.6 mW/cm <sup>2</sup> | 1.0 mW/cm <sup>2</sup> |
| public limit, ICNIRP        | $f/200$ W/m <sup>2</sup>             | 4.5 W/m <sup>2</sup>   | 9 W/m <sup>2</sup>     |

### Final answer: the comparison

- Every zone boundary scales with  $\lambda$ , so at 1800 MHz each zone is **half** the size of the 900 MHz one:



The comparison drawn to one scale: every 1800 MHz boundary is half the 900 MHz one. From `zonefig.py`.

- Both antennas are electrically small, so the near-field region is tiny: anyone more than a metre away is in the far field of either, and exposure there is set by power and distance ( $S = P_t G / 4\pi R^2$ ), not by zone.
- **900 MHz**: longer wavelength penetrates deeper into tissue (several cm) and is nearer the body-resonance region → **tighter limit** (0.6 mW/cm<sup>2</sup>).

- **1800 MHz:** absorbed more superficially → limit relaxes to  $1.0 \text{ mW/cm}^2$ . Higher path loss → smaller cells, more sites, lower power per site.

The same break-down hits the deck's second example (150 MHz,  $D = 0.5 \text{ m}$ ,  $\lambda = 2 \text{ m}$ ): it prints reactive  $< 0.155 \text{ m}$  and far field  $> 0.25 \text{ m}$ , but  $0.25 \text{ m}$  is inside  $\lambda/2\pi = 0.32 \text{ m}$ . The small-antenna rule gives reactive  $< 0.32 \text{ m}$ , far field  $> 4 \text{ m}$ . The formulas cross over at  $D = 0.28\lambda$  (`ant.py`).

## Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short   | Means                                | Short    | Means                                   |
|---------|--------------------------------------|----------|-----------------------------------------|
| RF      | radio frequency                      | MW, M/W  | microwave                               |
| EM, EMR | electromagnetic (radiation)          | TEM      | transverse electromagnetic mode         |
| TE      | transverse electric mode             | TM       | transverse magnetic mode                |
| WG      | waveguide                            | TL       | transmission line                       |
| VSWR    | voltage standing wave ratio          | SWR      | standing wave ratio                     |
| VNA     | vector network analyzer              | SA       | spectrum analyzer                       |
| DUT     | device under test                    | S-params | scattering parameters                   |
| RL      | return loss                          | IL       | insertion loss                          |
| ML      | mismatch loss                        | NF       | noise figure                            |
| BW      | bandwidth                            | SNR      | signal to noise ratio                   |
| Tx, TX  | transmitter                          | Rx, RX   | receiver                                |
| BTS     | base transceiver station             | GSM      | global system for mobile communications |
| EIRP    | effective isotropic radiated power   | ERP      | effective radiated power                |
| SAR     | specific absorption rate             | SARPs    | standards and recommended practices     |
| HERP    | hazards of EM radiation to personnel | RADAR    | radio detection and ranging             |
| ASR     | airport surveillance radar           | CW       | continuous wave                         |

| Short     | Means                                                       | Short   | Means                                                   |
|-----------|-------------------------------------------------------------|---------|---------------------------------------------------------|
| TWT, TWTA | traveling wave tube (amplifier)                             | BWO     | backward wave oscillator                                |
| CFA       | cross-field amplifier                                       | LNA     | low noise amplifier                                     |
| HPA       | high power amplifier                                        | SWS     | slow wave structure                                     |
| O-type    | linear beam tube (axial $B$ -field)                         | M-type  | crossed-field tube ( $E \perp B$ )                      |
| MASER     | microwave amplification by stimulated emission of radiation | LASER   | light amplification by stimulated emission of radiation |
| MESFET    | metal-semiconductor FET                                     | HEMT    | high electron mobility transistor                       |
| GaAs      | gallium arsenide                                            | FET     | field effect transistor                                 |
| BJT       | bipolar junction transistor                                 | PIN     | p-type, intrinsic, n-type diode                         |
| IMPATT    | impact ionization avalanche transit-time                    | TRAPATT | trapped plasma avalanche triggered transit              |
| LPF       | low pass filter                                             | HPF     | high pass filter                                        |
| BPF       | band pass filter                                            | BSF     | band stop filter                                        |
| MAG       | maximum available gain                                      | MSG     | maximum stable gain                                     |
| MIC       | microwave integrated circuit                                | MMIC    | monolithic microwave integrated circuit                 |
| ADS       | advanced design system                                      | IF      | intermediate frequency                                  |
| LO        | local oscillator                                            | RFIC    | radio frequency integrated circuit                      |
| PCB       | printed circuit board                                       | SMA     | subminiature version A connector                        |

| Short    | Means                | Short  | Means                            |
|----------|----------------------|--------|----------------------------------|
| attr(s)  | attribute(s)         | no.    | number                           |
| freq     | frequency            | info   | information                      |
| ops      | operations           | config | configuration                    |
| rm       | remove               | vs, Vs | versus                           |
| w/, w/o  | with, without        | Eg     | example / for example            |
| +Fig     | draw a figure        | (I)    | importance / need / significance |
| →        | leads to / therefore | +      | and                              |
| =        | is / equals          | norm   | normalized                       |
| max, min | maximum, minimum     | calc   | calculation / calculate          |
| eq, eqn  | equation             | approx | approximately                    |
| pts      | points               | coeff  | coefficient                      |
| temp     | temperature          | params | parameters                       |

**Standard and component terms** (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ( $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ ) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

**Key mathematical symbols:**  $Z_0$  characteristic impedance ·  $Y_0$  characteristic admittance ·  $\Gamma$  reflection coefficient ·  $S$  scattering matrix ·  $K$  Rollett's stability factor ·  $\Delta$  determinant of S-matrix ·  $\mu$  single parameter stability metric ·  $G_T$  transducer gain ·  $G_{TU}$  unilateral transducer gain ·  $f_c$  cutoff frequency ·  $\lambda_g$  guide wavelength ·  $v_p$  phase velocity ·  $v_g$  group velocity.

**Exam-paper month codes** (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

**Bold** year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

**Tier chips** over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth  $\geq 6$  marks.

**Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.